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University Pierre et Marie Curie

Thesis submitted for the degree

Master in Mathématiques Fondamentales

∞-Operadic methods

towards Factorization Homology

by

Andrea Marino

Supervisor: **Benjamin Hennion**

June 2019

Abstract

∞-Operadic methods towards Factorization Homology

The aim of this thesis is to introduce factorization homology with coefficients in a $\mathbb{E}_n$ algebra, e.g. CDGAs. Take a manifold M and a properly defined algebraic structure A, over a ground category that admits a tensor product. One defines a sort of locally constant (preco)sheaf P_A - precisely, a locally constant Disk_M-algebra - that assigns to every disk the object A; for disjoint disks $U_1, \ldots, U_n$ in V, there is a map $A \otimes \ldots \otimes A \to A$ that is meant to encode the multiplication on A. Further constraint are imposed to encode the entire algebraic structure of A. Finally, one can extend P_A to every open, allowing sections on small opens to glue together via a general tensor product. (Derived) global sections of this extension are called factorization homology, or topological chiral homology.

The present work gives a solid formalization to the above ideas in an ∞ setting, where commuting diagrams are substitued by *homotopy coherent* diagrams. Instead of only having points and arrows, we have n-cells for every n and constraints between all of them. A reader interested in approaching the ∞ world - especially in operads - will take profit from the first two chapters as a springboard to more advanced material, e.g. Lurie's Higher Algebra. He will be introduced to fundamental tools like ∞ (operadic) Kan extensions, weak approximation, families of (reduced) operads. The third chapter is more specific, where a basic treatment of little cube operads over manifolds is given. Finally, the last chapter puts together all the developed theory to define factorization homology and study some simple properties.

Acknowledgment

This long and hard trip would not have been possible without the precious aid of Professors and friends. Firstly, I would like to thank Ginot and Harpaz for their clear classes, respectively on model categories and ∞ categories: they opened me the doors to higher structures. Also, they did not back off when I needed some further explanations.

Professor Hennion, my advisor, had the patience to discuss with me from the greater theorems to the little details, and constantly guided me with suggestions and directions. Let me say that our blackboard sessions are still central in my intuitive comprehension of the subject. Other informal discussions happened with Roy Magen on prefactorization algebras, and with Denis Nardin on general doubts about homotopy theory.

Last but not least, thanks to my friends and my family, that strongly supported me during the hard times spent in writing down the material.

To Davide Tittarelli,

my eternal soul friend

Contents

Abstract **ii**

Acknowledgment **iii**

Contents **v**

Introduction **1**

1 Infinity Categories **3**

1.1 Homotopy category 5

1.2 Fibrations 8

1.3 The ∞ category of ∞ categories 12

1.4 Transition functors 13

1.5 Join and slices 14

1.6 Relative colimits 16

1.7 Cofinality 18

1.8 Relative Kan extensions 20

1.9 Symmetric monoidal Categories 21

2 Operads **28**

2.1 Strict operads as multicategories 29

2.2 Fibration-like version 32

2.3 Infinity operads 33
2.4 Infinity Category of operads 39
2.5 Weak Approximation 42
2.6 Families of operads 50
2.7 Disintegrating unital operads 54
2.8 Relative operadic colimits 56
2.9 Relative operadic Kan extensions 59

3 Little cube operads 64

3.1 Definition 64
3.2 Homotopy groups of map spaces 66
3.3 The whole series 68
3.4 Examples and fundamental theorems 70
3.5 Generalization to manifolds 74

4 Factorization Homology 79

4.1 Strict prefactorization algebras 80
4.2 $\mathbb{E}_M$ algebras and Disk_M algebras 81
4.3 Factorization Algebras 84
4.4 Factorization homology 87
4.5 Hochschild homology 90

Bibliography 99

Introduction

The only prerequisite to read the present work is to have some confidence with simplicial sets and category theory. In some points some knowledge of CW complexes, manifold and topology is preferable. All the rest is meant to be self contained, from basic ∞ category theory to factorization homology; somewhere we use homotopy colimits in chain complexes and spaces, but a newbie can read them as "colimits with some weird construction more" without no risk.

The first chapter deals with the fundamental tools of ∞ category theory. We illustrate how to pass from categories to ∞ categories and viceversa, and we give a fast introduction to the main models of ∞ categories. We define various notions of fibrations, with the intention to go towards Grothendieck Lurie strainghtening-unstraightening. Relative colimits and Relative Kan extensions are studied as the main technical tool for the rest of the work, with some tecnhiques to reduce the complexity of computation (e.g. cofinality). At the end, we talk about symemtric monoidal categories as a springboard to the next chapters.

The second chapter is about operads. We decided to start from the definition of classical operad, that gives some intuition about what we do in the ∞ setting. An operad abstractly encodes the instructions to build up a certain kind of algebraic structure; for example, there exist an operad Ass that regulates the construction of associative algebras. The main theorem we show is the weak approximation theorem, that permits to compare

different operads and relate the structure they define. Relative operadic colimits and relative operadic Kan extension must be thought as generalizations of colimits and Kan extensions, where the $\oplus$ that appears in computing a finite discrete colimit is essentially substituted by a $\otimes$. Last but not least, we explain how to unpack an operad in a family of operads with trivial unary operations.

The third chapter illustrate the most famous example of ∞ operads, i.e. $\mathbb{E}_n$ operads. They are a new phenomena of the ∞ setting. In the ordinary setting, an associative algebra is commutative or it isn't at all. Here, there is an increasing level of possible commutativity: so $\mathbb{E}_1$ are associative algebras, and $\mathbb{E}_\infty$ are commutative algebras, but we have an entire chain $\mathbb{E}_1 \subset \mathbb{E}_2 \subset \ldots \mathbb{E}_\infty$. A generalization $\mathbb{E}_M$ to manifolds $M^{(n)}$ is presented: this can be thought as a family of $\mathbb{E}_n$ operads, one for every disk $\simeq \mathbb{R}^n$ living on the manifold, with some compatibility conditions when we have an inclusion of disks.

The last chapter is geometrically and algebraically more significant, and we have the opportunity to see the efficiency of the formalism developed in the previous chapters. We show that $\mathbb{E}_M$ algebras A are equivalent to locally constant Disk_M algebras F; the latter being thought as assigning to every disk a vector space, with multiplication maps $F(U) \otimes F(V) \to F(W)$ whenever U, V are disjoint and contained in W. We can think of extending somehow the disk algebra to an algebra defined over all the opens of the manifold. Roughly speaking, the value of F extended on an open U is given by the colimit of $F(U_1) \otimes \ldots \otimes F(U_n)$, as $(U_1, \ldots, U_n)$ varies in the tuples of disjoint opens contained in U. In particular, taking the global sections, we obtain an invariant $\int_M F$ called *factorization homology* or *topological chiral homology*. We show the significance of this new invariant by recovering the Hochschild Homology in case $M = S^1$ and A an associative algebra.

Well, enjoy the long trip!

Chapter 1

Infinity Categories

Infinity categories arise as an attempt to model both categories and (nice) spaces with the same kind of structure. In this spirit, we can say that they are at the intersection of algebra and homotopy theory. Some purely categorical constructions remain unvaried, while others are modified so that they acquire an homotopical meaning. A straightforward example of this phenomena is the creation of homotopy (co)limits, which takes into account the fact that we deal with an homotopical context. In spaces, for example, the homotopy pushout of X, Y along Z is the gluing of X, Y along the cylinder of Z. Referring to the figure, maps from the homotopy pushout to a fourth space W are the pairs of maps $f : X \to W, g : Y \to W$ such that $f|Z$ is *homotopic* to $g|Z$.

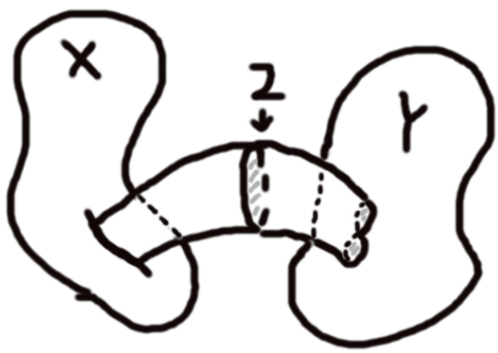


Figure 1.1: Homotopy pushout along a disk, with collapsed endings

There are mainly five models for infinity categories: Segal spaces, Segal complete spaces, topological categories, locally Kan simplicial categories, quasi categories (simplicial sets with some lifting property). We will not use the Segal models; thus we only give an account for the latter three models. All the model categories these objects fit into are Quillen equivalent; roughly speaking, there is a way to pass from one to another without changing the homotopy story going on.

Topological categories are categories with hom-sets being topological spaces, and composition being continuous. This is the easiest generalization of both spaces and categories, but it turns out to be not comfortable for a number of applications.

We can take the singular chain space of all the hom-spaces, and we get a category where the hom sets are Kan complexes, composition being simplicial. This is the second model for infinity categories. One can take the coherent nerve $\mathcal{N}_{coh}$ of a simplicial category X to get a simplicial set, with some additional properties; precisely, one gets a quasicategory. To define the coherent nerve, one consider a fancy cosimplicial simplicial category $\{\mathfrak{C}[n]\}$, a modification of the poset $[n]$, and set $\mathcal{N}_{coh}(X)_n := \mathrm{Hom}_{sCat}(\mathfrak{C}[n], X)$.

However, this is not the best way to introduce quasi categories. To get a first insight, let's take a category C and try to get a simplicial set. We somehow have a one skeleton (objects and arrows); to take into account the composition inside the category, we declare that a n-simplex is a commutative diagram which has the shape of a n-simplex (objects $0 \dots n$, increasing arrows, ...). Formally, the latter category is the poset $[n] = \{0, \dots, n\}$ of $n+1$ totally ordered objects, and we define $\mathcal{N}(C)_n = \mathrm{Hom}_{Cat}([n], C)$. The $[n]$ is a cosimplicial category, thus $\mathcal{N}$ is a functor from C to simplicial sets which has a left adjoint.

We verify that $\mathcal{N}(C)$ has the filling property with respect to inner horns $\Lambda^n_k \to \Delta^n$; indeed, a n- simplex is completely determined by the 1-dimensional spine $0 \to 1 \to \dots \to n$, hence inner horns, that contains the spine, can always be (uniquely) filled.

On the other hand, we can take the singular chain space of a topological space and end up with a simplicial set that have horn filling property for *all* the horns, but not in a unique way. The property that is exactly in the middle of the two is to have inner horn fillings, not necessarily unique. This is the definition of a quasi-category, and it will be the main model we adopt in the thesis.

1.1 Homotopy category

We set up the definition of an ∞ category once and for all. Recall that Λ^n_k is Δ^n without the interior and the face opposite to k (try to picture Λ^2_1, Λ^3_2).

Definition 1.1.1. Let C be a simplicial set. It is said to be a ∞ category if for every $n, 0 < k < n$ and diagram

$$\begin{tikzcd} \Lambda^n_k \arrow[r] \arrow[d] & C \\ \Delta^n \arrow[ur, dotted] & \end{tikzcd}$$

the dotted lift exist.

Let X be an ∞ category. We can define the map simplicial set between two objects x, y as the pullback

$$\{x\} \times_{\Delta^{\{0\}}} \mathrm{Map}_{sSets}(\Delta^1, X) \times_{\Delta^{\{1\}}} \{y\}$$

Moreover, we say that two arrows $f, g : x \to y$ are homotopic ($\simeq$) if they enters a two cell σ such that:

$$\sigma(01) = f, \sigma(02) = g, \sigma(12) = 1 := s_0(y)$$

We claim that if we define $\tau(X)$ as the graph that has for objects X_0 and for hom sets $\mathrm{Map}(x, y)/\simeq$ we get a category. Given $f : x \to y, g : y \to z$ two arrows,we say that a map h is a composite of f, g if it enters a two cell σ such that

$$\sigma(01) = f, \sigma(12) = g, \sigma(02) = h$$

The composition is well defined because of this lemma:

Lemma 1. *Let X be an infinity category. Let $f : x \to y$, $g : y \to z$ two arrows, and let h be a composite of f, g.*

1. *If h' is another composite of f, g, then it is homotopic to h.*
2. *Suppose $f' : x \to y$ and $g' : y \to z$ are homotopic respectively to f, g. Then h is also a composite of f', g'.*
3. *Define $[f] \circ [g]$ as $[h]$. Then the composition is associative in $\tau(X)$.*

Proof. We start with (1). Take the 3 horn σ given by

$$\sigma(01) = f,\ \sigma(12) = g,\ \sigma(02) = h$$

$$\sigma(13) = g,\ \sigma(23) = 1, \sigma(03) = h'$$

Where all the 2-cells are naturally filled, beside the one corresponding to the vertex 1. The horn filling condition gives us the remaining two cell, that determines the homotopy between h, h'.

Let's prove point (2). It is enough to show that h is a composite of g, f'; the method to pass to f', g' is completely analogous. Take a the 3-horn τ given by

$$\tau(02) = f',\ \tau(23) = g,\ \tau(03) = h$$

$$\tau(12) = f,\ \tau(01) = 1,\ \tau(13) = h$$

All faces are filled but the 1-th face. Filling the horn yield the desired 2-cell composition between f', g, h.

We conclude with (3). Consider maps f, g, h and composites α, β of f, g and g, h respectively. Finally, consider a composite γ of α, h. We organize these data in a 3-horn

ρ such that

$$\rho(01) = f,\ \rho(12) = g,\ \rho(23) = h$$

$$\rho(02) = \alpha,\ \rho(13) = \beta,\ \rho(03) = \gamma$$

As above, all the 2-cells are filled beside the 2-th face. Filling the horn yields that γ is also a composite of f, β, which is the thesis. $\square$

The category $\tau(X)$ is called the homotopy category of X. We say that an arrow is invertible, or an equivalence, in X if it is an iso in the homotopy category $\tau(X)$.

Now we notice that if we have a map of simplicial sets from an infinity category X to the nerve $\mathcal{N}(C)$ of an ordinary category, two homotopic maps are sent to the same arrow, and a composite is sent to a composite. Thus we have an induced map $\tau(X) \to C$. More precisely, we have the following:

Lemma 2. *The functors $\tau : \infty Cat \to Cat : \mathcal{N}$ forms a pair of adjoint functors. Moreover, the counit is an equivalence, and express Cat as a full subcategory of ∞Cat.*

Proof. By general nerve-realization machinery [9], it is enough to show that $\tau(\Delta^n) = [n]$ and that τ is cocontinuous. Indeed, the adjunction is completely determined by $\tau|\Delta$. Moreover, the only cocontinuous functor that extend $\tau|\Delta$ is its Kan extension.

It is evident from the definition that $\tau(\Delta^n) = [n]$: it has the same objects, and there is a unique arrow $x \to y$ for every $x \leq y$, and a unique two cell $x \to y \to z$ for every $x \leq y \leq z$.

Also, τ preserves disjoint unions. It remains to show that τ preserve coequalizers. Let $f, g : X \to Y$ be maps of infinity categories, and let Q be their coequalizer. As colimits are computed point-wise in simplicial sets, Q_0, Q_1, Q_2 are the coequalizers of $(f_0, g_0), (f_1, g_1), (f_2, g_2)$ respectively. The homotopy category is defined in terms of cells up to dimension 2, so we are done.

For the last point, note that the counit $\tau\mathcal{N}(C) \to C$ is the identity on objects. On arrows, note that two maps $f, g; x \to y$ in the nerve are homotopic iff they are equal, so that

$$\mathrm{Hom}_{\tau\mathcal{N}(C)}(x, y) = \mathrm{Map}_{\mathcal{N}(C)}(x, y)/\simeq = \mathrm{Map}_{\mathcal{N}(C)}(x, y)_0 =$$

$$= \{x\} \times_{\Delta^{\{0\}}} \mathrm{Hom}(\Delta^1, \mathcal{N}(C)) \times_{\Delta^{\{1\}}} \{y\} \simeq \{x\} \times_0 \mathrm{Hom}([1], C) \times_1 \{y\} \simeq \mathrm{Hom}_C(x, y)$$

and we conclude. □

It follows in particular that the nerve is continuous. Moreover, we have the following continuity of the homotopy category:

Lemma 3. *Suppose X, Y are infinity categories. Then $\tau(X \times Y) \simeq \tau(X) \times \tau(Y)$.*

Proof. It comes from the very definition that $(f, g) \simeq (f', g')$ iff $f \simeq f', g \simeq g'$. □

Furthermore, the objects $[n]$ are compact, meaning that every morphism

$$[n] \to \mathrm{colim}_{k \in \mathbb{N}} X_k$$

factors through an X_k. This gives us the following cocontinuity property of the nerve:

Corollary 1.1.1.1. *Suppose J is a filtered category and $J \to \mathrm{Cat}$ is a diagram. Then*

$$\mathcal{N}(\mathrm{colim}_{j \in J} C_j)_n \simeq Hom([n], \mathrm{colim} C_j) \simeq \mathrm{colim} Hom([n], C_j) \simeq \mathrm{colim} \mathcal{N}(C_j)$$

1.2 Fibrations

Definition 1.2.1. An inner fibration $C \to D$ of simplicial sets is a simplicial map that has the right lifting property with respect to all the inner horn inclusions $\Lambda^n_k \to \Delta^n$.

Remark 1. An ∞ category is exactly a simplicial set inner fibrant over the point. As maps with some right lifting property are stable for pullback, we see that all the fibers of

an inner fibration are infinity categories. In this sense, an inner fibration can be thought as a family of infinity categories.

The following definition is sometimes useful, but will be refined by the one of (co)cartesian morphisms.

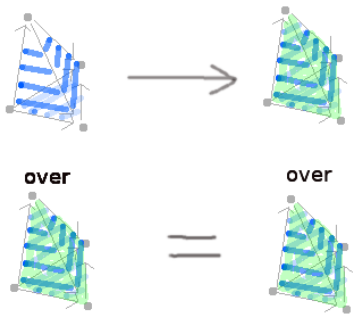


Figure 1.2: (3,2) horn filling

Definition 1.2.2. A right fibration is a map having the right lifting property with respect to inner horns and the right outer horn $\Lambda^n_n \to \Delta^n$. A left fibration has the right lifting property with respect to inner horns and left outer horn $\Lambda^n_0 \to \Delta^n$.

The notion of (co)cartesian fibrations relies on the definition of a (co)cartesian morphism f.

Definition 1.2.3. Let $p : C \to D$ be an inner fibration. We say that an arrow $f \in C$ is p-cocartesian if every left horn inclusion $\Lambda^n_0 \to \Delta^n$ can be filled in $C \to D$ whenever the 01 arrow of Λ^n_0 maps to f.

Dually, we have:

Definition 1.2.4. Let $p : C \to D$ be an inner fibration. We say that an arrow $f \in C$ is p-cartesian if every right horn inclusion $\Lambda^n_n \to \Delta^n$ can be filled in $C \to D$ whenever the $(n-1)n$ arrow of Λ^n_n maps to f.

We simply say (co)cartesian when the fibration p is evident from the context. Finally, we can define what a cocartesian fibration is:

Definition 1.2.5. A cocartesian fibration $p : C \to D$ is an inner fibration such that there is a sufficient supply of cocartesian lifts. Namely, for every $f : x \to y$ in D, and for every lift $\bar{x}$ of the domain, there exist a cocartesian lift $\bar{f} : \bar{x} \to \bar{y}$ of f.

Dually, we have:

Definition 1.2.6. A cartesian fibration $p : C \to D$ is an inner fibration such that there is a sufficient supply of cartesian lifts. Namely, for every $f : x \to y$ in D, and for every lift $\bar{y}$ of the codomain, there exist a cartesian lift $\bar{f} : \bar{x} \to \bar{y}$ of f.

We collect here a few lemmas that will be useful in the following.

Lemma 4. *(Joyal). Let $C \to D$ be an inner fibration, and $f \in C$ a lift of the identity. Then f is cocartesian iff it is an equivalence iff it is cartesian.*

We have two straightforward corollaries:

Corollary 1.2.6.1. *Let C be an infinity category. Then $f \in C$ is cocartesian iff it is an equivalence iff it is cartesian.*

Proof. Just apply the lemma to the case in which $D = *$. □

Corollary 1.2.6.2. *Let C be an infinity category. Then C is a Kan complex iff all the arrows are invertible, i.e. it is an ∞ groupoid.*

Proof. In fact, left horns with first arrow f can be filled iff f is an equivalence by the previous corollary; analogously, right horns with last arrow f can be filled iff f is an equivalence. On balance, outer horns can be filled iff all the arrow are equivalences. □

Lemma 5. *Suppose $p : C \to D$ is an inner fibration, $f : x \to y \in D$ an arrow, and $g : \bar{x} \to z, g' : \bar{x} \to z'$ two cocartesian lifts. Then they are equivalent.*

Proof. Drawings are strongly encouraged to follow the proof. Take the $\sigma : \Lambda^2_0 \to C$ horn such that $\sigma(01) = g, \sigma(02) = g'$. The image $p\sigma$ can be filled by a degenerate 2-cell, so by

hp (g is cocartesian) we can fill σ with a two cell $\bar{\sigma}$ such that $\bar{\sigma}(12) = \alpha$.

Roughly speaking, we found an α such that g' is a composite of α, g. In the same way, we can find a β such that g' is a composite of β, g'. Build the following $\theta : \Lambda^3_0 \to C$ horn such that:

- $\theta(01) = g', \theta(02) = g, \theta(03) = g'$;
- $\theta(13) = 1, \theta(12) = \beta, \theta(23) = \alpha$;
- $\theta(013)$ is the degenerate 2-cell corresponding to g';
- $\theta(012), \theta(023)$ are the 2-cell obtained above, that relates α, β with g, g'.

The image $p\theta$ of this horn can be filled, via the 0 degenerate 3-cell corresponding to f. By hp (g' is cocartesian) we can fill this horn, yielding in particular that 1 is a composite of β, α.

In the same fashion, exchanging the roles of g, g', we get a 2-cell exhibiting 1 as a composite of α, β also; thus they are invertible.

On balance, g, g' are obtained one from each other by composing with an equivalence, thus they are equivalent. □

We quote the last result from [11], §2.4.1.7, in its dual version:

Proposition 1.2.7. *Let $p : C \to D$ be an inner fibration, and $\rho : \Delta^2 \to C$ a 2-cell. Suppose $\rho^{\{0,1\}}$ is p-cocartesian. Then $\rho^{\{1,2\}}$ is p cocartesian iff $\rho^{\{0,2\}}$ is p-cocartesian.*

Let's call x, y, z the three vertices of ρ, in increasing order. Note that the cocartesian map $\rho^{\{0,1\}}$ yield an equivalence

$$\mathrm{Map}_C(y, z) \times_{\mathrm{Map}_D(py,pz)} \{p\rho^{\{1,2\}}\} \to \mathrm{Map}_C(x, z) \times_{\mathrm{Map}_D(px,pz)} \{p\rho^{\{0,2\}}\}$$

Obtained by composition. The cocartesian condition gives an inverse. What proposition 1.2.7 says is that this equivalence preserve cocartesian maps.

1.3 The ∞ category of ∞ categories

Let us see how infinity categories fit themselves into an infinity category. We say that an infinity category G is an infinity groupoid if all its arrows are invertible. It follows form the Joyal lemma that an infinity groupoid is a Kan complex.
Given an infinity category C, we set $C^{\simeq}$ as the full sub infinity category (in fact, a infinity groupoid) spanned by invertible arrows.

In particular, given two infinity categories C, D we can form a Kan complex of maps $\mathrm{Fun}(C, D) := \mathrm{Map}(C, D)^{\simeq}$ between them. This turn qCat into a locally Kan simplicial category qCat^{Δ}, and we can get as usual an infinity category setting $\mathrm{Cat}_{\infty} := \mathcal{N}_{coh}(\mathrm{qCat}^{\Delta})$

We turn our attention to equivalences of infinity categories.

Definition 1.3.1. Let $F : C \to D$ be a functor of infinity categories. We say that:

- F is fully faithful if the induced map $\mathrm{Map}(x, y) \to \mathrm{Map}(Fx, Fy)$ is an equivalence of simplicial sets for every $x, y \in C$.
- F is essentially surjective if the induced functor $\tau(F) : \tau(C) \to \tau(D)$ is essentially surjective.

As in ordinary categories, we have the following

Lemma 6. *$F : C \to D$ is an equivalence in Cat_{∞} iff it is fully faithful and essentially surjective.*

We will see that a particular class of functors, namely the ones towards Cat_{∞}, admit a correspondence that will shed light on the previous chapter of fibrations.

Let's take a cocartesian fibration $\phi : C \to S$. Since it is an inner fibration, the fiber $\phi^{-1}(s)$ is an infinity category for any s. In next lemmas we show that if $s \to t$ is a

map, then there exist a *transition functor* $\phi^{-1}(s) \to \phi^{-1}(t)$ lifting $s \to t$. This leads to think that one can obtain a functor $\mathrm{St}(\phi) : S \to \mathrm{Cat}_\infty$. This is the case. In [11], §3.2, a full account of this correspondence is given. We will only describe how to construct the transition functors, that is sufficient for main applications.

1.4 Transition functors

Suppose we have a cocartesian fibration $C \to D$. We want to show that, in some sense, homotopies in D lifts to homotopies in C. Formally:

Lemma 7. *Let K be a simplicial set, and consider a lifting problem of the form*

$$\begin{tikzcd} K \times \Delta^{\{0\}} \arrow[r, "\tilde{f}"] \arrow[d] & C \arrow[d, "p"] \\ K \times \Delta^1 \arrow[r, "f"] \arrow[ur, dotted] & D \end{tikzcd}$$

If p is a cocartesian fibration, then the lifting problem admits a solution. Moreover, any pair of liftings are equivalent.

Proof. After choosing a cocartesian lift for any $f(\{k\} \times \Delta^1)$ based at $\tilde{f}(\{k\} \times \Delta^{\{0\}}$, the result follows from [6], §2.6.17. The equivalence between any two functors can be proved in a similar way. One can consider the point-wise degenerate two cell $T : K \times \Delta^2 \to D$, and the horn $\tilde{T} : K \times \Lambda^2_0 \to D$. Lifting T to $\bar{T}$ simplex by simplex, we choose a an arrow $\bar{T}(\{k\} \times \Delta^{12})$ that connects two cocartesian liftings of the same arrow; by Joyal Lemma, this is an equivalence. □

In particular, we can apply the lemma to get functors between the fibers of p associated to arrows in the base. Namely, let $f : p^{-1}(d) \times \Delta^1 \to \Delta^1 \to D$ the map pointing at an arrow $g : d \to e$. Let $\tilde{f} : p^{-1}(d) \times \Delta^{\{0\}} \to C$ be the inclusion of the fiber. The existence of a lifting yields precisely the existence of a functor $g^! : p^{-1}(d) \to p^{-1}(e)$, that is called a transition functor. Every pair of transition functors are moreover equivalent.

On the other hand, the *relative nerve* construction (see [6], §2.6.21), furnishes a way to pass from functors to Cat_∞ to cocartesian fibrations, in the case when the base is an ordinary category. See also [11], §3.2.5, for an exhaustive treatment, or 23 for an example of application.

1.5 Join and slices

In order to generalize the slice categories to the infinity setting, we have to introduce what is the join of two simplicial sets. Roughly, the idea is that the geometric realization $|X \star Y|$ should be thought informally as the space of formal linear combinations $\{tx + (1-t)y : t \geq 0, x \in |X|, y \in |Y|\}$. We thus will have a $n+m-1$ simplex for every $n-1$ simplex in X and $m-1$ simplex in Y (think about the case $n = 1, m = 2$). Formally we define

$(X \star Y)_n := \coprod_{i=-1}^{n} X_i \times Y_{n-1-i}$

where we agree that $X_{-1} = \{*\}$. Note that in particular the operation is not commutative, because arrows in the middle are oriented from the first simplicial set to the second. This definition is useful for actual proofs, but conceptually is easier to think at $\star$ as the only bifunctor such that:

- $\Delta^{n-1} \star \Delta^{m-1} \simeq \Delta^{n+m-1}$ via some fixed ψ_{nm};
- $T \mapsto T \star S, T \mapsto S \star T$ are cocontinuous as functors from simplicial sets to simplicial sets over S.

We have the following result due to Joyal:

Lemma 8. *Let C, C' be ∞ categories. Then $C \star C'$ is an infinity category.*

Proof. Take an inner horn in $C \star C'$. If vertices are contained in one of the two sSets, we are done. Otherwise, if $0.. \ldots i$ are in the first sset and $i+1 \ldots n$ in the second, there are two cases:

- both $0..i$ and $i+1..n$ are filled; then their join defines the desired fill;

- if they miss one face, one of them should be $n-1$ dimensional; wlog, $i=0$. But in this case the missing $n-1$ face is not $1\dots n$ because the horn is inner, thus we are done.

$\square$

We are ready to define slice categories.

Definition 1.5.1. Let $p : K \to C$ be a functor. Define the over category $C_{/p}$ as $(C_{/p})_n := \mathrm{Hom}_p(\Delta^n \star K, C)$, where $\mathrm{Hom}_p(X, Y)$ with $K \to X, K \to Y$ are functions that preserve the copy of K in the two sSets.

The definition, by the Yoneda Lemma, can be rephrased as

$$\mathrm{Hom}(\Delta^n, C_{/p}) \simeq \mathrm{Hom}_p(\Delta^n \star K, C)$$

By cocontinuity in Δ^n, this extends to an adjunction

$$\mathrm{Hom}(X, C_{/p}) \simeq \mathrm{Hom}_p(X \star K, C)$$

Furthermore, we have a canonical projection $C_{/p} \to C$ given by $f \in \mathrm{Hom}_p(\Delta^n \star K, C)$ is mapped to $f|\Delta^n$.

Dually, we have:

Definition 1.5.2. Let $p : K \to C$ be a functor. Define the under category $C_{p/}$ as $(C_{p/})_n := \mathrm{Hom}_p(K \star \Delta^n, C)$

As above, we have the adjunction

$$\mathrm{Hom}(X, C_{/p}) \simeq \mathrm{Hom}_p(K \star X, C)$$

When $p : \Delta^0 \to C$ corresponds to a point x, we denote $C_{p/}$ by $C_{x/}$. The two main lemmas we will need about slice categories are the following:

Lemma 9. *Let C be an ordinary category, $x \in C$. Then $\mathcal{N}(C_{x/}) \simeq \mathcal{N}(C)_{x/}$ and analogously for the over category. Here $C_{x/}$ is the category of maps from x.*

Proof. Left as an exercise. □

Lemma 10. *Let $x \in C$ be an element of an infinity category. Then $C_{x/} \to C$ is a left fibration.*

Proof. The data of an horn $\Lambda^n_k \to C_{x/}$ corresponds by adjunction to a map from $\Delta^0 \star \Lambda^n_k \to C$ This object is Δ^{n+1} with four missing faces: the old two $1 \dots n+1, 1 \dots k \hat{+} 1 \dots n+1$ and the new two $0 \dots n+1, 0 \dots k \hat{+} 1 \dots n+1$. The additional data of $\Delta^n \to C$ amounts to fill the two faces in the base of the cone, i.e. $1 \dots n+1$ and $1 \dots k \hat{+} 1 \dots n+1$.

On balance, we have now a map from $\Lambda^{n+1}_{k+1} \to C$. This can be filled for $0 \leq k < n$, yielding by adjunction the desired lifting. □

Remark 2. Note that left fibrations, being in particular cocartesian fibrations, should correspond by Lurie-straightening to functors towards Cat_∞. The formula is roughly given by taking fibers. In particular, $C_{x/} \to C$ corresponds to the functor $y(\mapsto C_{x/})_y$; the latter is an equivalent model for $\mathrm{Map}_C(x, y)$, so that these fibrations correspond to *representable functors*.

1.6 Relative colimits

The first step to generalize colimits to the ∞ categorical setting is to define a good notion of initial object. This is the easier type of colimit, but it contains all the informations to define a colimit.

Definition 1.6.1. Let C be an ∞ category, $x \in C$ an object. x is said to be initial if for every $y \in C$ the sSet $\mathrm{Map}_C(x, y)$ is weakly contractile.

Remark 3. The definition evidently dualizes for final objects. In the following, we will only deal with "left" version of definitions and proofs, and we leave to the reader the dualization of statements.

The definition of colimit now follows, because it is an initial object in the category of cocones.

Definition 1.6.2. Let $p : K \to C$ be a diagram. A colimit of p is an initial object of $C_{p/}$.

Indeed, you can imagine that elements of $C_{p/}$ are cocones over p, and $\bar{p} \in C_{p/}$ being initial barely means that for every other cocone q there exist an essentially unique morphism of cocones $\bar{p} \to q$. This motivates the following proposition.

Proposition 1.6.3. *Let $\bar{p} : K^{\triangleright} \to C$ be a map, and set $p = \bar{p}|K$. Then $\bar{p}$ is a colimit of p if and only if*

$$C_{\bar{p}/} \to C_{p/}$$

is a trivial Kan fibration.

Proof. Suppose that $C_{\bar{p}/} \to C_{p/}$ is a trivial Kan fibration. Then it is in particular an equivalence, and the image of the initial object $1_{\bar{p}} \in C_{\bar{p}/}$ yields an initial object of $C_{p/}$. On the other side, note that $C_{\bar{p}/} \simeq (C_{p/})_{\bar{p}/}$. The result now follows from [11], corollary 1.2.12.5. It is a reasonable fact, because it is always a left fibration (see 10) and fibers are weakly contractile by the hypothesis of being initial. □

We now introduce the notion of *relative* colimit, that generalizes the above proposition.

Definition 1.6.4. Let $q : C \to D$ be an inner fibration, and $\bar{p} : K^{\triangleright} \to C$ a map of simplicial sets. Set $p = \bar{p}|K$. $\bar{p}$ is said to be a colimit diagram if

$$C_{\bar{p}/} \to C_{p/} \times_{D_{qp/}} D_{q\bar{p}/}$$

is a trivial Kan fibration.

Remark 4. By Remark [11], 4.3.1.2, the definition is equivalent to the map being a categorical equivalence.

Intuitively, this means that a map of cones $\bar{p} \to c$ is essentially determined by its projection $q\bar{p} \to qc$.

Remark 5. Note that for $D = *$, the fiber product on the right boils down to $C_{p/}$, so a $*$-relative colimit is just a colimit.

We finish this section with the following criterion:

Lemma 11. *Given an inner fibration $q : C \to D$, a map $p : K^{\triangleright} \to C$ is a q-colimit diagram iff*

$$\begin{tikzcd} C_{\bar{p}/} \arrow[r] \arrow[d] & C_{p/} \arrow[d] \\ D_{q\bar{p}/} \arrow[r] & D_{qp/} \end{tikzcd}$$

is homotopy cartesian.

This is [11], 4.3.1.2.

1.7 Cofinality

In this section we introduce a technique to reduce the difficulty of computation of (relative) colimits.

Definition 1.7.1. Let $v : K' \to K$ be a map of simplicial sets. v is said to be left cofinal if for every ∞ category C and every map $p : K \to C$

$$C_{p/} \to C_{p'/}$$

is an equivalence, where $p' = pv$.

A map $f : C \to D$ is right cofinal if f^{op} is left cofinal (explicitly, one has to turn all the over categories in under categories). In the following we will only deal with left

cofinality, simply called cofinality, and we leave to the reader to dualize them. The main use of cofinality comes from the following lemma:

Lemma 12 (Cofinal maps determine the colimit)**.** . *Suppose $q : C \to D$ is an inner fibration. Let $v : K' \to K$ be a cofinal map, and $\bar{p} : K^{\triangleright} \to C$ a map. Then $\bar{p}$ is a colimit diagram iff $\bar{p}v^{\triangleright}$ is a colimit diagram.*

Proof. By remark 11, $\bar{p}$ is a q-colimit diagram iff

$$\begin{tikzcd} C_{\bar{p}/} \arrow[r] \arrow[d] & C_{p/} \arrow[d] \\ D_{q\bar{p}/} \arrow[r] & D_{qp/} \end{tikzcd}$$

is homotopy cartesian. Since both $v, v^{\triangleright}$ are cofinal maps, and homotopy cartesian diagrams are stable by equivalence, this is equivalent to

$$\begin{tikzcd} C_{\bar{p}v^{\triangleright}/} \arrow[r] \arrow[d] & C_{pv/} \arrow[d] \\ D_{q\bar{p}v^{\triangleright}/} \arrow[r] & D_{qpv/} \end{tikzcd}$$

being homotopy cartesian, which is the definition of $\bar{p}v^{\triangleright}$ being a q-colimit diagram. □

Since cofinal maps are apparently difficult to find, we expose a criterion due to Joyal, generalizing a result of Quillen:

Theorem 1.7.2 (Quillen Theorem A for ∞ categories)**.** *Suppose $v : C \to D$ is a map of simplicial sets, where D is an ∞ category. The following conditions are equivalent:*

1. *The functor v is cofinal;*

2. *For every object $d \in D$, the simplicial set $C \times_D D_{d/}$ is weakly contractile.*

This is [11], 4.1.3.1.

We finally turn our attention to relative Kan extensions.

1.8 Relative Kan extensions

The general theory of (relative) left Kan extensions in the ∞ setting require the introduction of the ∞ category of left extensions to D of a functor $F : C \to E$, along a given functor $G : C \to D$. This is meant to model the pairs (F', δ), where $F' : D \to E$ and $\delta : F \to F'G$. The theory simplifies a lot when G is a full and faithful inclusion, the only case we will need; that's why we make this assumption for the rest of the section.

Definition 1.8.1. Suppose we have an inner fibration $q : D \to D'$, and full and faithful inclusion of infinity categories $C^0 \to C$. A functor $f : C \to D$ is said to be a q-left Kan extension of $f|C^0$ if, for every $X \in C$, the diagram

$$\begin{tikzcd} C^0_{/X} \arrow[r, "f^0_X"] \arrow[d, hook] & D \arrow[d, "q"] \\ (C^0_{/X})^{\triangleright} \arrow[ur, "f_X"] & D' \end{tikzcd}$$

express f_X as the q-colimit of f^0_X. Here:

- $C^0_{/X} := C_{/x} \times_C C^0$;
- $f^0_X : C^0_{/X} \to C_0 \to D$;
- f_X is the join of f^0_X and the map sending the cone point to X.

Remark 6. Note that if $X \in C^0$, the condition is automatically satisfied. Indeed, $C^0_{/X}$ has a final object given by 1_X, so that its colimit is exactly X.

A natural question on Kan extensions is the following: what does it happen to Kan extensions in a tower of inclusion of ∞ categories $C^0 \subset C^1 \subset C$?

Lemma 13 (Pasting lemma)**.** *Suppose $q : D \to D'$ is a categorical fibration, $C^0 \subset C^1 \subset C$ full inclusions of ∞ categories, and $f : C \to D$ a functor. Moreover, suppose that $f|C^1$ is a q-left Kan extension of $f|C^0$.*

Then f is a q-left Kan extension of $f|C^1$ iff it is a q left Kan extension of $f|C^0$.

A proof can be found in [11], 4.3.2.8.

The second key question is the following: given a functor $F_0 : C^0 \to D$, how many possible Kan extensions to C there are? It turns out that the extension is essentially unique, given that a "point-wise" extension exist, and it is detailed in the next result.

Lemma 14. *Let $C^0 \subset C$ be a full inclusion of ∞ categories, and suppose we are given a diagram of ∞ categories*

$$C \to D' \xleftarrow{q} D$$

Where q is a categorical fibration. Let $\mathcal{K} \subset Map_{D'}(C, D)$ be the sSet of functors F that are q-left Kan extensions of $F|C_0$. Let $\mathcal{K}' \subset Map_{D'}(C^0, D)$ be the sSet of functors $F : C^0 \to D$ such that, for every $X \in C$, the map $F : C^0_{/X} \to D$ admits a q colimit.

Then the restriction functor $\mathcal{K} \to \mathcal{K}'$ is a trivial Kan fibration.

A proof can be found in [11], 4.3.2.15.

1.9 Symmetric monoidal Categories

Symmetric monoidal categories are meant to endow a category with a general bifunctor $\otimes : C^2 \to C$ that behaves like a tensor product, i.e. it is commutative and associative. For example, we will have a one arrow

$$X_i \otimes (X_j \otimes X_k) \to (X_i \otimes X_j) \otimes X_k$$

relating the two posible three-variable operations. In an ordinary category setting, this natural transformation is called the associator, and the one swapping the two variables the commutator. These transformations must be isomorphisms and respect some compatibility axioms. For a precise definition, see [8], VII, 1.

Being in an infinity setting, we should also control higher homotopies. That's really unpleasant: we do not want to deal with an infinite amount of polyhedra of axioms.

To deal with this, the idea is the following: instead of having one tensor product, we have a contractile Kan complex of tensor product functors. Moreover, for every choice of tensor products $\otimes_1, \otimes_2, \otimes_3$ there exist a fourth $\otimes_4$ such that

$$(X_i \otimes_1 X_j) \otimes_2 X_k = X_i \otimes_4 (X_j \otimes_3 X_k)$$

Without any involved equivalence. In other words, if C is the ∞ category we are working on, the following diagram is commutative (in Cat_∞):

$$\begin{tikzcd} C^3, \arrow[r, "\otimes_1 \times Id"] \arrow[d, "Id \times \otimes_3"] & C^2 \arrow[d, "\otimes_4"] \\ C^2 \arrow[r, "\otimes_2"] & C \end{tikzcd}$$

In this way, the "infinity stuff" is moved to organize these contractile Kan complexes of functors in a coherent structure. As we will see in the next chapter, we can think to extend this operation to an operad encoding also n-ary operations like

$$(X_1, \ldots, X_4) \to (((X_1 \otimes X_2) \otimes X_3) \otimes X_4)$$

This organizes in the following category.

Definition 1.9.1. Let Fin_* be the skeleton of pointed finite sets. Explicitly:

- Objects are $\langle n \rangle := \{0 \ldots n\}$, for $n \geq 0$;
- Maps are functions that preserve the zero.

Denote by $\langle n \rangle^0 := \{1, \ldots, n\}$ the set without the base point.

Remark 7. As you will see in the following, the base-point is there for this reason. It gives the possibility to define functions $\langle n \rangle^0 \to \langle m \rangle^0$*not everywhere*, by sending an undesired element to the base-point.

We say that $\alpha : \langle n \rangle \to \langle m \rangle$ is inert if for every $i \in \langle m \rangle^0$ we have $|\alpha^{-1}(i)| = 1$. An

inert map correspond to an embedding $\alpha^{-1} : \langle m\rangle^0 \to \langle n\rangle^0$. We say that α is active if $\alpha^{-1}(0) = 0$ (everywhere defined functions).

For $n \in \mathbb{N}, 1 \leq i \leq n$, define the i-Kronecker map $\rho_i : \langle n\rangle \to \langle 1\rangle$ the inert map given by $\rho_i(j) = 0$ if $j \neq i$, $\rho_i(i) = 1$.

We can finally state what is an infinity symmetric monoidal category.

Definition 1.9.2. An ∞ symmetric monoidal category is a cocartesian fibration $p : C^{\otimes} \to \mathcal{N}(\mathrm{Fin}_*)$ such that transition functors $\rho_i^! : C^{\otimes}_{\langle n\rangle} \to C^{\otimes}_{\langle 1\rangle}$ over the Kronecker maps yield an equivalence

$$(\rho_1^!, \ldots, \rho_n^!) : C^{\otimes}_{\langle n\rangle} \to (C^{\otimes}_{\langle 1\rangle})^n$$

Remark 8. Note that we can equivalently require that "there exist transition functors" or "for every transition functor", as every two transition functors over the same map are equivalent.

We call $C := C^{\otimes}_{\langle 1\rangle}$ the underlying category. Transition functors over $\langle 2\rangle \to \langle 1\rangle$ which sends 2,1 to 1 yield a contractile family of functors $\otimes : C^2 \to C$, as we wanted. Moreover, the property that for every choice of three tensor products there exist a fourth making the diagram commute comes from the following reasoning. Choose transition functors

- F_1 over $\delta^{23} : \langle 3\rangle \to \langle 2\rangle$ sending $3, 2$ to 1, and identity on the rest. It is meant to model $(X, Y, Z) \to (X, Y \otimes Z)$.
- F_3 over $\delta^{12} : \langle 3\rangle \to \langle 2\rangle$ sending $1, 2$ to 1 and 3 to 2. It is meant to model $(X, Y, Z) \to (X \otimes Y, Z)$.
- F_2 over $\delta : \langle 2\rangle \to \langle 1\rangle$ sending $2, 1$ to 1.

By construction of transition functors, for every object (X_1, X_2, X_3) of C^3 there exist cocartesian liftings d^{23}, d^{12}, d of $\delta^{23}, \delta^{12}, \delta$ such that

$$\begin{tikzcd}
(X, Y, Z) \arrow[r, "d^{12}"] \arrow[d, "d^{23}"] & F_3(X, Y, Z) \\
F_1(X, Y, Z) \arrow[r, "d"] & F_2F_1(X, Y, Z)
\end{tikzcd}$$

By firstly extending to the diagonal arrow (which is cocartesian as composition of cocartesian, see 1.2.7), and then getting the right vertical map by cocartesian condition, we get a cocartesian map d' (again by 1.2.7) fitting into the commutative diagram. Making X, Y, Z vary in a proper way, this family of cocartesian maps can be promoted to a cocartesian functor by the routine of homotopy lifting along cocartesian maps (7, [6] §2.6.17), yielding the same commutative diagram with functors.

The main source of examples are ordinary symmetric monoidal categories. Indeed, let C be an ordinary symmetric monoidal category. Define a new category $C^{\otimes}$ in the following way:

- Objects are tuples $(\langle n\rangle, X_1, \dots, X_n)$ where $\langle n\rangle \in \mathrm{Fin}_*$;
- Morphism are given by

$$\mathrm{Hom}_{C^{\otimes}}((\langle n\rangle, X_1, \dots, X_n), (\langle m\rangle, Y_1, \dots, Y_m)) = \coprod_{\alpha:\langle n\rangle\to\langle m\rangle} \prod_{i\in\langle m\rangle^0} \mathrm{Hom}_C(\otimes_{\alpha(j)=i} X_j, Y_i)$$

In this context, we have the following:

Proposition 1.9.3. *Given a symmetric monoidal category $(C, \otimes)$, the ∞ category $\mathcal{N}(C^{\otimes})$ yield an ∞ symmetric monoidal category.*

The proof relies on the same ideas of 2.3.1; details are left to the reader.

Remark 9. One can also give a definition of simplicial symmetric monoidal categories. We can perform the $\otimes$ construction on the latter by substituting Hom_C with Map_C, obtaining a simplicial category. Taking the coherent nerve we get an infinity category: it is a ∞ symmetric monoidal category.

This provides a lot of examples: modules/sheaves with tensor product, chain complexes with tensor product, spaces with disjoint union, simplicial sets with product, ecc.

In general, one can see the following:

Example 1. *An infinity category C that admit finite products can be turned into a symmetric monoidal category.*

We here develop the formalism for the *cartesian structure* on C that makes the work.

Definition 1.9.4. Let $\Gamma^\times$ be the following category:

- objects are pairs $(\langle n\rangle, I)$, where $I \subset \langle n\rangle^0$;
- maps between $(\langle n\rangle, I)$ and $(\langle m\rangle, J)$ are $\alpha : \langle n\rangle \to \langle m\rangle$ such that $\alpha^{-1}(J) \subseteq I$

We have an obvious forgetful functor $\Gamma^\times \to \mathrm{Fin}_*$. We define a simplicial set $\bar{C}^\times$ in the following way. Its n-simplices are pairs (σ, τ) with $\sigma : \Delta^n \to \mathcal{N}(\mathrm{Fin}_*)$ and $\tau : \Delta^n \times_{\mathcal{N}(\mathrm{Fin}_*)} \mathcal{N}(\Gamma^\times) \to C$, where the fibered product is taken with respect to σ.

To give an idea of who $\bar{C}^\times$ is, let's see what are its points. It is the data of a $\langle n\rangle \in \mathcal{N}(\mathrm{Fin}_*)$, and a map from $\mathcal{N}(\Gamma^\times_{\langle n\rangle}) \simeq \mathcal{N}(\Gamma^\times)_{\langle n\rangle} \to C$. The latter fiber can be computed explicitly. Its points are tuples $(\langle n\rangle, I)$, $I \subset \langle n\rangle^0$ (n is fixed). Its arrows are $\alpha : \langle n\rangle \to \langle n\rangle | \alpha^{-1}(J) \subset I$, with α inducing the identity on $\langle n\rangle$. In other words, $\Gamma^\times_{\langle n\rangle}$ is equivalent to $\mathcal{N}(\mathcal{P}_n)$, the nerve of the poset $\mathcal{P}_n$ of subsets $I \subset \langle n\rangle^0$ with reverse order.

Generally, a n simplex consists of a functor $\sigma : [n] \to \mathrm{Fin}_*$, i.e. n increasing maps $\langle q_0\rangle \xrightarrow{f_1} \dots \xrightarrow{f_n} \langle q_n\rangle$ generating a subcategory $\mathcal{Q}_\sigma \subset \mathrm{Fin}_*$, and a functor from $\mathcal{N}(\mathcal{P}_\sigma) \to C$. The latter poset $\mathcal{P}_\sigma$ consists of subsets of $\langle q_i\rangle^0$, and $I \leq J$ iff there exist a map $f \in \mathcal{Q}_\sigma$ such that $f^{-1}(J) \subseteq I$. Note that $\mathcal{P}_\sigma$ is generated by the relations $I \leq J$ whenever $f_i^{-1}(J) \subseteq I$ (they belong to two adjacent $q_i's$) and $J \subseteq I$ (they belong to the same q_i).

Definition 1.9.5. The cartesian symmetric monoidal category $C^\times$ is defined as the full subsimplicial subset of $\bar{C}^\times$ of vertices $x : \mathcal{N}(\mathcal{P}_n) \to C$ such that the maps $x(I) \to x(\{i\})$ for $i \in I$ express $x(I)$ as the product of the $x(\{i\})$'s. It is a non trivial fact that $C^\times$ is symmetric monoidal, see [10] 2.4.1.5, point 4.

In a fairly easier way, one can define $C^{\amalg}$ for C endowed with finite coproducts.

Define Γ^* as the following category:

- Objects are pairs $(\langle n\rangle, i)$, with $\langle n\rangle \in \mathrm{Fin}_*$ and $i \in \langle n\rangle^0$;
- maps $(\langle n\rangle, i) \to (\langle m\rangle, j)$ are morphisms $\langle n\rangle \to \langle m\rangle$ in Fin_* that sends i to j.

Definition 1.9.6. Let C be an ∞-category. We define an ∞ category $C^{\amalg}$ with n simplices given by pairs

$$(\sigma : \Delta^n \to \mathcal{N}(\mathrm{Fin}_*), f : \Delta^n \times_{\mathcal{N}(\mathrm{Fin}_*)} \mathcal{N}(\Gamma) \to C)$$

This has an evident projection to $\mathcal{N}(\mathrm{Fin}_*)$ by projecting on the first component.

Remark 10. For every map of simplicial sets $\Delta^n \to \mathcal{N}(\mathrm{Fin}_*)$, we have by definition

$$\mathrm{Hom}_{/\mathcal{N}(\mathrm{Fin}_*)}(\Delta^n, C^{\amalg}) \xrightarrow{\simeq} \mathrm{Hom}_{\mathrm{sSet}}(\Delta^n \times_{\mathcal{N}(\mathrm{Fin}_*)} \mathcal{N}(\Gamma^*), C)$$

This is cocontinous in Δ^n, so that we get an adjunction

$$\mathrm{Hom}_{/\mathcal{N}(\mathrm{Fin}_*)}(K, C^{\amalg}) \xrightarrow{\simeq} \mathrm{Hom}_{\mathrm{sSet}}(K \times_{\mathcal{N}(\mathrm{Fin}_*)} \mathcal{N}(\Gamma^*), C)$$

For every $K \to \mathcal{N}(\mathrm{Fin}_*)$.

With the following three proposition, we show that if C admits coproducts and a neutral element for coproducts, $C^{\amalg}$ is an ∞ symmetric monoidal category. If there are not coproducts, it is simply an ∞ operad (see the next chapters). We have separated the lifting of inert maps to show that in this case coproducts are not necessary. We encourage the reader to see what is the fiber of $C^{\amalg}$ over a vertex $\langle n\rangle$, and what is the fiber over a a map $\langle n\rangle \to \langle m\rangle$ before getting into the proofs.

Proposition 1.9.7. *For any $\alpha : \langle n\rangle \to \langle m\rangle$ inert and $(X_1, \ldots, X_n) \in C^n$ over $\langle n\rangle$,there exist a cocartesian lift (α, f) from $X \to Y$.*

Proof. Suppose that α corresponds to an embedding $\beta : \langle m \rangle^0 \to \langle n \rangle^0$. Note that maps over α departing from X are collections of maps $f_i : X_{\beta(j)} \to Y_j$ for $(Y_1, \ldots, Y_m) \in C^m$. By choosing $Y_j = X_{\beta(j)}, f_j = Id$, one can verify that it is a cocartesian lifting (essentially because is initial among such maps). □

Proposition 1.9.8. *Let $\bar{\rho}_i : X \to X_i$ be cocartesian liftings of ρ_i. Then the transition functor $\prod \rho_i^! : C^n \to C^n$ is an equivalence.*

Proof. Making reference to the cocartesian liftings of the above proposition, we have that $\rho_i^!$ are exactly the projection on the i-th component, yielding the desired equivalence. □

Proposition 1.9.9. *Suppose that C admits coproducts and a neutral element for $\oplus$. Then for any $\alpha : \langle n \rangle \to \langle m \rangle$ and $(X_1, \ldots, X_n) \in C^n$ over $\langle n \rangle$,there exist a cocartesian lift (α, f) from $X \to Y$.*

Proof. Define $Y_j = \oplus_{\alpha(i)=j}$, where 0 is meant where the $\oplus$ is empty. Then we have a map $X \to Y$ over α, where the components $f_i : X_i \to Y_j$ are the inclusions of X_i in the coproduct. One can show that this lifting is cocartesian, basically because it is initial in the set of liftings. □

See [10], 2.4.3 for a complete account.

Chapter 2

Operads

In the following we give an account of what an operad is. Essentially, an operad abstractly encodes the data of a certain algebraic structure: we have an operad which models associate algebras, lie algebras, commutative algebras... We refer to these operads as *strict* operads, i.e. relations in the structure they define holds with a real "=" sign.

As we are working in an "infinity" setting, there are analogues of ordinary operads that encodes for example associative algebras up to *coherent homotopy*. We point out here the subtle difference between "a relation holds up to homotopy" and "a relation holds up to coherent homotopy". In the latter, indeed, we consider not only homotopies between the two members of a relation, but also homotopies between homotopies, homotopies between homotopies between homotopies...

We will develop some tools to compare different infinity operads; in particular, we are interested in whether or not the algebra structures they define are equivalent. This goes under the name of weak approximation theorem.

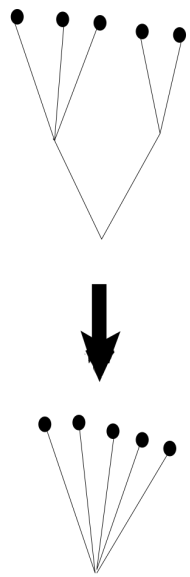
We will see how one can build up a family of ∞ operads; unlike the categorical case, this will no longer be an operad itself, and so we will cook up a new definition for operads

parameterized by an ∞ category. Then, we show that any operad with trivial nullary operations can be seen as a family of reduced operads, i.e. with trivial unary operations too.

In the end, we develop a generalization of colimits and Kan extensions to the operadic case; intuitively, one can think that the operadic colimit of a finite discrete diagram in a symmetric monoidal category is given by the tensor product in place of the coproduct.

2.1 Strict operads as multicategories

The definition we are going to give is commonly referred as "symmetric colored operad". What is commonly referred as "operad" is our operad with a single object (see below).



Definition 2.1.1. An operad $\mathcal{O}$ consists of:

- A set of object Ob $\mathcal{O}$;
- For every $x, y_1, \ldots, y_n \in \mathcal{O}$ a set

$$\mathrm{Mul}_{\mathcal{O}}(\{y_i\}_{i=1,\ldots,n}, x)$$

  of multimaps from $\{y_i\}_{i=1,\ldots,n}$ to x.
- A composition law: for every $x \in \mathcal{O}, \{y_i\}_{i\in I} \subset \mathcal{O}, \{z_j\}_{j\in J} \subseteq \mathcal{O}$ and a function $\alpha : J \to I$, a composition map $\mathcal{O}_\alpha$ along α:

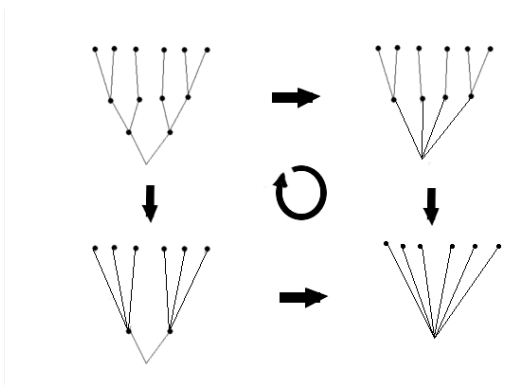
$$\prod_{i\in I} \mathrm{Mul}_{\mathcal{O}}(\{z_j\}_{\alpha(j)=i}, y_i) \times \mathrm{Mul}_{\mathcal{O}}(\{y_i\}_{i\in I}, x) \quad \to \quad \mathrm{Mul}_{\mathcal{O}}(\{z_j\}_{j\in J}, x)$$

  such that it is *associative*. This means that given $\beta : K \to J, \alpha : J \to I$, if K_i denotes $\beta^{-1}\alpha^{-1}(i)$ then $\mathcal{O}_{\alpha\beta}(\{\mathcal{O}_{\beta|K_i}(\cdot,\cdot)\},\cdot) =$

$\mathcal{O}_\beta(\cdot, \mathcal{O}_\alpha(\cdot,\cdot))$. We leave to the reader to fill in the dots with proper variables.

- a preferred object $1_x \in \mathrm{Mul}_{\mathcal{O}}(x,x)$ called the identity, which is neutral wrt composition.

Remark 11. Beside the ugly definition, there is a fancy way to depict operads which is way more clear. For a multimap $f : \{y_i\} \to x$, draw n arrows from the y_i's to x. We call this a *corolla*. Take n corollas based at y_i, and a corolla from the y_i to x: composition yields a bigger corolla. Associativity tells us that there is essentially one way to compose consecutive corollas. As an exercise, try to interpret a tree rooted in x and with leafs $y_1, \ldots, y_n$ as a multimap from $y_1, \ldots, y_n$ to x.



Remark 12. Given a monoidal category $(C, \otimes)$, we can define an operad $\mathcal{O}$ over C as an operad where multimaps are objects of C, and composition is a morphism in C, where product is substituted by $\otimes$. This must not be confused with $\mathcal{O}-$algebras over a symmetric monoidal category, see example 2.

We have seen what is the structure of an operad. Let's see what is a morphism of operads.

Definition 2.1.2. A morphism of operads $F : \mathcal{O} \to \mathcal{P}$ is a map from the objects of $\mathcal{O}$ to the objects of $\mathcal{P}$, and for every $x_1, \ldots, x_n, y \in \mathcal{O}$ a map $\mathrm{Mul}_{\mathcal{O}}(x_1, \ldots, x_n, y) \to \mathrm{Mul}_{\mathcal{P}}(F(x_1), \ldots, F(x_n), F(y))$. It must preserve the structure, i.e.:

- it is functorial with respect to composition, namely $F(\mathcal{O}_\alpha(\cdot,\cdot)) = \mathcal{P}_\alpha(F(\cdot), F(\cdot))$;
- It preserves identities, namely $F(1_x) = 1_{F(x)}$;

Here we have an important example of operad, which plays the same role of the general linear group with respect to representations of groups.

Example 2 (Strict symmetric monoidal category)**.** *Given $\mathcal{C}$ a strict smc, we can set*

$$Mul(\{x_1, \dots, x_n\}, y) := Hom(x_1 \otimes \dots \otimes x_n, y)$$

and composition of corollas is defined by taking the tensor product of maps. In particular, sets equipped with products admit a structure of operad.

The above example is important for the following reason. We want to think that an algebraic structure on a set is the same as a morphism of operads from some operad $\mathcal{O}$ encoding the algebraic structure to Sets equipped with product. In the same fashion, a ring should be a morphism from some "associative operad" Ass to abelian groups equipped with tensor product. Indeed, this would yield a bilinear associative operation on some abelian group. Let's see a couple of concrete examples.

Example 3 (Commutative operad)**.** *Let Comm be a one object operad such that multimaps are given by*

$$Mul((*)_{i=1,\dots,n}, *) := \{*\}$$

Then a functor of operads $F : Comm \to Sets$ picks one object $X \in Sets$, and for every n a unique $\mu_n : X^n \to X$. Take a permutation σ of n elements. The composition along σ of n identities with a multimap $f : \{*\}_{i=1,\dots,n} \to \{*\}$ is the same multimap f because multimaps in Comm have just one element. Applying the functor F, this gives that μ_n are symmetric with respect to permutation of variables. In particular, $\mu_2(x, y) = \mu_2(y, x)$. Moreover, $\mu_2(x, \mu_2(y, z)), \mu_2(\mu_2(x, y), z)$ are ternary operation, hence they coincide with $\mu_3(x, y, z)$. We got that the functor F defines an associative and commutative operation μ_2.

On the other side, we can reconstruct $\mu_n(x_1, \dots, x_n)$ *as* $\mu_2(\dots \mu_2(\mu_2(x_1, x_2), x_3) \dots, x_n)$. *On balance, a functor* $F : Comm \to Sets$ *is the same as a commutative associative operation on a set.*

Exercise 1. *Formalize the one object operad Lie over vector spaces equipped with tensor product.*

2.2 Fibration-like version

Given two tuples of objects $\{y_i\}_{j \in J}, \{x_j\}_{i \in I}$, we can define

$$\mathrm{Mul}_{\mathcal{O}}(\{y_j\}_{j \in J}, \{x_i\}_{i \in I}) = \coprod_{\alpha : J \to I} \prod_{i \in I} \mathrm{Mul}_{\mathcal{O}}(\{y_j\}_{\alpha(j)=i}, x_i)$$

In this way the somewhat unpleasant definition of associative composition is just a collection of maps

$$\mathrm{Mul}(\{z_k\}, \{y_j\}) \times \mathrm{Mul}(\{y_j\}, \{x_i\}) \to \mathrm{Mul}(\{z_k\}, \{x_i\})$$

such that is associative when we plug into three Mul's. This hints at the fact that an operad can be thought as a category of tuples, with some extra conditions. Moreover, in order to pass to an infinity setting, we should find a way of defining an operad which is easy to "relax" homotopically. For example, it would be pleasant to have a definition in terms of functors of categories and then nervify.

Recall the definition of Fin_*, Kronecker maps, inert-active maps from 1.9.1. Given an operad $\mathcal{O}$, we can build a category $\mathcal{O}^{\otimes}$ over Fin_* in the following way:

- objects are pairs $(\langle n \rangle, x_1, \dots x_n)$, with $x_i \in \mathcal{O}$;

- Morphisms are pairs

$$(\alpha, (g_j)_{j\in\langle m\rangle^0}) : (\langle n\rangle, x_i) \to (\langle m\rangle, y_j)$$

  where $\alpha : \langle n\rangle \to \langle m\rangle$ and $g_j \in \mathrm{Mul}(\{x_i\}_{\alpha(i)=j}, y_j)$. In the "tree-drawing" of operads, this corresponds to a union of corollas rooted at the x_i's, where α is the function "who is my root". Composition is defined obviously by this interpretation, using $\mathcal{O}$ composition of corollas.

- The map $\pi : \mathcal{O}^{\otimes} \to \mathrm{Fin}_*$ is given by the forgetful $(\langle n\rangle, y_1, \ldots, y_n) \mapsto \langle n\rangle$.

We denote by $\mathcal{O}^{\otimes}_{\langle n\rangle}$ the fiber of π over $\langle n\rangle$. In the next section we introduce infinity operads, and we will show how the nerve of $\mathcal{O}^{\otimes} \to \mathrm{Fin}_*$ actually defines an infinity operad.

2.3 Infinity operads

An infinity operad is an infinity category $\mathcal{O}^{\otimes}$ [1] over $\mathcal{N}(\mathrm{Fin}_*)$ such that:

1. For every inert map in the base $\alpha : \langle n\rangle \to \langle m\rangle$ and $x \in \mathcal{O}^{\otimes}_{\langle n\rangle}$, there exist a cocartesian lift $f : x \to y$ of α.

2. There exist a transition functor $\rho_i^! : \mathcal{O}^{\otimes}_{\langle n\rangle} \to \mathcal{O}^{\otimes}_{\langle 1\rangle}$ over $\rho_i : \langle n\rangle \to \langle 1\rangle$ such that the map

$$\prod_{i=1}^{n} \rho_i^! : \mathcal{O}^{\otimes}_{\langle n\rangle} \to \left(\mathcal{O}^{\otimes}_{\langle 1\rangle}\right)^n$$

   is an equivalence.

3. For every $x \in \mathcal{O}^{\otimes}_{\langle n\rangle}$ there exist cocartesian lifts $\tilde{\rho}_i : x \to y_i$ over the kronecker maps, such that the $\tilde{\rho}_i$'s express x as the relative product of y_i over $\mathcal{N}(\mathrm{Fin}_*)$, i.e. we have

[1] We use $\otimes$ notation also for operads which does not come from $\otimes$ construction, to remember that the underlying category of elements should be thought as $\mathcal{O}^{\otimes}_{\langle 1\rangle}$.

the following homotopy pullback diagram for all $z \in \mathcal{O}^{\otimes}_{\langle m \rangle}$:

$$\begin{tikzcd} \mathrm{Map}(z, x) \arrow[r] \arrow[d] & \prod_i \mathrm{Map}(z, y_i) \arrow[d] \\ \mathrm{Map}(\langle m \rangle, \langle n \rangle) \arrow[r] & \prod_i \mathrm{Map}(\langle m \rangle, \langle 1 \rangle) \end{tikzcd}$$

Cocartesian lifts of inert maps are called *inert*, while lifts of active maps are called active.

Remark 13. Referring to our prototypical $\otimes$ construction, we briefly explain the three conditions. The first is quite technical, e.g. needed for the second condition. The second says that elements over $\langle n \rangle$ are essentially a tuple of elements over $\langle 1 \rangle$. The third condition, in the language of Remark 11, says that a multimap from a tuple to another tuple is a union of corollas, where the map in the left-bottom Hom is the "who is my root" function.

Remark 14. As every two transition functors over the same map are equivalent, the second condition implies that *for every* transition functors $\rho_i^!$ over the kronecker maps the functor $\prod \rho_i^!$ is an equivalence.

Remark 15. As every two cocartesian lifts of the same map are equivalent, and that Map sSets are equivalent if we change one argument by an equivalent one, the third condition implies that *for every* cocartesian lift $\bar{\rho}_i$ the diagram is a pullback diagram.

We prove now that ordinary operads are ∞ operads.

Proposition 2.3.1. *Let $\mathcal{O}$ be a strict operad. Then $\mathcal{N}(\mathcal{O}^{\otimes}) \to \mathcal{N}(Fin_*)$ is an ∞ operad.*

Proof. Let's begin from condition (1). Fix an inert map $\alpha : \langle m \rangle \to \langle n \rangle$, corresponding to an embedding $\beta : \langle n \rangle^0 \to \langle m \rangle^0$. Let $x \in \mathcal{O}^{\otimes}_n$ be $(\langle m \rangle, x_1, \ldots, x_m)$. Define $y = (\langle n \rangle, x_{\beta(1)}, \ldots, x_{\beta(n)})$ and $f := (\alpha, 1_{x_{\beta(1)}}, \ldots, 1_{x_{\beta(n)}}) : x \to y$. We claim that it is cocartesian. Indeed, being nerves of ordinary categories, it is enough to check the two dimensional horn inclusion.

Namely, let $\gamma : \langle n \rangle \to \langle q \rangle$ be a map and $g = (\gamma\alpha, g_i) : x \to z := (\langle q \rangle, z_1, \ldots, z_q)$ a map

over $\gamma\alpha$. Recall that $g_i \in \mathrm{Mul}(\{x_j\}_{j\in\alpha^{-1}\gamma^{-1}(i)}, z_i)$. We should find an $h : y \to z$ over γ, i.e. $h_i \in \mathrm{Mul}(\{x_{\beta(j)}\}_{j\in\gamma^{-1}(i)}, z_i)$ such that $hf = g$.

If you watch closely, this is tautological. Indeed, by definition $\beta(i)$ is the only element in $\alpha^{-1}(i)$. This implies $\alpha^{-1}(\text{subset}) = \beta(\text{subset})$; in particular $\{x_{\beta(j)}\}_{j\in\gamma^{-1}(i)} = \{x_j\}_{j\in\alpha^{-1}\gamma^{-1}(i)}$. This allows to set $h_i := g_i$, and the commutation $hf = g$ is straightforward, being f made of identities.

Let's check condition (2). Being the nerve functor a right adjoint, it commutes with taking fibers. Hence $\mathcal{N}(\mathcal{O}^{\otimes})_{\langle n\rangle} \simeq \mathcal{N}(\mathcal{O}^{\otimes}_{\langle n\rangle})$. Furthermore the nerve takes equivalences to equivalences, so it is enough to check that the involved map is an equivalence at the level of ordinary categories.
Let $\tilde{\rho}$ be the transition functor associated with the cocartesian lifts we found in checking condition (1). Explicitly, we have that

$$\tilde{\rho}(\langle n\rangle, x_1, \dots, x_n) = ((\langle 1\rangle, x_1), \dots, (\langle 1\rangle, x_n))$$

$$\tilde{\rho}(1_{\langle n\rangle}, h_i : x_i \to y_i) = ((1_{\langle 1\rangle}, h_1), \dots, (1_{\langle 1\rangle}, h_n))$$

And indeed this is an iso of categories with trivial inverse.
Let's finish with condition (3). Recall that map ssets are defined as

$$\mathrm{Map}_{\mathcal{N}(\mathcal{O}^{\otimes})}(x, y)_n := \{x\}_n \times_{\Delta^{\{0\}}} \mathrm{Map}(\Delta^1, \mathcal{N}(\mathcal{O}^{\otimes}))_n \times_{\Delta^{\{1\}}} \{y\}_n \simeq$$

$$\simeq \{x\}_n \times_{\Delta^{\{0\}}} \mathrm{Hom}(\Delta^1 \times \Delta^n, \mathcal{N}(\mathcal{O}^{\otimes})) \times_{\Delta^{\{1\}}} \{y\}_n \simeq \{x\}_n \times_{\{0\}} \mathrm{Hom}([1]\times[n], \mathcal{O}^{\otimes}) \times_{\{1\}} \{y\}_n \simeq$$

$$\simeq \mathcal{N}(x)_n \times_{\{0\}} \mathrm{Hom}([n], \mathrm{Hom}([1], \mathcal{O}^{\otimes})) \times_{\{1\}} \mathcal{N}(y)_n \simeq$$

$$\simeq \mathcal{N}(x) \times_{\{0\}} \mathcal{N}(\mathrm{Hom}([1], \mathcal{O}^{\otimes}))_n \times_{\{1\}} \mathcal{N}(y)_n \simeq \mathcal{N}(\{x\} \times_{\{0\}} \mathrm{Hom}([1], \mathcal{O}^{\otimes}) \times_{\{1\}} \{y\})_n$$

Being the equivalence natural in n, we conclude that the two simplicial sets are equivalent.

So, as expected, the map sets in the case of a nerve are exactly the nerve of the intuitive (discrete!) equivalent for ordinary categories which appears at last. As pullbacks commute with the nerve, it is enough to show the thesis at the level of ordinary categories. Interpreting hom sets as categories, i.e. $\mathrm{Hom}(x,y) = \{x\} \times_{\{0\}} \mathrm{Hom}([1], C) \times_{\{1\}} \{y\}$, we are left with showing

$$\begin{tikzcd}
\mathrm{Hom}_{\mathcal{O}^\otimes}(z,x) \arrow[r] \arrow[d] & \prod_i \mathrm{Hom}_{\mathcal{O}^\otimes}(z,y_i) \arrow[d] \\
\mathrm{Hom}_{\mathcal{O}^\otimes}(\langle m\rangle, \langle n\rangle) \arrow[r] & \prod_i \mathrm{Hom}_{\mathcal{O}^\otimes}(\langle m\rangle, \langle 1\rangle)
\end{tikzcd}$$

And this is the very definition of hom sets in $\mathcal{O}^\otimes$: a morphism from $(z_1, \dots, z_m)$ to $(x_1, \dots, x_n)$ is the data of a function $\alpha : \langle m\rangle \to \langle n\rangle$ plus a bunch of multifunctions $\{z_j\}_{\alpha(j)=i} \to x_i$. Note that the categorical structure is discrete, so that one only have to check the set-theoretic pullback. $\square$

As strict symmetric monoidal categories are an example of strict operads (see example 2), we expect the same phenomena appearing in ∞ context. Indeed, the following holds:

Proposition 2.3.2. *Let $p : C^\otimes \to \mathcal{N}(Fin_*)$ be a ∞ symmetric monoidal category. Then it is an ∞ operad.*

Proof. Condition (1) is obviously satisfied, because p is a cocartesian fibration. Condition (2) appears also in the definition of symmetric monoidal ∞ category. The only non trivial part comes from (3). Recall the diagram we are working with:

$$\begin{tikzcd}
\mathrm{Map}(z,x) \arrow[r] \arrow[d] & \prod_i \mathrm{Map}(z,y_i) \arrow[d] \\
\mathrm{Map}(\langle m\rangle, \langle n\rangle) \arrow[r] & \prod_i \mathrm{Map}(\langle m\rangle, \langle 1\rangle)
\end{tikzcd}$$

** Firstly, assume that z, x belong to $C^\otimes_{\langle n\rangle}$. As the underlying map in Fin_* is the identity,

the lower map ssets are just points. The thesis is equivalent to

$$\mathrm{Map}(z,x)_{id} \to \prod_i \mathrm{Map}(z,x_i)_{\rho_i}$$

being an equivalence. Subscripts denote the fiber with respect to the projection to $\mathcal{N}(\mathrm{Fin}_*)$. Call $(z_1,\dots,z_n)$ the image of z under the correspondence $C^{\otimes}_{\langle n\rangle} \simeq (C^{\otimes}_{\langle 1\rangle})^n$. Recall that x corresponds to $(x_1,\dots,x_n)$. This part of the proof is concluded with the aid of lemma **??**:

1. The cocartesian lifting $z \to z_i$ yields an equivalence $\mathrm{Map}(z,x_i)_{\rho^i} \to \mathrm{Map}(z_i,x_i)_{id}$
2. The equivalence of $C^{\otimes}_{\langle n\rangle}$ with $(C^{\otimes}_{\langle 1\rangle})^n$ yields an equivalence

$$\mathrm{Map}(z,x)_{id} \simeq \prod_i \mathrm{Map}(z_i,x_i)_{id}$$

And we are done.

** For the second part, take an arbitrary z, with image $\langle m\rangle$ in $\mathcal{N}(\mathrm{Fin}_*)$. From [1], 3.3.18, we can equivalently show that, for every map $\alpha : \langle m\rangle \to \langle n\rangle$, the map of homotopy fibers

$$\mathrm{Map}(z,x)_{h\alpha} \to \prod_i \mathrm{Map}(z,x_i)_{h\rho_i\alpha}$$

is an homotopy equivalence. Note that in the nerve of an ordinary category, map sets are discrete: if we have a one arrow $\sigma : f \to g$, where $f,g : a \to b$, it can be depicted as a commutative diagram

$$\begin{tikzcd} a \arrow[r, "f"] \arrow[d, equal] & b \arrow[d, equal] \\ a \arrow[r, "g"] & b \end{tikzcd}$$

And this implies that $f = g$ and that $\sigma = s_0(f)$ is made of the degenerate 2-cells. On balance, we have no non trivial arrows.

From the description of the homotopy fiber $X \to Y$ at a point y as $X \times_Y Y^{\simeq}_{/y}$, we get that if Y is discrete the homotopy fiber coincide with the fiber.

We are reduced to show that there is an equivalence

$$\mathrm{Map}(z,x)_\alpha \to \prod_i \mathrm{Map}(z,x_i)_{\rho_i\alpha}$$

Take a cocartesian lifting β of α starting from z and ending at z'. As cocartesian lifting over collapses encodes tensor product, you can imagine that z' has the tensor product on all the components that map to the same index in $\langle n\rangle$. From the description of strict symmetric monoidal maps (see example 2), we expect now to have an essentially unique component-by-component morphism $z' \to x$.

Formally, take the two cell τ in $\mathcal{N}(\mathrm{Fin}_*)$ such that:

$$\tau(01) = \alpha, \tau(02) = \alpha, \tau(12) = 1$$

The arrow β over 01 of this cell, induces by composition an equivalence

$$\mathrm{Map}(z,x)_\alpha \simeq \mathrm{Map}(z',x)_{id}$$

Putting together with the first part:

$$\mathrm{Map}(z,x)_\alpha \simeq \mathrm{Map}(z',x)_{id} \simeq \prod_i \mathrm{Map}(z',x_i)_{\rho_i} \simeq \mathrm{Map}(z,x_i)_{\rho_i\alpha}$$

Where in the last passage we used the same $z - z'$ argument over the two cell

$$\tau'(01) = \alpha, \tau(12) = \rho_i, \tau(02) = \rho_i\alpha$$

And we are done!

□

Now that we have a definition of what an ∞ operad is, let's see how they fits into an infinity category.

2.4 Infinity Category of operads

Given two ∞ operads $\mathcal{O}^{\otimes}, \mathcal{P}^{\otimes}$, define a morphism of ∞ operads as an infinity functor $F : \mathcal{O}^{\otimes} \to \mathcal{P}^{\otimes}$ over $\mathcal{N}(\mathrm{Fin}_*)$, such that takes inert maps to inert maps. Morphisms can be made into a Kan simplicial set by setting $\mathrm{Map}(\mathcal{O}^{\otimes}, \mathcal{P}^{\otimes}) := \mathrm{Alg}_{\mathcal{O}^{\otimes}}(\mathcal{P}^{\otimes})^{\simeq}$, where $\mathrm{Alg}_{\mathcal{O}^{\otimes}}(\mathcal{P}^{\otimes})$ is the subsimplicial set of $\mathrm{Fun}(\mathcal{O}^{\otimes}, \mathcal{P}^{\otimes})$ spanned by maps of operads. Thus the operads fit into a locally Kan simplicial category Op^{Δ}, and we can define the infinity category of operads by setting $\mathrm{Op}_{\infty} := \mathcal{N}(\mathrm{Op}^{\Delta})$

Furthermore, we have the following

Proposition 2.4.1. *Let C be an infinity symmetric monoidal category, and $\mathcal{O}^{\otimes}$ an infinity operad. Then $Alg_{\mathcal{O}^{\otimes}}(C^{\otimes})$ is an infinity symmetric monoidal category.*

See [10] §3.2.4.4 for the precise definition. The symmetric monoidal structure on algebras is meant to model the componentwise tensor product.

Although the definition of Op_{∞} is easy to formulate, it can be a bit unpleasant because it is quite indirect. In particular, it is not easy to see which maps are weak equivalences. For this reason, let us introduce marked simplicial sets.

Definition 2.4.2. A marked simplicial set is a pair (X, Γ) where X is a simplicial set and Γ is a subset of arrows that contains degenerate edges. We call Γ marked edges. A map of marked simplicial set is a simplicial map that preserve marked edges. Denote by sSet^{+} the simplicial category of marked simplicial sets, where map sets are given by

$$\mathrm{Map}^{\sharp}(X, Y)_n := \mathrm{Hom}_{\mathrm{sSet}^{+}}((\Delta^n)^{\sharp} \times X, Y)$$

Given a simplicial set X, we can always form two privileged marked simplicial sets:

1. $X^{\sharp}$, where marked edges are all edges; it is easily a right adjoint to the forgetful functor.

2. $X^{\flat}$, where marked edges are only degenerate edges; it is easily a left adjoint to the forgetful functor.

In [2] you can find a precise account on these kind of constructions, and how they interact with the standard model category on simplicial sets. Let us define $\mathcal{N}(\mathrm{Fin}_*)^\natural$ as the simplicial set where marked edges are inert maps. More generally, for an infinity operad $\mathcal{O}^\otimes$ define $\mathcal{O}^{\otimes,\natural}$ as the marked simplicial set with inert maps marked. We obviously have that $\mathcal{O}^{\otimes,\natural}$ is a marked simplicial set over $\mathcal{N}(\mathrm{Fin}_*)^\natural$. Moreover, we have the following strong fact:

Fact 1. *There exist a simplicial model category on $(sSet^+)_{/\mathcal{N}(Fin_*)^\natural}$, called ∞-preoperadic, such that*

- *cofibrations are monomorphisms;*
- *weak equivalences are* ***operadic equivalences****, i.e. maps $f : X \to Y$ such that for every operad $\mathcal{O}^{\otimes,\natural}$ restriction along f yields an equivalence*

$$Map^\sharp_{/\mathcal{N}(Fin_*)^\natural}(Y, \mathcal{O}^{\otimes,\natural}) \to Map^\sharp_{/\mathcal{N}(Fin_*)^\natural}(Y, \mathcal{O}^{\otimes,\natural})$$

- *fibrant objects are $\mathcal{O}^{\otimes,\natural}$, for $\mathcal{O}^\otimes$ an infinity operad;*
- *The underlying infinity category is equivalent to Op_∞.*

Indeed, we have the following proposition about equivalences:

Proposition 2.4.3. *Let $f : \mathcal{O}^\otimes \to \mathcal{P}^\otimes$ be a morphism of ∞ operads. Then the following are equivalent:*

1. *f is an operadic equivalence;*
2. *f is an equivalence in Op_∞;*
3. *For every $\mathcal{Q}^\otimes$, restriction along f yields an equivalence*

$$Alg_{\mathcal{P}^\otimes}(\mathcal{Q}^\otimes) \to Alg_{\mathcal{O}^\otimes}(\mathcal{Q}^\otimes)$$

Proof. Facts 1 and 2 are equivalent because Op_∞ is equivalent to the infinity category underlying marked simplicial sets over $\mathcal{N}(\mathrm{Fin}_*)^\natural$. So, f is an equivalence in the latter iff it is an iso in the homotopy category iff it is an equivalence in the underlying infinity category. To show the equivalence of 1 and 3, it is enough to prove that there is an equivalence natural in O of

$$\mathrm{Map}^{\#}_{/\mathcal{N}(\mathrm{Fin}_*)^\natural}(\mathcal{O}^\otimes, \mathcal{Q}^\otimes) \simeq \mathrm{Alg}_{\mathcal{O}^\otimes}(\mathcal{Q}^\otimes)$$

This mainly amounts to unwinding the definitions; the only mathematical fact we use is that a cocartesian lift of the identity is an equivalence.

The n simplices of the LHS are maps $\tau : (\Delta^n)^\# \times \mathcal{O}^{\otimes,\natural} \to \mathcal{Q}^{\otimes,\natural}$ of marked simplicial sets over $\mathcal{N}(\mathrm{Fin}_*)$. This means that:

- $\tau(x, f)$ is inert for x vertex in Δ^n and f inert map in $\mathcal{O}^\otimes$;
- $\tau(\delta, y)$ is inert for δ arrow in Δ^n and y vertex in $\mathcal{O}^\otimes$.
- If $\beta \in \mathcal{O}^\otimes$ lies over $\sigma \in \mathcal{N}(\mathrm{Fin}_*)$, then $\tau(\alpha, \beta)$ lies over σ.

Mixed maps (δ, f) will be then automatically sent to inert maps because they are a composition of some "separated" maps $(x, f), (\delta, y)$, and inert maps are stable for composition. Note that if $y \in \mathcal{O}^\otimes$ lies over $\langle n \rangle$, then $\tau(\delta, y)$ is an inert arrow over $\langle n \rangle$. This means that it is a cocartesian lift of the identity, and so it is an equivalence.

The RHS is $\mathrm{Map}_{inert}(\mathcal{O}^\otimes, \mathcal{Q}^\otimes)^\simeq$. Its n simplices are maps $\tau : \Delta^n \times \mathcal{O}^\otimes \to \mathcal{Q}^\otimes$ such that:

- $\tau(x, f)$ is inert for f arrow in $\mathcal{O}^\otimes$ and x is a vertex in Δ^n; this explain the "inert"-preserving condition.
- $\tau(\delta, y)$ is an equivalence for δ an arrow in Δ^n and y a vertex in $\mathcal{O}^\otimes$. Indeed, in this way all the restrictions (1-simplices) $\Delta^1 \times \mathcal{O}^\otimes \to \Delta^n \times \mathcal{O}^\otimes \to \mathcal{Q}^\otimes$ are equivalences

as arrows in $\mathrm{Map}(\mathcal{O}^{\otimes}, \mathcal{Q}^{\otimes})$. Recall in fact that generally the maximal subgroupoid $C^{\simeq}$ of C is defined as the subcategory spanned by invertible arrows.

So the two simplicial sets have really the same definition, and in particular they are naturally equivalent. □

On balance, two operads are equivalent iff they define the same kind of homotopical algebra structure. That's why this is interesting. It is hard to show that two operads are equivalent. In the next section we introduce the main tool to show that two operads define the same algebra structure, namely the approximation theorem; with weaker hypothesis, one can still show that algebras of one operad correspond to "locally constant" algebras of the other, in an appropriate sense.

2.5 Weak Approximation

We begin with the main definition of the section.

Definition 2.5.1. Let $p : \mathcal{O}^{\otimes} \to \mathcal{N}(\mathrm{Fin}_*)$ be an ∞ operad. We say that a categorical fibration $f : C \to \mathcal{O}^{\otimes}$ is an *approximation* if the following holds:

1. Let $p' = pf$, $X \in C, \langle n \rangle = p'(X)$. Then for every $1 \leq i \leq n$, there exist a p-cocartesian morphism $\alpha_i : C \to C_i$ such that $p(\alpha_i)$ is inert and $p'(\alpha_i)$ is the i-th kronecker map $\langle n \rangle \to \langle 1 \rangle$.

2. Let $X \in C$, $\alpha : Y \to f(X)$ active in $\mathcal{O}^{\otimes}$. Then there exist an f-cartesian lift $\bar{\alpha} : \bar{Y} \to X$ of α.

The map f is said to be a *weak approximation* if (1) is satisfied together with the following condition:

Let $\alpha : Y \to f(X)$ be a morphism in $\mathcal{O}^{\otimes}$. Then the subcategory $\mathcal{E} \subset C_{/X} \times_{\mathcal{O}^{\otimes}_{/f(X)}} \mathcal{O}^{\otimes}_{\alpha//f(X)}$ spanned by inert $\beta : Y \to f(X')$ in the second component, is weakly contractible.

If $f : C \to \mathcal{O}^{\otimes}$ is not a categorical fibration, we say that f is a (weak) approximation if it factors as $C \xrightarrow{f'} C' \xrightarrow{f''} \mathcal{O}^{\otimes}$ such that f' is a categorical equivalence and f'' is a categorical fibration that is a (weak) approximation.

Let us give a coherence proposition that justifies the weakening of the hypothesis.

Proposition 2.5.2. *Let $f : C \to \mathcal{O}^{\otimes}$ be an approximation. Then it is a weak approximation.*

Proof. Fix an $\alpha : Y \to f(X)$. As inert-active is a factorization system on $\mathcal{O}^{\otimes}$, α can be factored as $Y \to Y' \xrightarrow{\beta} f(X)$ via a 2-cell σ. By condition (2) of approximation, there exist an f cartesian morphism β' lifting β. We claim that the element (σ, β') is a final object of $\mathcal{E}$. Indeed, recall that inert-active maps are a *factorization system* on $\mathcal{O}^{\otimes}$ [2]

In particular left orthogonality informally yields that for any diagram

$$\begin{tikzcd} Y \arrow[r] \arrow[d] & Y' \arrow[d, "\beta"] \\ Z \arrow[r] \arrow[ur, dotted] & f(X) \end{tikzcd}$$

The space of dotted lifts is contractible. This is the precise definition of (σ, β') being final in $\mathcal{E}$, so the latter is weakly contractible. □

We now define what is a locally constant algebra.

Definition 2.5.3. Let $p : \mathcal{O}^{\otimes} \to \mathcal{O}'^{\otimes}$ be a morphism of operads, i.e. an element of $\mathrm{Alg}_{\mathcal{O}^{\otimes}}(\mathcal{O}'^{\otimes})$. It is said to be locally constant if, for every arrow $f \in \mathcal{O} = \mathcal{O}^{\otimes}_{\langle 1 \rangle}$, the image $p(f)$ is an equivalence in $\mathcal{O}' = \mathcal{O}'^{\otimes}_{\langle 1 \rangle}$. The infinity category spanned by locally constant algebras is denoted by $\mathrm{Alg}^{loc}_{\mathcal{O}^{\otimes}}(\mathcal{O}'^{\otimes})$.

Remark 16. This definition is saying that a locally constant algebra is an algebra such that all unary operations are sent to equivalences.

The main result we are interested in, mainly based on "essentially-unique-Kan-extension" lemma (14), is the following.

[2] See [11] 5.2.8.8 for the definition, and [10] 2.1.2.4 for the proposition about inert-active factorizations.

Theorem 2.5.4. *Let $p' : \mathcal{C}^{\otimes} \to \mathcal{N}(Fin_*), p : \mathcal{O}^{\otimes} \to \mathcal{N}(Fin_*), q : \mathcal{O'}^{\otimes} \to \mathcal{N}(Fin_*)$ be infinity operads, and $f : \mathcal{C}^{\otimes} \to \mathcal{O}^{\otimes}$ be a weak approximation.*

1. *If $f_{\langle 1 \rangle} : \mathcal{C} \to \mathcal{O}$ is an equivalence on underlying categories, then*

$$Alg_{\mathcal{O}^{\otimes}}(\mathcal{O'}^{\otimes}) \simeq Alg_{\mathcal{C}^{\otimes}}(\mathcal{O'}^{\otimes})$$

   via composition with f.

2. *If $f_{\langle 1 \rangle} : \mathcal{C} \to \mathcal{O}$ is a weak equivalence on underlying categories, then*

$$Alg_{\mathcal{O}^{\otimes}}(\mathcal{O'}^{\otimes}) \simeq Alg^{loc}_{\mathcal{C}^{\otimes}}(\mathcal{O'}^{\otimes})$$

Proof. The thesis is invariant under categorical equivalence in $\mathcal{C}^{\otimes}$, so that we can suppose that f is a categorical fibration. Consider a cartesian fibration $u : \mathcal{M} \to \Delta^1$ associated to the functor $f : \mathcal{C}^{\otimes} \to \mathcal{O}^{\otimes}$. In particular, we have

$$\mathcal{M} \times_{\Delta^1} \Delta^{\{0\}} \simeq \mathcal{O}^{\otimes}, \mathcal{M} \times_{\Delta^1} \Delta^{\{1\}} \simeq \mathcal{C}^{\otimes}$$

Recall indeed that cartesian fibrations are associated to contravariant functors to Cat_{∞}, so that fibers are the other way around. The idea of the proof is the following: define a certain subset $\mathcal{X} \subset \mathrm{Fun}(\mathcal{M}, \mathcal{O'}^{\otimes})$ such that restriction to the two endings $\mathcal{C}^{\otimes}, \mathcal{O}^{\otimes}$ will give an equivalence with $\mathrm{Alg}_{\mathcal{C}^{\otimes}}(\mathcal{O'}^{\otimes}), \mathrm{Alg}_{\mathcal{O}^{\otimes}}(\mathcal{O'}^{\otimes})$. The inverse to these restrictions will be given by Kan extensions.

Let's define the subsset $F \in \mathcal{X}$:

- The restriction $F|\mathcal{O}^{\otimes}$ is in $\mathrm{Alg}_{\mathcal{O}^{\otimes}}(\mathcal{O'}^{\otimes})$;
- For every u-cartesian morphism α in $\mathcal{M}$, $F(\alpha)$ is an equivalence.

Notice that the second condition is equivalent to F being a q-left Kan extension of $F|\mathcal{O}^{\otimes}$. Indeed, u-cartesian morphisms over the identities of Δ^1 are equivalence by Joyal

lemma 4, so their image are automatically equivalences. Concerning an u-cartesian lifting $\beta : Y \to X$ of $0 \to 1$, note that it is a final object of $\mathcal{O}_{/X}$ because of cartesianity. Being a Kan extension means that $F(X)$ is the q-colimit of $\mathcal{O}_{/X}$; but the latter has a final object, so it is equivalent to $F(\beta)$ being an equivalence.

Furthermore, note that by the same reasoning we can say that every $G \in \mathrm{Alg}_{\mathcal{O}^\otimes}(\mathcal{O}'^\otimes)$ is "ready to be extended", namely every functor from a comma category $G_X : \mathcal{O}_{/X} \to \mathcal{O} \xrightarrow{G} \mathcal{O}'^\otimes$ admits a q-colimit, because $\mathcal{O}_{/X}$ has a final object.

We conclude, by 14, that $\mathcal{X} \to \mathrm{Alg}_{\mathcal{O}^\otimes}(\mathcal{O}'^\otimes)$ is a trivial Kan fibration.

For the other restriction, we have to show that:

- $F \in \mathcal{X}$ if and only if it is the q-right Kan extension of $F|\mathcal{C}^\otimes$ and $F|\mathcal{C}^\otimes \in \mathrm{Alg}_{\mathcal{C}^\otimes}(\mathcal{O}'^\otimes)$ (or $F|\mathcal{C}^\otimes \in \mathrm{Alg}^{loc}_{\mathcal{C}^\otimes}(\mathcal{O}'^\otimes)$ if we are in case (2));
- For every $X \in \mathcal{O}^\otimes$, the map $F_X : \mathcal{C}^\otimes_{X/} \to \mathcal{O}'^\otimes$ admits a q-colimit.

We firstly prove the second claim. This is achieved by reducing, via cofinality or Kan extensions, the complexity of the diagram $\mathcal{C}^\otimes_{/X}$.

Reduction to inerts . Let $\mathcal{C}'_{X/}$ be the set of maps $X \to C$ that correspond to inerts $X \to f(C)$. More precisely, take a cartesian morphism $f(C) \to C$, and lift by cartesianity $X \to C$ to $X \to f(C)$. If the latter is inert, $X \to C \in \mathcal{C}'_{X/}$.

Let's prove that $\mathcal{C}'_{X/} \subset \mathcal{C}_{X/}$ is right cofinal. By using the dual of 1.7.2, it is enough to show that for any $\alpha : X \to C$, $\mathcal{C}'_{X/} \times_{\mathcal{C}_{X/}} \mathcal{C}_{X//\alpha}$ is weakly contractible. Note that

the latter is the sset of two-cells

$$\begin{tikzcd} X \arrow[r, "\beta"] \arrow[d, "\alpha"] & C' \arrow[dl] \\ C & \end{tikzcd}$$

Such that β correspond to an inert arrow. By the dual of the homotopy lifting lemma 7, the cartesian arrows $f(C') \to C', f(C) \to C, X = X$ gives us a two-cell

$$\begin{tikzcd} X \arrow[r, "\beta'"] \arrow[d, "\alpha'"] & f(C') \arrow[dl] \\ f(C) & \end{tikzcd}$$

such that β' is inert, and by projecting on $\mathcal{C}^{\otimes}_{C/}$ from $\mathcal{C}'_{X/} \times_{\mathcal{C}_{X/}} \mathcal{C}_{X//\alpha} \simeq \mathcal{C}'_{X/} \times_{\mathcal{C}_{X/}} \mathcal{C}_{\alpha//C}$ we get an arrow $C' \to C$. On balance, we have a map

$$\mathcal{C}'_{X/} \times_{\mathcal{C}_{X/}} \mathcal{C}_{X//\alpha} \to \{(\beta, \sigma) \in \mathcal{C}^{\otimes}_{/C} \times_{\mathcal{O}^{\otimes}_{/f(C)}} \mathcal{O}^{\otimes}_{\alpha//f(C)} : X \to f(C') \text{ is inert } \} = \mathcal{E}$$

We can construct an homotopical inverse to this map. Namely, given $\gamma : C' \to C$ and a 2-cell

$$\begin{tikzcd} X \arrow[r, "\beta'"] \arrow[d, "\alpha'"] & f(C') \arrow[dl] \\ f(C) & \end{tikzcd}$$

We can build a 2 cell in $\mathcal{C}'_{X/} \times_{\mathcal{C}_{X/}} \mathcal{C}_{X//\alpha}$. Drawings are strongly encouraged. The recipe is the following:

- Take cartesian arrows $f(C') \to C', f(C) \to C$;
- By composing $X \to f(C) \to C$ and $X \to f(C') \to C'$, obtain the maps $X \to C, X \to C'$;

- By composing $f(C') \to f(C) \to C$, obtain a map $f(C') \to C$;
- $X \to f(C') \to f(C) \to C$ determines a Λ^3_2 horn. The filling yields a two cell $X \to f(C') \to C$;
- $X \to f(C') \to C' \to C$ determines a Λ^3_1 horn. The filling yields a two cell $X \to C' \to C$, as desired.

With bigger diagrams, one can indeed show that the two maps are homotopy inverses. In fact, all the choices that we have done vary in contractible Kan complexes. By assumption $\mathcal{E}$ is weakly contractible, and so it is $\mathcal{C}'_{X/} \times_{\mathcal{C}_{X/}} \mathcal{C}_{X//\alpha}$.

Reduction to underlying category . Let's define $\mathcal{C}''_{X/}$ the subcategory of $\alpha : X \to C \in \mathcal{C}'_{X/}$ such that $pf(C) = \langle 1 \rangle$. We claim that $F_X|\mathcal{C}'_{X/}$ is a q-right Kan extension of $F_X|\mathcal{C}''_{X/}$. To do this, we must show that the comma categories $\mathcal{D}_X := \mathcal{C}''_{X/} \times_{\mathcal{C}'_{X/}} (\mathcal{C}'_{X/})_{\alpha/}$ enters a q-colimit diagram given by the natural map $\to \mathcal{C} \to \mathcal{M} \to \mathcal{O}^{\otimes\prime}$. Note that $\mathcal{D}_X$ can be described as the two cells

$$\begin{tikzcd} X \arrow[r, "\alpha"] \arrow[d, "\beta"] & C \arrow[dl, "\gamma"] \\ C' & \end{tikzcd}$$

such that α is fixed and correspond to an inert, β correspond to an inert, C' is over $\langle 1 \rangle$. Note that γ correspond to an inert: indeed, denoting $\bar{\alpha} : X \to f(C)$ the map corresponding to $X \to C$ and analogously for β, γ

$$1 = |p\bar{\beta}^{-1}(\{1\})| = |p\bar{\alpha}^{-1}p\bar{\gamma}^{-1}(\{1\})| = p\bar{\gamma}^{-1}(\{1\})|$$

Using inertia of $\bar{\alpha}, \bar{\beta}$. Furthermore, C' is over $\langle 1 \rangle$, so $p\bar{\gamma}$ is of the form ρ^i_n for some i (n is fixed by α).

We understand that $\mathcal{D}$ writes as a disjoint union of $\mathcal{D}(i)$, where $\mathcal{D}(i)$ is the full subcategory of two cells such that γ is ρ^i. By assumption, there exist a pf cocartesian lift $C \to C_i$ of ρ_i starting at C. By composition with $X \to C$, we get a two cell in $\mathcal{D}(i)$. We claim that this is initial in $\mathcal{D}(i)$. Indeed, given a diagram $X \to C \to C'$ and considering our diagram $X \to C \to C_i$, we can find by cocartesianity of $C \to C_i$ an arrow $C_i \to C'$; this choice vary in a contractible Kan complex. The two cells we have fits into a Λ^3_1 horn $X \to C \to C_i \to C'$, yielding a filling that is a morphism from $X \to C \to C_i$ to $X \to C \to C'$. We do not prove formally that the sset of morphisms is contractible, but it should be evident from the fact that fillings are essentially uniquely determined.

On balance, we have found that the maps $F(C) \to F(C_i)$ forms a q-limit diagram, namely that $F(C)$ is the q-product of $F(C_i)$. But $C \to C_i$ are inert maps covering ρ_i, and by the assumption that $F|\mathcal{C}$ is an algebra, it sends them to inert maps over ρ_i. By assumption of $\mathcal{O}'$ being an operad, condition 3, this express $F(C)$ as a q product of $F(C_i)$.

Case 1, Reduction to product . We place in case 1 of the theorem. Using [11], 4.3.2.7, we are reduced to show that $\mathcal{C}''_{X/} \to \mathcal{O}'$ extend to a q-limit diagram. We use the same technique as above, i.e. splitting $\mathcal{C}''_{X/}$ in disjoint categories with an initial object, so that we will have to show that $F(X)$ is a product of some elements.

Precisely, denote by $\mathcal{C}''(i)_{X/}$ the sset of $X \to C$ that corresponds to $X \to f(C)$ over ρ^i. Then $\mathcal{C}''_{X/}$ is the disjoint union of $\mathcal{C}''(i)_{X/}$. Choose cocartesian maps $X \to X_i$ over ρ_i. By assumption, $\mathcal{C}_{\langle 1 \rangle} \to \mathcal{O}$ is an equivalence, so that there exist C_i with $X_i \simeq f(C_i)$. We then claim that $X \to C_i$ is initial in $\mathcal{C}''(i)_{X/}$. Indeed, $X \to C \in \mathcal{C}''(i)_{X/}$ correspond to $X \to f(C)$, and there is a contractible space of

maps from $X \to f(C_i)$ to $X \to f(C)$, by initiality. Translating along the cartesian morphisms $f(C) \to C, f(C_i) \to C_i$, we get the contractibility of the space of maps from $X \to C_i$ to $X \to C$.

We are reduced to show that the $\{F(C_i)\}_{1 \leq i \leq n}$ admits a q-coproduct. This follows from the assumption that $\mathcal{O}^{\otimes\prime}$ is an ∞ operad, condition on transition functors.

For the case 2, look at [10], 2.3.3.23.

We now prove (b). Suppose $F \in \mathcal{X}$. Note that we have showed that $\mathcal{C}''_{X/}$ is cofinal in $\mathcal{C}''_{X/}$, and that the latter has a set of initial-in-their-component objects $X \to C_i$, that corresponds to $X \to f(C_i) \simeq X_i$, where $X \to X_i$ is a cocartesian lift of ρ^i. So we have that F is a q-right Kan extension of $F|C$ iff $F(X \to C_i)$ exhibits $F(X)$ as the product of $F(C_i)$. Being over the ρ^i, this is equivalent to $F(C \to C_i)$ being inert.

By the second condition in the definition of $\mathcal{X}$, $F(f(C_i) \to C_i)$ is an equivalence. Moreover, $F(X \to f(C_i)) \simeq F(X \to X_i)$ is cocartesian. We conclude that $F(X \to C_i) \simeq F(X \to f(C_i))$ is cocartesian, so it is an inert map. Moreover, it is evident that $F|\mathcal{C}^{\otimes} = (F|\mathcal{O}^{\otimes}) \circ f$ is an algebra; it is locally constant in case of (2) because every unary operation of $\mathcal{C}^{\otimes}$ is firstly sent to something in $\mathcal{O}$, which is a Kan complex, and so it is an equivalence.

On the other hand, suppose that $F|\mathcal{C}^{\otimes} \in \mathrm{Alg}_{\mathcal{C}^{\otimes}}(\mathcal{O}^{\otimes\prime})$ (or $\mathrm{Alg}^{loc}_{\mathcal{C}^{\otimes}}(\mathcal{O}^{\otimes\prime})$) and that it is a right Kan extension of $F|\mathcal{C}^{\otimes}$. We firstly show that F satisfies the second condition of being in $\mathcal{X}$. Take a cartesian morphism $f(C) \to C$, and inert morphisms $C \to C_i$ over ρ^i. The fact that $F|C \in \mathrm{Alg}_{\mathcal{C}^{\otimes}}(\mathcal{O}^{\otimes})$ implies that $F(C \to C_i)$ are inerts. It then suffice to show that $F(f(C) \to C_i)$ are inerts, and this follows from the fact the cofinality in $\mathcal{C}^{\otimes}_{/f(C)}$ we mentioned above.

It is easy to show that a map of infinity categories preserves inerts iff preserve inerts over ρ_i. Take $X \in \mathcal{O}^{\otimes}$ and $X \to X_i$ inert over ρ_i. By reasoning as above, we have that

$X_i \simeq f(C_i)$, and that there is a cartesian morphism $f(C_i) \to C_i$. By what we have shown, cartesian morphisms are sent to equivalences, and so it is $F(f(C_i) \to C_i)$. Now because of q-right Kan extension condition, $F(X \to X_i) \simeq F(X \to C_i)$ are inert, and we are finally done. $\square$

The next sections deal with a generalization of operads and of Kan extensions to the operadic case. Due to space and time needs, they are of expository nature, mainly without proofs. The only additional contribution to Higher Algebra ([10]) is some sparse comments.

2.6 Families of operads

We are interested in parameterizing an operad with index in an infinity category. Unlike other parts of mathematics, the resulting structure is no longer an operad, and so we have to introduce the notion of

Definition 2.6.1 (Generalized Operad)**.** Let $q : \mathcal{O}^{\otimes} \to \mathcal{N}(\mathrm{Fin}_*)$ a map of infinity categories. We will say that $\mathcal{O}^{\otimes}$ is a generalized operad of the following hold:

1. For every inert map $\alpha : \langle n \rangle \to \langle m \rangle$ and X such that $p(X) = \langle n \rangle$, there exist a q-cocartesian lift $\bar{\alpha} : X \to Y$.

2. For every diagram of inert morphisms

$$\begin{tikzcd} \langle m \rangle \arrow[r] \arrow[d] & \langle m' \rangle \arrow[d] \\ \langle n \rangle \arrow[r] & \langle n' \rangle \end{tikzcd}$$

   That induces by counterimage a pushout square

$$\begin{tikzcd} \langle m \rangle^0 \arrow[r, leftrightarrow] \arrow[d, leftrightarrow] & \langle m' \rangle^0 \\ \langle n \rangle^0 & \langle n' \rangle^0 \arrow[l] \arrow[u] \end{tikzcd}$$

We have a pullback square

$$\begin{tikzcd} \mathcal{O}^{\otimes}_{\langle m\rangle} \arrow[r] \arrow[d] & \mathcal{O}^{\otimes}_{\langle m'\rangle} \arrow[d] \\ \mathcal{O}^{\otimes}_{\langle n\rangle} \arrow[r] & \mathcal{O}^{\otimes}_{\langle n'\rangle} \end{tikzcd}$$

3. Given the diagram of inert morphisms above, suppose that it lifts to a diagram made of q-cocartesian morphisms

$$\begin{tikzcd} X \arrow[r] \arrow[d] & X' \arrow[d] \\ Y \arrow[r] & Y' \end{tikzcd}$$

   Then it is a q-limit diagram (q-pullback).

We call a cocartesian lift of an inert morphism an inert map, as in the case of operads. We briefly explain the given conditions. The first condition is a straightforward generalization of the inert-lifting condition for operads. The second must be thought as a relative version of $\mathcal{O}^{\otimes}_{\langle m\rangle} \simeq (\mathcal{O}^{\otimes}_{\langle 1\rangle})^m$: indeed, if this was the case, the condition would be satisfied, but here we don't have a common brick to build up all the fibers. As we can't reduce everything to the fiber over 1, we must also generalize the condition of x being the q-relative product of its "components". So, whenever we have two elements that "share some components", the concatenation of their components is the relative pullback of the two along the intersection. To get confidence with, take the case of

$$\begin{tikzcd} \langle m+1\rangle \arrow[r] \arrow[d] & \langle 2\rangle \arrow[d] \\ \langle m\rangle \arrow[r] & \langle 1\rangle \end{tikzcd}$$

for $\mathcal{O}^{\otimes}$ a strict operad (here we are adding a new "component" via $\langle 1\rangle^0 \to \langle 2\rangle^0$).

Note that in particular a generalized operad fits into the preoperadic model structure, with marked edges being the inert morphisms. A map of generalized operads is thus a

preoperadic map (i.e. of marked simplicial sets over $\mathcal{N}(\mathrm{Fin}_*)$).

There is another model structure on preoperads where fibrant objects are *generalized* infinity operads:

Proposition 2.6.2. *There exist a left proper combinatorial simplicial model structure on $sSet^+_{/\mathcal{N}(Fin_*)}$, called the generalized preoperadic model structure, such that:*

- *Cofibrations are monomorphisms of underlying simplicial sets;*
- *An object is fibrant iff it is a generalized infinity operad.*
- *Weak equivalences are $\alpha : S \to T$ such that, for any generalized ∞ operad $\mathcal{O}^\otimes$*

$$\alpha_* : Map_{sSet^+_{/\mathcal{N}(Fin_*)}}(T, \mathcal{O}^\otimes) \to Map_{sSet^+_{/\mathcal{N}(Fin_*)}}(S, \mathcal{O}^\otimes)$$

is an homotopy equivalence.

This can be proven by the theory of categorical patterns devolved by Lurie in Higher Algebra. For a precise reference, see [10], §2.3.2.4 together with §2.3.2.6. We are particularly interested in the following kind of generalized operads:

Definition 2.6.3. Let C be an infinity category. A C-family of operads is a categorical fibration $p : \mathcal{O}^\otimes \to C \times \mathcal{N}(\mathrm{Fin}_*)$ such that:

- For every $c \in C$, the map $\mathcal{O}^\otimes_c \to \mathcal{N}(\mathrm{Fin}_*)$ is an ∞ operad.
- For every $X \in \mathcal{O}^\otimes_c$ and inert $\alpha : p(X) \to \langle m \rangle$, there exist a p-cocartesian lift $X \to Y$ of α in $\mathcal{O}^\otimes_c$;
- For every $X \in \mathcal{O}^\otimes_C$, $p(X) = \langle n \rangle$ p-cocartesian lifts $X \to X_i$ of the Kronecker maps ρ^i determine a p-limit diagram $(\langle n \rangle^0)^\triangleleft \to \mathcal{O}^\otimes$.

A few comments on the definition. The first condition is the most natural to ask for a family of operads. The last two are generalizations of operad conditions, where relative

concepts like cocartesianity and limit are taken with respect to the entire family p, and not only the operadic projection $p_c : \mathcal{O}_c^{\otimes} \to \mathcal{N}(\mathrm{Fin}_*)$. To understand the interaction between the concept of generalized operads and families, we recall the following proposition due to Lurie ([10], §2.3.2.11):

Proposition 2.6.4. *Let C be an ∞ category, and $p : \mathcal{O}^{\otimes} \to C \times \mathcal{N}(Fin_*)$ a categorical fibration of simplicial sets. The following are equivalent:*

1. *The map p is a fibration of generalized operads, and $\mathcal{O}^{\otimes}_{\langle 0 \rangle} \to C$ is a trivial Kan fibration;*

2. *The map p exhibits $\mathcal{O}^{\otimes}$ as a C family of operads.*

In particular, a family of operads is a generalized operad.

We do not prove the result, but we made a few comments. Firstly, note that $C \times \mathcal{N}(\mathrm{Fin}_*)$ is a generalized operad: indeed, any lifting can be taken constant in the first variable, and so we are reduced to show that $\mathcal{N}(\mathrm{Fin}_*)$ is an operad. Moreover, a direct corollary is the following:

Corollary 2.6.4.1. *Let C be an infinity category, and consider the homotopy fiber product $Op_\infty^C := Op_\infty^{gn} \times_{h\mathrm{Cat}_\infty} \{C\}$, where Op_∞^{gn} is the infinity category of generalized operads and the fiber product is taken with respect to the functor $F : \mathcal{O}^{\otimes} \mapsto \mathcal{O}^{\otimes}_{\langle 0 \rangle}$. Then Op_∞^C can be identified with a full subcategory of $p \in (Op_\infty^{gn})_{/\mathcal{N}(Fin_*) \times C}$ of C-families of operads. In other words, we can think of a generalized operad as a pair $(C, \mathcal{O}^{\otimes})$ where C is an infinity category and $\mathcal{O}^{\otimes}$ is a C-family.*

A proof can be found in [10], §2.3.2.13.

We define the last concept, that allows to find an ∞ operad that is operadically equivalent to a family of operads.

Definition 2.6.5. Suppose $\gamma : \mathcal{O}^{\otimes} \to \mathcal{O}'^{\otimes}$ is a map of generalized operads, where $\mathcal{O}^{\otimes}$ is a family of ∞ operads and $\mathcal{O}'^{\otimes}$ is an ∞ operad. We will say that γ assembles $\mathcal{O}^{\otimes}$ into

$\mathcal{O}'^{\otimes}$ if γ is an operadic equivalence. In other words, for every ∞ operad $\mathcal{P}^{\otimes}$ we have that

$$\gamma_* : \mathrm{Map}_{\mathrm{Op}_\infty^{gn}}(\mathcal{O}'^{\otimes}, \mathcal{P}^{\otimes}) \to \mathrm{Map}_{\mathrm{Op}_\infty^{gn}}(\mathcal{O}^{\otimes}, \mathcal{P}^{\otimes})$$

is an equivalence.

In particular, note that the fibrant replacement functor of the preoperadic model structure restricts to a functor $\mathrm{Assem} : \mathrm{Op}_\infty^{gn} \to \mathrm{Op}_\infty$, furnishing an assembly for any given family of operads.

The main reason why we introduced this generalized concept comes from the next two sections. For a complete account, see "Assembly and Disintegration", [10].

2.7 Disintegrating unital operads

We start with the definition of a *unital* operad.

Definition 2.7.1. An ∞ operad $\mathcal{O}^{\otimes}$ is unital if there exist an element $1 \in \mathcal{O}^{\otimes}_{\langle 0 \rangle}$ such that, for every $X \in \mathcal{O}^{\otimes}_{\langle 1 \rangle}$, the space $\mathrm{Map}(1, X)$ is contractible.

Remark 17. As $\mathcal{O}^{\otimes}_{\langle 0 \rangle} \simeq (\mathcal{O}^{\otimes}_{\langle 1 \rangle})^0 \simeq \Delta^0$, every two elements in $\mathcal{O}^{\otimes}_{\langle 0 \rangle}$ are equivalent. It follows that "there exist" can equivalently be substituted by "for every".

Furthermore, the situation of topological operad admits an intuitive criterion, which is the motivation for the definition. We also conjecture that the following proposition is an if and only if, despite we don't have a proof.

Proposition 2.7.2. *Let $\mathcal{O}$ be a topological operad, and let $\mathcal{O}^{\otimes}$ be its associated $\otimes$ construction. If $Mul_{\mathcal{O}}(\emptyset, X)$ is contractible, then the coherent nerve of the singular chain of $\mathcal{O}^{\otimes}$ is unital.*

Proof. Firstly, we note that the functor Sing preserves the homotopy type of map spaces, because $|\mathrm{Sing}(X)| \simeq X$. Henceforth, it is enough to show that if C is a simplicial category

and $\mathrm{Map}_C(x,y)$ is contractible, then $\mathrm{Map}_{\mathcal{N}_{coh}(C)}(x,y)$ is contractible. Calling $\mathfrak{C}$ the left adjoint to the coherent nerve, we have

$$\begin{aligned} \mathrm{Map}_{\mathcal{N}_{coh}(C)}(x,y)_n &\simeq \{x\} \times_{\Delta^{\{0\}}} \mathrm{Map}_{sSet}(\Delta^1, \mathcal{N}_{coh}(C))_n \times_{\Delta^{\{1\}}} \{y\} \simeq \\ \{x\} \times_{\Delta^{\{0\}}} \mathrm{Hom}(\mathfrak{C}(\Delta^1 \times \Delta^n), C) \times_{\Delta^{\{1\}}} \{y\} &\simeq \{x\} \times_{\Delta^{\{0\}}} \mathrm{Hom}(\mathfrak{C}(\Delta^1 \times \Delta^n), \mathrm{Map}_C(x,y)) \times_{\Delta^{\{1\}}} \{y\} \end{aligned}$$

Where we mean $\mathrm{Map}_C(x,y)$ to be the simplicial category that has two objects, $\mathrm{Map}(x,y)$ between the two, and just identities as endomorphisms. The last equation comes from the fact that a functor $\mathfrak{C}(\Delta^1 \times \Delta^n) \to C$ that restricts to $1_x, 1_y$ on the boundary of Δ^1, of course factors through $\mathrm{Map}(x,y)$ in a unique way. It follows that if $\mathrm{Map}_C(x,y)$ is contractible all the hom sets are naturally equivalent to a point, yielding contractibility of $\mathrm{Map}_{\mathcal{N}_{coh}(C)}(x,y)$. □

We now introduce the second main concept of this section, the one of reduced families of operads. If unital operads model the ones where nullary operations are trivial, reduced operads model the ones where also unary operations are trivial.

Definition 2.7.3. An ∞ operad $\mathcal{O}^\otimes$ is said to be reduced if it is unital and $\mathcal{O} := \mathcal{O}^\otimes_{\langle 1\rangle}$ is a contractible Kan complex.

This generalizes to families of operads in the following way:

Definition 2.7.4. A family of infinity operads $\mathcal{O}^\otimes$ is said to be reduced if $\mathcal{O}^\otimes_{\langle 0\rangle}$ is a Kan complex and for every $X \in \mathcal{O}^\otimes_{\langle 0\rangle}$, the infinity operad $\mathcal{O}^\otimes_{/X}$ is reduced.

It is not clear a priori that $\mathcal{O}^\otimes_{/X}$ is an infinity operad, and why it is the "right" thing to declare reduced. That's why we make the following

Remark 18. Let $X \in \mathcal{O}^\otimes_{\langle 0\rangle}$ and $c = p(X)$ the image in C of X. We claim that $\mathcal{O}^\otimes_{/X}$ is equivalent to the infinity operad $\mathcal{O}^\otimes_c$. Indeed, being X q-final by hypothesis, we have a trivial Kan fibration $\mathcal{O}^\otimes_{/X} \to C_{/c} \times \mathcal{N}(\mathrm{Fin}_*)_{/\langle 0\rangle} \simeq \mathcal{N}(\mathrm{Fin}_*)$. Moreover, $C_{/c}$ is a contractible Kan complex: there always exist a trivial $C_{/c} \times \Delta^1 \to C_{/c}$ given by contracting to 1_c,

and by the assumption that C is a Kan complex it is componentwise an equivalence. In particular, the condition of $\mathcal{O}^{\otimes}_{/X}$ being reduced can be substituted by $\mathcal{O}^{\otimes}_{c}$ reduced for every $c \in C$.

We are ready to state the main theorem. It basically tells us that any unital operad can be assembled as a reduced family of infinity operads. The complexity of unary operations is thus transferred to the complexity of the family, but each operad in the family has trivial unary operations.

Theorem 2.7.5. *Let $Op_{\infty}^{gn,rd}$ be the full subcategory of Op_{∞}^{gn} spanned by reduced families of operads. Then the essential image of the replacement functor* $\mathrm{Assem} : Op_{\infty}^{gn,rd} \to Op_{\infty}$ *is the full subcategory of unital operads $\mathcal{O}^{\otimes}$ such that $\mathcal{O}$ is a Kan complex. Moreover,* $\mathcal{O}^{\otimes}_{\langle 1\rangle} \to \mathrm{Assem}(\mathcal{O}^{\otimes})_{\langle 1\rangle}$ *is a homotopy equivalence.*

A proof is given in [10], §2.3.4.4, §2.3.4.5.

The theorem tells us that, given a unital operad, we can "disintegrate" it into a family of reduced of operads. This will be of aid in the chapter on little cube operads. We now pass to generalize the concept of Kan extension when it inter-plays with the theory of operads.

2.8 Relative operadic colimits

To introduce the notion of colimits that respect the operadic structure, we firstly have to define a symmetric monoidal structure on the active part $\mathcal{O}^{\otimes}_{act}$ of an ∞ operad. Informally, if X is classified by $(X_1, \ldots, X_n)$ and Y is classified by $(Y_1, \ldots, Y_m)$, the tensor product of X, Y is classified by the concatenation $(X_1, \ldots, X_n, Y_1, \ldots, Y_m)$. In particular, we have a functor $\oplus : \mathcal{C}^{\otimes}_{act} \times \mathcal{C}^{\otimes}_{act} \to \mathcal{C}^{\otimes}_{act}$ that we call *concatenation*. We won't need a precise definition of this notion. For a formal treatment using the theory of monoidal envelopes, see [10], §2.2.4.

Definition 2.8.1. For $f : X \to \mathcal{C}^{\otimes}_{act}$, denote by $\mathcal{C}^{act}_{f/}$ the pullback $\mathcal{C} \times_{\mathcal{C}^{act}} (\mathcal{C}^{\otimes}_{act})_{f/}$.

Definition 2.8.2. Let $q : \mathcal{C}^{\otimes} \to \mathcal{O}^{\otimes}$ be a fibration of ∞ operads. A diagram $\bar{p} : K^{\triangleright} \to \mathcal{C}^{\otimes}_{act}$ is said to be a weak operadic q colimit diagram if

$$\mathcal{C}^{act}_{\bar{p}/} \to \mathcal{C}^{act}_{p/} \times_{\mathcal{O}^{act}_{qp/}} \mathcal{O}^{act}_{q\bar{p}/}$$

is a trivial Kan fibration, where $p = \bar{p}|K$. In other words, an (active) map of (active) cocones $\bar{p} \to f$ is essentially determined by its image in $\mathcal{O}^{\otimes}$.

We are ready to define what a relative operadic colimit is:

Definition 2.8.3. Let $q : \mathcal{C}^{\otimes} \to \mathcal{O}^{\otimes}$ be a fibration of ∞ operads. A diagram $\bar{p} : K^{\triangleright} \to \mathcal{C}^{\otimes}_{act}$ is said to be an operadic q colimit diagram if for every $C \in \mathcal{C}^{\otimes}_{act}$

$$K^{\triangleright} \xrightarrow{\bar{p}} \mathcal{C}^{\otimes}_{act} \xrightarrow{\oplus C} \mathcal{C}^{\otimes}_{act}$$

is a weak operadic q-colimit diagram.

Remark 19. Note that if a diagram is an operadic q-colimit, in particular (taking C to be the unit of the symmetric monoidal structure on $\mathcal{C}^{\otimes}_{act}$) it is a weak operadic q-colimit. In the next sections, we will use an existence theorem for operadic colimits. In this way we can actually compute it via the weak operadic formula, without making any reference to the symmetric monoidal structure on $\mathcal{C}^{\otimes}_{act}$.

Remark 20. In the same fashion of relative colimits, one can show that if $j : L \to K$ is left cofinal and $p : K^{\triangleright} \to \mathcal{C}^{\otimes} \to \mathcal{O}^{\otimes}$ is a weak operadic q-colimit diagram iff $pj^{\triangleright}$ is.

Remark 21. Again, one can show with the aid of proposition 2.1.2.1, [11], that the map in the weak operadic colimit definition is trivial Kan fibration iff it is a categorical equivalence.

The main result we will use is a special case of the theory developed from Lurie, formulated ad-hoc in this way by the author.

Proposition 2.8.4 (Formula for absolute operadic colimits in symmetric monoidal categories)**.** . *Let $q : \mathcal{C}^{\otimes} \to \mathcal{N}(Fin_*)$ be a symmetric monoidal category. Let $\bar{p} : K^{\triangleright} \to \mathcal{C}^{\otimes}_{act}$ be a diagram. Then there exist an explicitly computable functor $\bar{p^{\otimes}} : K^{\triangleright} \to \mathcal{C}$ such that $\bar{p}$ is a weak operadic q-colimit diagram iff $\bar{p^{\otimes}}$ is a colimit diagram. Informally, if p is given by $p(k) = (\langle n \rangle(k), X_1(k), \ldots, X_n(k))$, $p^{\otimes}$ is given by*

$$r(k) = X_1(k) \otimes \ldots X_n(k)$$

In particular, if $\bar{p}$ is an operadic colimit, it is given by the above formula.

Proof. Let $\bar{p}^{\otimes} : K^{\triangleright} \xrightarrow{\bar{p}} \mathcal{C}^{\otimes} \xrightarrow{\otimes} \mathcal{C}$ be the functor built in the following way. For every $k \in K^{\triangleright}$, we have a cocartesian lifting $s(k)$ of the unique active map $qp(k) \to \langle 1 \rangle$ starting from $\bar{p}(k)$ and ending in $\bar{p}^{\otimes}(k) \in \mathcal{C}$. Notice that $q\bar{p} : K^{\triangleright} \to \mathcal{C}^{\otimes}_{act} \to \mathcal{N}(\mathrm{Fin}_*)_{act}$ has an homotopy to the constant functor $\langle 1 \rangle$, because $\langle 1 \rangle$ is final in $\mathcal{N}(\mathrm{Fin}_*)_{act}$. Choose a lifting H of this homotopy via the cocartesian morphisms $s(k)$ (see [6], §2.6.17). By evaluating at 1, we get a functor $\bar{p}^{\otimes} : K^{\triangleright} \to \mathcal{C}$. By proposition [10], §3.1.1.16, note that $\bar{p}$ is a weak operadic colimit diagram iff $\bar{p}^{\otimes}$ is so. Moreover, note that everything in $C^{act}_{p^{\otimes}/}$ lies over $\langle 1 \rangle$, and conversely everything in $C_{p^{\otimes}/}$ is active. Thus we have that the inclusion $C_{p^{\otimes}/} \to C^{act}_{p^{\otimes}/}$ is an isomorphism. Analogously, $C_{\bar{p}^{\otimes}/} \to C^{act}_{\bar{p}^{\otimes}/}$ is an equivalence. Thus we have a diagram

$$\begin{tikzcd}
\mathcal{C}^{act}_{\bar{p}^{\otimes}/} \arrow[r, "\simeq"] \arrow[d] & \mathcal{C}_{\bar{p}^{\otimes}} \arrow[d] \\
\mathcal{C}^{act}_{p^{\otimes}/} \arrow[r, "\simeq"] & \mathcal{C}_{p^{\otimes}}
\end{tikzcd}$$

From which we deduce that $\bar{p}^{\otimes}$ is a weak operadic q-colimit diagram seen as a functor $\to \mathcal{C}^{\otimes}_{act}$ iff it is a q-colimit diagram seen as a functor $\to \mathcal{C}$.

□

In the case of proof above, we call $p^{\otimes}$ the *associated* $\otimes$ diagram. We have the following stronger corollary:

Corollary 2.8.4.1. *Let $p : K^{\triangleright} \to \mathcal{C}^{\otimes}_{act}$ an active diagram, where $q : \mathcal{C}^{\otimes} \to \mathcal{N}(Fin_*)$ is a symmetric monoidal category. Let $p^{\otimes}$ be the associated $\otimes$ diagram. Then p is an operadic q-colimit diagram iff $p^{\otimes} \otimes C$ is a colimit diagram for every $C \in \mathcal{C}^{\otimes}$.*

Proof. It is enough to note that $(p \oplus C)^{\otimes} \simeq p^{\otimes} \otimes C$, see also the introduction to relative operadic colimits. The result follows from proposition 2.8.4. □

2.9 Relative operadic Kan extensions

Let $\mathcal{M}^{\otimes} \to \mathcal{N}(\mathrm{Fin}_*) \times \Delta^1$ be a Δ^1-family of operads. We will call this family a *correspondence* from $\mathcal{A}^{\otimes} := \mathcal{M}^{\otimes} \times_{\mathcal{N}(\mathrm{Fin}_*)\times\Delta^1} \mathcal{N}(\mathrm{Fin}_*) \times \Delta^{\{0\}}$ to $\mathcal{B}^{\otimes} := \mathcal{M}^{\otimes} \times_{\mathcal{N}(\mathrm{Fin}_*)\times\Delta^1} \mathcal{N}(\mathrm{Fin}_*) \times \Delta^{\{1\}}$.

We give the definition of an operadic left Kan extension.

Definition 2.9.1. Let $\mathcal{M}^{\otimes} \to \mathcal{N}(\mathrm{Fin}_*) \times \Delta^1$ be a correspondence from $\mathcal{A}^{\otimes}$ to $\mathcal{B}^{\otimes}$. Let $q : \mathcal{C}^{\otimes} \to \mathcal{O}^{\otimes}$ be a fibration of ∞ operads. For $B \in \mathcal{B}^{\otimes}$, define $K_B := (\mathcal{M}^{\otimes}_{act})_{/B} \times_{\mathcal{M}^{\otimes}} \mathcal{A}^{\otimes}$. The map $K_B \to (\mathcal{M}^{\otimes}_{act})_{/B}$ induces by adjunction a map $K_B^{\triangleright} \to \mathcal{M}^{\otimes}_{act}$. We say that $\bar{F} : \mathcal{M}^{\otimes} \to \mathcal{C}^{\otimes}$ is an operadic q-left Kan extension of $F = \bar{F}|\mathcal{A}^{\otimes}$ if for every B the map

$$K_B^{\triangleright} \to \mathcal{M}^{\otimes} \to \mathcal{C}^{\otimes}$$

is an operadic q-colimit diagram.

Remark 22. Notice that a diagram of infinity operads $\mathcal{N}(D) \to \mathrm{Cat}_{\infty}$ induces, by the relative nerve construction, a $\mathcal{N}(D)$ family of operads $\mathcal{X}^{\otimes} \to \mathcal{N}(\mathrm{Fin}_*) \times \mathcal{N}(D)$. In particular, if one takes $D = [n]$, we obtain a Δ^n family.

Remark 23. Taking [6] §2.6.2.1 as a model for the relative nerve, one can show that if $\mathcal{M}^{\otimes} \to \Delta^1 \times \mathcal{N}(\mathrm{Fin}_*)$ is the relative nerve of a functor $F : \mathcal{A}^{\otimes} \to \mathcal{B}^{\otimes}$ interpreted as $\mathcal{F} : \Delta^1 \to \mathrm{Cat}_{\infty}$, then for every $B \in \mathcal{B}^{\otimes}$:

$$\mathcal{M}^{\otimes}_{/B} \times_{\mathcal{M}^{\otimes}} \mathcal{A}^{\otimes} \equiv \mathcal{B}^{\otimes}_{/B} \times_{\mathcal{B}^{\otimes}} \mathcal{A}^{\otimes}$$

Meaning that they are isomorphic. Recall that

$$\mathcal{M}_n := (\sigma : [n] \to [1], \{\tau_S : \Delta^S \to \mathcal{F}(\max \sigma(S)\}_{S \subset [n]} \text{ such that for every } S \subset S' :$$

$$\begin{tikzcd} \Delta^S \arrow[r, "\tau_S"] \arrow[d] & \mathcal{F}(\max \sigma(S)) \arrow[d, "\mathcal{F}(\max \sigma(S) \max \sigma(S')"] \\ \Delta^{S'} \arrow[r, "\tau_{S'}"] & \mathcal{F}(\max \sigma(S')) \end{tikzcd}$$

With boundaries given by the restriction of the family τ_S. We can give the following description of n-simplices above:

$$(\mathcal{M}^{\otimes}_{/B} \times_{\mathcal{M}^{\otimes}} \mathcal{A}^{\otimes})_n = \{(\sigma : [n{+}1] \to [1], \tau_S) : \sigma(0 \dots n) = 0, \sigma(n{+}1) = 1, \tau_{\{n+1\}} = B \text{ and commutation}\}$$

We have a natural map towards $\mathcal{B}^{\otimes}_{/B} \times_{\mathcal{B}^{\otimes}} \mathcal{A}^{\otimes}$ given by $(\sigma, \tau_S) \to (\tau_{\{0,\dots,n+1\}}, \tau_{\{0,\dots,n\}})$. We have an explicit inverse given by $(\alpha, \beta) \mapsto (\sigma, \tau_S^{\alpha\beta})$, where:

- σ is given by $\sigma(i) = 0$ for $0 \leq i \leq n$, and $\sigma(n+1) = 1$;
- $\tau_S^{\alpha\beta} = \alpha|\Delta^S$ if $n+1 \in S$, otherwise $\tau^{\alpha\beta} = \beta|S$.

By the commutation constraint on the τ_S, this is an isomorphism.

Remark 24. By the previous remark, we can formulate the notion of operadic q-left Kan extension along a functor $F : \mathcal{A}^{\otimes} \to \mathcal{B}^{\otimes}$ without making reference to the correspondence. Explicitly, we say that $G : \mathcal{B}^{\otimes} \to \mathcal{C}^{\otimes}$ is an operadic left Kan extension relative to $q : \mathcal{C}^{\otimes} \to \mathcal{O}^{\otimes}$ if, for every B, the map

$$(\mathcal{B}^{\otimes}_{act})_{/B} \times_{\mathcal{B}^{\otimes}} \mathcal{A}^{\otimes})^{\triangleright} \to \mathcal{B}^{\otimes} \to \mathcal{C}^{\otimes}$$

is an operadic q-colimit diagram.

Remark 25. Note that if $\alpha : \mathcal{A}^{\otimes} \subset \mathcal{B}^{\otimes}$ is an inclusion of ∞ operads, then for $A \in \mathcal{A}^{\otimes}$ the diagram

$$(\mathcal{B}^{\otimes}_{act})_{/A} \times_{\mathcal{B}^{\otimes}} \mathcal{A}^{\otimes})^{\triangleright}$$

has a final object given by $(1_A, A)$. The requirement that the functor is an operadic colimit diagram is automatically satisfied, because the arrow is an identity. In particular, we get that if $F : \mathcal{B}^{\otimes} \to \mathcal{C}^{\otimes}$ is an operadic Kan extension of $G : \mathcal{A}^{\otimes} \to \mathcal{C}^{\otimes}$, then $\alpha^* F = G$.

We expose here the main property that operadic extensions enjoy:

Theorem 2.9.2 (Transitivity formula for Operadic Kan Extension)**.** *Let $\mathcal{M}^{\otimes} \to \Delta^2 \times \mathcal{N}(Fin_*)$ be a flat Δ^2 family of ∞ operads, $q : \mathcal{C}^{\otimes} \to \mathcal{O}^{\otimes}$ a fibration of ∞ operads, and $F : \mathcal{M}^{\otimes} \to \mathcal{C}^{\otimes}$ a map of generalized operads. Suppose that $F|\mathcal{M}^{\otimes} \times_{\Delta^2} \Delta^{\{0,1\}}, F|\mathcal{M}^{\otimes} \times_{\Delta^2} \Delta^{\{1,2\}}$ are q-left operadic Kan extensions. Then $F|\mathcal{M}^{\otimes} \times_{\Delta^2} \Delta^{\{0,2\}}$ is an operadic q-left Kan extension.*

A proof can be found in [10], §3.1.4.1.

Remark 26. Suppose we have two ordinary functors $M : C^{\otimes,op} \times D^{\otimes} \to \text{Sets}, N : D^{\otimes,op} \times E^{\otimes} \to \text{Sets}$, where $C^{\otimes}, D^{\otimes}, E^{\otimes}$ are the $\otimes$ constructions for ordinary operads. These are called distributors and are meant to generalize functors. Indeed, a functor $F : X \to Y$ gives a distributor by the formula $\text{Hom}_Y(Fx, y)$. As well explained in [10] B.3, a distributor yields a correspondence, and consecutive distributors yield a flat [2] family. Taking the nerve, we get a flat Δ^2 family of operads and we can apply the above theorem for transitivity. Note that the composition of two distributors $M : C^{\otimes,op} \times D^{\otimes} \to \text{Sets}, N : D^{\otimes,op} \times E^{\otimes} \to \text{Sets}$, generalizing the composition of two functors, is given by the coend formula

$$(N \circ M)(C, E) = \int_{D \in D^{\otimes}} N(C, D) \times M(D, E)$$

To talk about operadic extensions in next chapters, we need an existence theorem, see also 19. Let us firstly give a definition.

Definition 2.9.3. Let $\mathcal{K}$ be a collection of sSets, $q : \mathcal{C}^{\otimes} \to \mathcal{N}(\text{Fin}_*)$ a symmetric monoidal category. We say that $\otimes$ preserves $\mathcal{K}$-indexed colimits if, for every $C \in \mathcal{C}^{\otimes}$ and colimit diagram $p : K^{\triangleright} \to \mathcal{C}$, $p \otimes C$ is still a colimit diagram.

A symmetric monoidal category is said to be *sifted*-complete if sifted colimits exist and $\otimes$ preserves sifted colimits. Recall that a diagram K is said to be sifted if the diagonal $K \to K \times K$ is left cofinal.

Example 4. Chain$(k)^{\otimes}$, *for k a field, is sifted complete.*

We are ready to state the existence theorem, making reference to the definition of operadic Kan extension:

Theorem 2.9.4 (Existence for Operadic Kan extensions)**.** *Suppose that $q : \mathcal{C}^{\otimes} \to \mathcal{N}(Fin_*)$ is sifted complete. Take a correspondence $\mathcal{M}^{\otimes}$ from $\mathcal{A}^{\otimes}$ to $\mathcal{B}^{\otimes}$ and a functor $F : \mathcal{A}^{\otimes} \to \mathcal{C}^{\otimes}$. Suppose that for every $B \in \mathcal{B}$ there exist a factorization $K_B \to K'_B \to \mathcal{C}^{\otimes}$ such that:*

1. *K'_B is sifted;*

2. *$K_B \to K'_B$ is left cofinal;*

Then there exist an operadic q-left Kan extension of F.

Proof. In virtue of [10], §3.1.2.3 (A), it is enough to show that for any $B \in \mathcal{B}^{\otimes}$ the diagram $K_B^{\triangleright} \to \mathcal{C}^{\otimes}$ is an operadic colimit diagram. Note that if $B \simeq (B_1, \dots B_k)$, then

$$(\mathcal{M}^{\otimes}_{act})_{/B} \simeq \prod_{i=1}^{k} (\mathcal{M}^{\otimes}_{act})_{/B_i}$$

because the components express B as the q-product of B_i. It follows that

$$K_B = (\mathcal{M}^{\otimes}_{act})_{/B} \times_{\mathcal{M}^{\otimes}_{act}} \mathcal{A}^{\otimes} \simeq (\prod_{i=1}^{k} (\mathcal{M}^{\otimes}_{act})_{/B_i}) \times_{\mathcal{M}^{\otimes}_{a}ct} \mathcal{A}^{\otimes}) \simeq \prod_{i=1}^{k} K_{B_i}$$

Note that by the absolute formula for operadic Kan extensions in smc's, it is enough to show that there exist the colimit of a K_B diagram in $\mathcal{C}$ (the associated $\otimes$ diagram), and that it is preserved by tensoring with an arbitrary element. By hypothesis, $K_{B_i} \to \mathcal{C}^{\otimes}$ factors through K'_{B_i} in a cofinal way, so that we have a commutative diagram of associated

$\otimes$ constructions:

$$K_B \to \prod_{i=1}^{k} K'_{B_i} \to \mathcal{C}$$

where the first is cofinal. Note that a product of sifted diagrams is sifted, because product of cofinal maps is cofinal. Indeed, one can use the Joyal criterion 1.7.2: if $C \to D, C' \to D'$ are cofinal, then for every (d, d')

$$C \times C' \times_{D \times D'} (D \times D')_{/(d,d')} \simeq (C \times_D D_{/d}) \times (C' \times_{D'} D'_{/d'}$$

and the latter is weakly contractible by cofinality of $C \to D, C' \to D'$. By the hypothesis the colimit over $\prod K_{B_i}$ exist and it is preserved by the tensor product; so it is its cofinal restriction to K_B. By the 2.8.4.1 criterion, $K_B \to \mathcal{C}$ is an operadic colimit diagram. $\square$

Chapter 3

Little cube operads

The little cube operads are the first interesting example of infinity operad that is essentially different from ordinary operads. As we have mentioned in the operad chapter, they are a series of infinity operads

$$\mathbb{E}_1 \subset \ldots \subset \mathbb{E}_n \subset \ldots \subset \mathbb{E}_\infty$$

such that $\mathbb{E}_1$ is equivalent to the associative operad, and $\mathbb{E}_\infty$ is equivalent to the commutative operad. They are meant to encode increasingly higher commutativity between k-homotopies. This will be manifest from the fact that multimaps spaces in $\mathbb{E}_n$ are $(n-2)$ connected. In the end, we will give some examples and quote some important theorems about $\mathbb{E}_n$ algebras.

3.1 Definition

Fix a natural number $n \geq 1$. Define:

- $\square^n := (-1,1)^n$ the n-cube;
- A rectilinear embedding $\square^n \to \square^n$ as a composition of shrinks and translations;

namely a function that has the form

$$(x_1, \ldots, x_n) \mapsto (a_1 x_1 + b_1, \ldots, a_n x_n + b_n)$$

where b_i and $a_i > 0$ are real numbers.

- Given a finite set S, $\mathrm{Rect}(\square^n \times S, \square^n)$ as the space of embeddings which are rectilinear on each component. Note that this is not equal to $\prod_S \mathrm{Rect}(\square^n, \square^n)$, because the function must be globally an embedding (so the choice of the image of the S squares is not independent).

Let $\mathbb{E}_n^t$ be the topological operad that has one object and multimaps spaces given by

$$\mathrm{Mul}(\{*\}_{s \in S}, \{*\}) = \mathrm{Rect}(\square^n \times S, \square^n)$$

We can turn this into a locally Kan simplicial operad by setting $\mathbb{E}_n^\Delta = \mathrm{Sing}\mathbb{E}_n^t$. The Sing functor in this case takes the singular chain functor Hom-wise. The simplicial $\otimes$ construction yields a simplicial category $\mathbb{E}_n^{\Delta,\otimes}$ over Fin_*. Finally, the coherent nerve of the latter $\mathbb{E}_n^\otimes = \mathcal{N}_{coh}(\mathbb{E}_n^{\Delta,\otimes})$ yields an infinity operad, called the *little n-cube operad*.

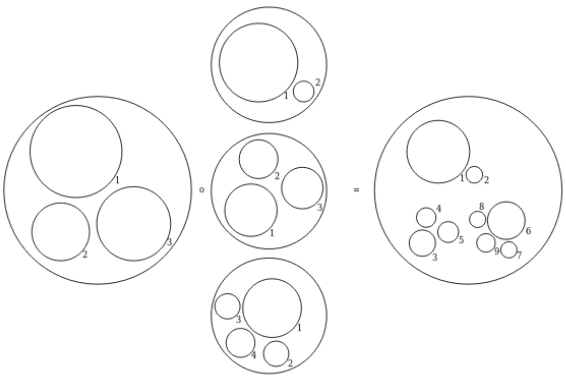


Figure 3.1: An example of composition in the little 2-cube operad (disks represent cubes!)

3.2 Homotopy groups of map spaces

We have in mind that an $\mathbb{E}_n^{\otimes}$ algebra is a map of operads $\mathcal{N}_{coh}(\mathbb{E}_n^{\Delta,\otimes}) \to \mathcal{O}^{\otimes}$. The latter corresponds to a map $\mathbb{E}_n^{\Delta,\otimes} \to \mathfrak{C}(\mathcal{O}^{\otimes})$ over Fin_* that preserve inerts, by adjunction (see ..). Thus, in order to study the homotopy type of k-ary operations in an $\mathbb{E}_n^{\otimes}$ algebra, it makes sense to study the spaces $\mathrm{Rect}(\square^n \times \{1,\dots,k\}, \square^n)$, which completely determines simplicial map spaces in the $\otimes$ construction $\mathbb{E}_n^{\Delta,\otimes}$.

For a space X, define the configuration spaces $\mathrm{Conf}_k(X)$ as the space of injective functions $\{1,\dots,k\} \to X$, topologized as a subspace of X^k. We now show the following

Lemma 15. *There is an homotopy equivalence*

$$\mathrm{ev}_0 : \mathrm{Rect}(\square^n \times \{1,\dots,k\}, \square^n) \to \mathrm{Conf}_k(\square^n)$$

Given by evaluation at the centers of the k squares.

Proof. We define an explicit homotopy inverse. Given $b = (b_1,\dots,b_k) \in \mathrm{Conf}_k(\square^n)$, let

$$\delta(b) := \frac{1}{3} \min_{i \neq j} d(b_i, b_j)$$

be the "safety radius". Here d stands for the euclidean distance. We claim that the function $s : \mathrm{Conf}_k(\square^n) \to \mathrm{Rect}(\square^n \times \{1,\dots,k\}, \square^n)$ defined as

$$s(b)(y_1,\dots,y_k) = (\delta(b)y_1 + b_1, \dots, \delta(b)y_k + b_k)$$

is an deformation section to ev_0. Note that it is well defined because $\delta(b)$ is small enough: indeed, if $\delta(b)y_i + b_i = \delta(b)y_j + b_j$ for distinct indices, then

$$3\delta(b) \leq \|b_i - b_j\| = \delta(b)\|y_i - y_j\| \leq 2\delta(b)$$

which is a contradiction. We easily see from the definition that $\mathrm{ev}_0 \circ s(b) = b$. For the other side, define

$$H : \mathrm{Rect}(\square^n \times \{1, \dots, k\}, \square^n) \times I \to \mathrm{Rect}(\square^n \times \{1, \dots, k\}, \square^n)$$

as (the function f correspond to coefficients a_i, b_i):

$$H(f,t)(y_1, \dots, y_k) = ([t\delta(b) + (1-t)a_1]y_1 + b_1, \dots, [t\delta(b) + (1-t)a_k]y_k + b_k)$$

It is an embedding at every time t for the same reason as above. It is straightforward to check that $H(f,0) = f, H(f,1) = s \circ \mathrm{ev}_0(f)$, yielding the desired homotopy equivalence.

$\square$

Note that $\mathrm{SingRect}(\square^n \times \{1, \dots, k\}, \square^n)$ is weakly equivalent to $\mathrm{Rect}(\square^n \times \{1, \dots, k\}, \square^n)$; so it is enough to study $\mathrm{Conf}_k(\square^n)$ in order to determine the homotopy type of multimap spaces in $\mathbb{E}_n^{\Delta}$.

We proceed with our investigation thanks to the following observation:

Lemma 16. *For $n \geq 2$, there exist a serre fibration*

$$\bigvee_{i=1}^{k} S^{n-1} \to \mathrm{Conf}_{k+1}(\square^n) \to \mathrm{Conf}_k(\square^n)$$

Indeed, the fiber is locally $\mathrm{Conf}_1(\square^n \setminus J) = \square^n \setminus J$, where J are k points. If we assume wlog that the points lie on a circumference, it is easy to see that $\square^n \setminus J$ deforms onto a wedge of spheres. Indeed, split the cube in k consecutive radial regions each containing exactly one point of J. The punctured regions deforms to the boundary which is S^{n-1}. Collapsing the radiuses of this "wheel" one obtains the desired wedge of spheres.
We obtain an exact sequence of homotopy groups

$$\pi_\ell(\bigvee_{i=1}^{k} S^{n-1}) \to \pi_\ell(\mathrm{Conf}_{k+1}(\square^n)) \to \pi_\ell(\mathrm{Conf}_k(\square^n))$$

By induction on k, we can show that $\mathrm{Conf}_k(\square^n)$ is $(n-2)$ connective. For $k = 1$, the space is contractible and there is nothing to show. For higher k's, note that $\pi_\ell(\bigvee_{i=1}^k S^{n-1}) = 0$ for $\ell \leq n-2$. By inductive hp, the same holds for $\mathrm{Conf}_k(\square^n)$. By the exact sequence, we get the result for $k+1$.

3.3 The whole series

Firstly, we explain why $\mathbb{E}_n$ embeds into $\mathbb{E}_{n+1}$. Indeed, at the level of topological operads, the rectilinear embeddings $\square^n \times S \to \square^n$ are a subset of rectilinear embeddings $\square^{n+1} \times S \to \square^{n+1}$ via a certain homomorphism $\varphi_{n,n+1}$. Indeed, for $f : \square^n \times S \to \square^n$ rectilinear given by $(x_1, \ldots, x_n, i) \mapsto (a_1^{(i)} x_1 + b_1^{(i)}, \ldots, a_n^{(i)} x_n + b_n^{(i)})$, just set

$$\varphi_{n,n+1}(f)(x_1, \ldots, x_{n+1}, i) = (a_1^{(i)} x_1 + b_1^{(i)}, \ldots, a_n^{(i)} x_n + b_n^{(i)}, x_{n+1})$$

Taking Sing and the coherent nerve of $\varphi_{n,n+1}$ yields a morphism $\mathbb{E}_n^{\otimes} \to \mathbb{E}_{n+1}^{\otimes}$. As this morphism is injective, we can really see the n little cube operad as a subset of the $n+1$ little cube operad.

Let us now define the $\mathbb{E}_\infty^{\otimes}$ operad. The naive definition is just the colimit in the category of simplicial sets of $\mathbb{E}_n^{\otimes}$ via the link maps $\varphi_{n,n+1}$. It is straightforward to check that the properties of being an infinity operad are satisfied, because this colimit is in fact a union level-wise; so every property that involves a finite number of elements can be verified in one fixed operad.

We observe a simple property. Both Sing and the coherent nerve are functors of the form $\mathrm{Hom}(s_n, \cdot)$ for s_n some cosimplicial compact object. It is evident from this description that they commute with infinite unions (or more generally with filtered colimits). In particular, we can define the $\mathbb{E}_\infty$ at a topological level, then taking hom-wise sing, and finally the coherent nerve: we will end up with $\mathbb{E}_\infty^{\otimes}$ as we have defined it above.

This fact gives us that the mapping spaces of the topological $\mathbb{E}_\infty^t$ are weakly contractible, because of the increasing connectivity we gain in the $\mathbb{E}_n$'s. Consider the constant

functor from $\mathbb{E}^t_\infty \to \mathrm{Comm}$, the latter being the operad with constant multimaps spaces (see..). The space of rectilinear embeddings is defined as a finite union of spaces defined by linear inequalities, they are a CW complex, as well as their infinite union in $\mathbb{E}^t_\infty$. We take the topological $\otimes$ construction of both $\mathbb{E}_\infty$ and Comm. The constant functor yields a functor of topological categories $T : \mathbb{E}^\otimes_\infty \to \mathrm{Comm}^\otimes$ such that:

- It is the identity on objects, in particular essentially surjective;
- It is a weak homotopy equivalence between CW complexes at multimapping spaces level; so, by whitehead theorem, it is an homotopy equivalence.

For a variant of fully faithful and essentially surjective implies equivalence in the realm of topological categories, we have that the functor is an equivalence. Passing to the quasi category realm, we get that $\mathbb{E}^\otimes_\infty$ is equivalent to $\mathcal{N}(\mathrm{Comm}^\otimes)$ [1].

On balance, at the extremely right part of the $\mathbb{E}_n$ series we have an operad modeling up to equivalence the commutative operations (where all homotopies commute, also). Let's see what happens at the extreme left.

As with the infinite case, we start by investigating $\mathbb{E}_1$ at a topological level. We saw that multimap spaces are equivalent to $\mathrm{Conf}_k(\square^1)$. Every configuration naturally determines a linear ordering on $\{1, \ldots, k\}$, by pulling back the linear order on $\square^1$. We call this function r. Formally, we say that $i < j$ iff $x_i < x_j$. We can represent a linear order by a function $\phi : \{1, \ldots, k\} \to \{1, \ldots, k\}$, where $i \leq j$ iff $\phi(i) \leq \phi(j)$.

We have a section s of r given by $\phi \mapsto (-\frac{1}{2} + \frac{\phi(i)-1}{k-1})_{i=1\ldots k})$. For example, if ϕ is the identity, this is a fixed choice of a configuration inducing the identity order. We have that $rs(\phi) = \phi$ by the very definition. On the other side $sr(x)$ can be deformed via a linear path to x: there will be no collapses because they induce the same order on points.

[1] Usually, one drops the $\mathcal{N}$, as it is evident from the context whether relaxed or ordinary version is meant

We thus have a functor $\mathbb{E}_1^t \to \mathrm{Ass}$, the operad introduced in (..), that is an homotopy equivalence at multimap level. Reasoning as in the infinite case, we get that $\mathbb{E}_1^{\otimes}$ is equivalent to $\mathcal{N}(\mathrm{Ass}^{\otimes})$.

On balance, at the extreme left we have an operad modeling up to equivalence associative operations.

3.4 Examples and fundamental theorems

In the following, we collect some examples and theorems about little cube operads.

Example 5. *(n-loop spaces). This is the fundamental example of $\mathbb{E}_n$ spaces. Take a pointed space X and fix a number $n \geq 0$. Recall that ΩX are the functions from the interval to X such that the boundary goes to the point. The structure functions*

$$\mathrm{Rect}(\square^n \times \{1, \ldots, k\}, \square^n) \times (\Omega^n X)^k \to \Omega^n X$$

are given by following construction. We are given an embedding of k cubes into itself, k functions from the cube to X such that the boundary goes to the point, and we must produce a function from the cube to X such that the boundary goes to the point. That's easy! Define the function from the cube as one of the input functions on the image of the embedding, and the point otherwise. It is easy to see that this construction is compatible with the structure of operad.

Example 6. *(free $\mathbb{E}_k$ algebra) For a topological space Z and a natural number $k \geq 1$, consider the following topological space*

$$\mathcal{F}_k(Z) := \coprod_{n \in \mathbb{N}} \mathrm{Conf}_n(\square^k) \times Z^n$$

An element of $\mathcal{F}_k(Z)$ can be thought as a "decorated" configuration of points, i.e. an element of Z lives over every point. To give an operadic structure we must produce functions

$$\mathcal{F}_k(Z)^m \times \mathrm{Rect}(\square^k \times \{1, \dots, m\}, \square^k) \to \mathcal{F}_k(Z)$$

Given an element of the left hand side, represented by

$$(((p_1^1, z_1^1, \dots, p_{n_1}^1, z_{n_1}^1), \dots, (p_1^m, z_1^m, \dots, p_{n_m}^m, z_{n_m}^m)), (f_1, \dots, f_m))$$

We just output the decorated configuration

$$(f_1(p_1^1), z_1^1, \dots, f_1(p_{n_1}^1), z_{n_1}^1, \dots, f_m(p_1^m), z_1^m, \dots, f_m(p_{n_m}^m), z_{n_m}^m)$$

You can see how this is operadic? We are literally plugging a bunch of (decorated) configurations inside the cube via an embedding, and this is exactly how an operad is supposed to act (see the picture ..). The precise check that everything is well defined and continuous is left to the reader.

There is a sort of "converse" to example 3. Given an $\mathbb{E}_n$ space, i.e. a functor of topological categories $X : \mathbb{E}_n^t \to \mathrm{Top}$, we can firstly restrict to $\mathbb{E}_1^t$ and then post compose with the functor $\pi_0 : \mathrm{Top} \to \mathrm{Set}$. We will end up with a discrete functor $\pi_0(X) : \mathbb{E}_1^t \to \mathrm{Set}$, that exactly correspond to the notion of a monoid. If this monoid is a group, we say that X is *group-like*.

Note that loop spaces are group-like, because $\pi_0(\Omega^n X) \simeq \pi_n(X)$ is a group under the composition of loops. Moreover, the following theorem holds:

Theorem 3.4.1. *(May recognition principle) A group like $\mathbb{E}_n$ space is weakly equivalent to a n loop space.*

The proof is rather complicated; we defer the interested reader to [3] for further details.

Another important theorem in $\mathbb{E}_n$ theory is the following, that relates the levels of commutativity between the different $\mathbb{E}_n$'s, $n \in \mathbb{N}$. Recall that $\mathcal{O}^{\otimes}$-algebras over a symmetric

monoidal category are a symmetric monoidal category itself, with tensor product informally given by $(F \otimes G)(x) := F(x) \otimes G(x)$. Furthermore, $\mathcal{P}^{\otimes}$-algebras over $\mathrm{Alg}_{\mathcal{O}^{\otimes}}(C^{\otimes})^{\otimes}$ can be interpreted as a subset of $\mathrm{Fun}(\mathcal{P}^{\otimes} \times \mathcal{O}^{\otimes}, C^{\otimes})$. These functors are called *bifunctors*, in analogy with bilinear functors of vector spaces. We say that an operad $\mathcal{Q}^{\otimes}$ is the tensor product of $\mathcal{O}^{\otimes}, \mathcal{P}^{\otimes}$ if there exist a $\varphi : \mathcal{O}^{\otimes} \times \mathcal{P}^{\otimes} \to \mathcal{Q}^{\otimes}$ such that for every symmetric monoidal category $C^{\otimes}$ the restriction along φ

$$\varphi^* : \mathrm{Alg}_{\mathcal{Q}^{\otimes}}(C^{\otimes}) \to \mathrm{Fun}(\mathcal{O}^{\otimes} \times \mathcal{P}^{\otimes}, C^{\otimes})$$

is fully faithful and is essential image are bifunctors, i.e.

$$\mathrm{Alg}_{\mathcal{O}^{\otimes}}(\mathrm{Alg}_{\mathcal{P}^{\otimes}}(C^{\otimes}))$$

We have the following lemma:

Lemma 17. *The tensor product of two infinity operads always exist.*

Proof. (sketch) Take $\mathcal{O}^{\otimes}, \mathcal{P}^{\otimes}$ two ∞ operads. In particular, they are marked simplicial sets over $\mathcal{N}(\mathrm{Fin}_*)$, referring to fact 1. We denote this model category as $K := sSet^{+}_{/\mathcal{N}(\mathrm{Fin}_*)}$- Consider a fibrant replacement $\alpha : \mathcal{O}^{\otimes} \times \mathcal{P}^{\otimes} \to \mathcal{Q}^{\otimes}$ of the product. Note that

1. $\mathcal{Q}^{\otimes}$ is an infinity operad.

2. α is an operadic equivalence.

3. Bifunctors from $\mathcal{O}^{\otimes}, \mathcal{P}^{\otimes}$ can be identified with $\mathrm{Map}_K(\mathcal{O}^{\otimes} \times \mathcal{P}^{\otimes}, \mathcal{Q}^{\otimes})$. We haven't given a precise definition of the monoidal structure on $\mathrm{Alg}_{\mathcal{O}^{\otimes}}(\mathcal{P}^{\otimes})$, so we can't prove this formally; you can trust that it amounts just to unwinding the definitions.

4. As well, maps of ∞ operads are the same when regarded as maps of marked simplicial sets over $\mathcal{N}(\mathrm{Fin}_*)$.

Putting this together, we get that for any infinity operad $\mathcal{R}^{\otimes}$:

$$\mathrm{Alg}_{\mathcal{Q}^{\otimes}}(\mathcal{R}^{\otimes}) \simeq \mathrm{Map}_K(\mathcal{Q}^{\otimes}, \mathcal{R}^{\otimes}) \simeq \mathrm{Map}_K(\mathcal{O}^{\otimes} \times \mathcal{P}^{\otimes}, \mathcal{R}^{\otimes}) \simeq \mathrm{Alg}_{\mathcal{O}^{\otimes}}\mathrm{Alg}_{\mathcal{P}^{\otimes}}(\mathcal{R}^{\otimes})$$

Exhibiting $\mathcal{Q}^{\otimes}$ as the desired tensor product. □

Nevertheless, it is generally really complicated to compute explicitly a tensor product, because fibrant replacements are far from being explicit.

In the special case of $\mathbb{E}_n$ algebras we have the following stunningly simple theorem:

Theorem 3.4.2 (Dunn Additivity Theorem)**.** *For every $n, k \geq 1$ we have that*

$$\mathbb{E}_n \otimes \mathbb{E}_k \simeq \mathbb{E}_{n+k}$$

The proof is rather complicated, but we give a sketch. One can build an approximation of the tensor product in a general framework, called the wreath product, which is explicitly computable. However, this is just a generalized operad, whatever this means, so it does not yield the tensor product even if it has the right algebras (generalized in a proper way). Then one builds up an explicit map from the wreath product of $\mathbb{E}_n, \mathbb{E}_k$ to $\mathbb{E}_{n+k}$ and, thanks to the aid of some discrete models of the $\mathbb{E}_n$ operads, one manages to show that it is an approximation. The other technical result one needs to show the theorem is the following: given a covering $\mathcal{U}$ of X such that the nerve of the poset $\mathcal{U}_x := \{U : x \in U\}$ is weakly contractible, then

$$\mathrm{hocolim}_{U \in \mathcal{U}} U \simeq X$$

Using the theory we developed in this chapter, we can prove the following

Corollary 3.4.2.1 (Generalized Eckmann-Hilton Principle)**.** *For any $1 \leq k \leq \infty$, we have*

$$\mathrm{Comm} \otimes \mathbb{E}_k \simeq \mathrm{Comm}$$

For $k = 1$, we get

$$\mathrm{Comm} \otimes \mathrm{Ass} \simeq \mathrm{Comm}$$

This fact has the following vague and concise reformulation, that resembles the Eckmann-Hilton principle. Suppose an object $c \in \mathcal{C}^{\otimes}$ of a symmetric monoidal category admit two compatible structures, one (*) being associative up to coherent homotopy and the other $(\cdot)$ being commutative up to coherent homotopy. Moreover, they share the same unit. In "symbols":

$$x \cdot (y \cdot z) \simeq_{coh} (x \cdot y) \cdot z, \quad x \cdot y \simeq_{coh} y \cdot x, \quad 1 * x \simeq_{coh} x * 1 \simeq_{coh} x \simeq_{coh} 1 \cdot x \simeq_{coh} x \cdot 1$$

$$x * (y * z) \simeq_{coh} (x * y) * z, \quad (x * y) \cdot (z * w) \simeq_{coh} (x \cdot z) * (y \cdot w)$$

Then they coincide up to coherent homotopy.

Proof. It is enough to show that the tensor preserve the filtered colimit of $\mathbb{E}_n$'s. Indeed, for any $\mathcal{R}^{\otimes}$ we have

$$\begin{aligned}
\mathrm{Alg}_{\mathrm{Comm}}(\mathcal{R}^{\otimes}) &\simeq \mathrm{Alg}_{colim_n \mathbb{E}_{n+k}}(\mathcal{R}^{\otimes}) \simeq \\
\mathrm{Alg}_{colim_n \mathbb{E}_n \otimes \mathbb{E}_k}(\mathcal{R}^{\otimes}) &\simeq \lim_n \mathrm{Alg}_{\mathbb{E}_n \otimes \mathbb{E}_k}(\mathcal{R}^{\otimes}) \simeq \\
\lim_n \mathrm{Alg}_{\mathbb{E}_n} \mathrm{Alg}_{\mathbb{E}_k}(\mathcal{R}^{\otimes}) &\simeq \mathrm{Alg}_{\mathrm{Comm}} \mathrm{Alg}_{\mathbb{E}_k}(\mathcal{R}^{\otimes}) \simeq \\
&\simeq \mathrm{Alg}_{\mathrm{Comm} \otimes \mathbb{E}_k}(\mathcal{R}^{\otimes})
\end{aligned}$$

And by ∞ Yoneda we conclude $\mathrm{Comm} \simeq \mathrm{Comm} \otimes \mathbb{E}_k$. □

3.5 Generalization to manifolds

In this section we want to introduce a version of the little cube operad $\mathbb{E}_M$ that depends on a k dimensional manifold M. It can be thought as a family of $\mathbb{E}_k$ parameterized by the points of manifold.

Firstly, we relax the rectilinear embedding spaces by introducing a topological monoid $\mathrm{BTop}(k)$ in place of $\mathbb{E}_k$. This takes into account all the embeddings.

Definition 3.5.1. Let $k \geq 1$. The topological operad $\mathbb{E}^t_{\mathrm{Btop}(k)}$ is defined as follows:

- Objects are $\langle n \rangle \in \mathcal{N}(\mathrm{Fin}_*)$;
- Maps are

$$\mathrm{Hom}(\langle n \rangle, \langle m \rangle) = \coprod_{\alpha : \langle n \rangle \to \langle m \rangle} \prod_{i \in \langle m \rangle^0} \mathrm{Emb}(\mathbb{R}^k \times \alpha^{-1}(i), \mathbb{R}^k)$$

Where Emb denote the embeddings. $\mathrm{Btop}(k)^{\otimes}$ is thus the coherent of the singular space associated to $\mathbb{E}^t_{\mathrm{Btop}(k)}$.

Remark 27. Note that the underlying category of $\mathrm{Btop}(k)^{\otimes}$ - as nerve and sing commutes with taking fibers - is the nerve of the singular space associated to $(\mathbb{E}^t_{\mathrm{Btop}(k)})_{\langle 1 \rangle}$. The latter is a one object topological category such that the endomorphism monoid is $\mathrm{Emb}(\mathbb{R}^k, \mathbb{R}^k)$. By the Kister-Mazur Theorem, [10] §5.4.1.5, the inclusion $\mathrm{Emb}(\mathbb{R}^k, \mathbb{R}^k) \to \mathrm{Aut}(\mathbb{R}^k)$ is a weak homotopy equivalence, thus an homotopy equivalence after applying sing, and an equivalence after applying the coherent nerve. It follows that $\mathrm{Btop}(k)$ is a Kan complex, being equivalent as ∞ categories to a Kan complex. Indeed, all the arrows will be invertible. Putting it differently, note that being a kan complex is a lifting property in Cat_∞, so it is stable by equivalence.

Remark 28. We can see $\mathrm{Btop}(k)^{\otimes}$ as a sub infinity operad of $\mathrm{Btop}(k)^{\amalg}$, consisting of multiembeddings of disks such that the images of the disks are disjoint. Indeed, at a topological level, the latter have multimaps given by $\mathrm{Mul}_{\mathrm{Btop}(k)^{\amalg}}(\mathbb{R}^k \times S, \mathbb{R}^k) = \mathrm{Emb}(\mathbb{R}^k, \mathbb{R}^k) \times S$ which contains $\mathrm{Emb}(\mathbb{R}^k \times S, \mathbb{R}^k) = \mathrm{Mul}_{\mathrm{Btop}(k)^{\otimes}}(\mathbb{R}^k \times S, \mathbb{R}^k)$.

Note that by choosing a homeo $\mathbb{R}^k \simeq \square^k$, we have an inclusion $\mathbb{E}^{\otimes}_k \to \mathrm{Btop}(k)^{\otimes}$. Via the theory of germs and using that $\mathrm{Btop}(k)^{\otimes}$ is a Kan complex, one can show the following ([10], 5.4.2.8):

Proposition 3.5.2. *The inclusion* $\mathbb{E}_k^{\otimes} \to \mathrm{Btop}(k)^{\otimes}$ *is an approximation.*

Now we bring a manifold M of dimension k into the picture.

Definition 3.5.3. Let M be a k manifold. Define $\mathcal{C}_M$ to be the topological category having two objects, denoted M and $\mathbb{R}^k$, with hom sets

$$\mathrm{Hom}(M, \mathbb{R}^k) = \emptyset, \mathrm{Hom}(M, M) = 1_M$$

$$\mathrm{Hom}(\mathbb{R}^k, \mathbb{R}^k) := \mathrm{Emb}(\mathbb{R}^k, \mathbb{R}^k), \mathrm{Hom}(\mathbb{R}^k, M) := \mathrm{Emb}(\mathbb{R}^k, M)$$

In other words, the only non trivial hom sets starts from $\mathbb{R}^k$, and in this case are embeddings. We denote by $\bar{\mathcal{C}}_m$ the coherent nerve of the singular chain of this topological category. Note that $\mathrm{Btop}(k)$ can be seen as a subcategory of $\bar{\mathcal{C}}_M$. Set $B_M := \mathrm{Btop}(k) \times_{\bar{\mathcal{C}}_M} (\bar{\mathcal{C}}_M)_{/M}$. Unwinding the definition, it is the infinity category of embeddings $f : \mathbb{R}^k \to M$, with arrows $\mathrm{Hom}(f, g) := \{h \in \mathrm{Emb}(\mathbb{R}^k, \mathbb{R}^k) : f \simeq gh$ via specified isotopy $\}$ (and similarly for higher homotopies). Finally:

Definition 3.5.4. The little M-cube operad $\mathbb{E}_M^{\otimes}$ is the ∞-operad $\mathrm{Btop}(k)^{\otimes} \times_{\mathrm{Btop}(k)^{\amalg}} B_M^{\amalg}$.

Spelling out the definition and making reference to remark 28, objects of this operad are embeddings $\mathbb{R}^k \times S \to M$, with arrows similar to the description for B_M. We justify the fact that $\mathbb{E}_M^{\otimes}$ generalizes little cube operads with the followings facts.

Remark 29. Note that $\mathbb{E}_M \to \mathrm{Btop}(k)^{\otimes}$ is an approximation. Indeed, being a problem of finding certain lifts, and being the map the pullback of $B_M^{\amalg} \to \mathrm{Btop}(k)^{\amalg}$, it is enough to show that $B_M \to \mathrm{Btop}(k)$ is surjective on arrows at the topological level. Unwinding the definitions, this corresponds to the following: given an embedding $f : \mathbb{R}^k \to \mathbb{R}^k$, there exist embeddings $g, h : \mathbb{R}^k \to M$ such that $gf = h$. This is easy: take g to be the inclusion of a chart in M and set $h = gf$.

Proposition 3.5.5. $\mathbb{E}_M^{\otimes}$ *is an assembly of a* B_M*-family of reduced operads.*

Proof. In virtue of 2.7.5, we show that $\mathbb{E}_M^{\otimes}$ is unital and that $\mathbb{E}_M$ is a Kan complex.

Note that $\mathbb{E}_M^{\otimes}$ comes from a topological operad, so that we can apply proposition 2.7.2. Denote by $\mathrm{Btop}(k)^t$ the topological category with one object and self embeddings of $\mathbb{R}^k$ as morphisms. Recall that $\mathrm{Btop}(k)$ comes from $\mathrm{Btop}(k)^t$. Then $\mathbb{E}_M^{\otimes}$ comes from

$$\mathbb{E}^t_{\mathrm{Btop}(k)} \times_{(\mathrm{Btop}(k)^t)^{\amalg}} (\mathrm{Btop}(k)^t \times_{C_M} C_{M/M})^{\amalg}$$

Objects are embeddings $f : \mathbb{R}^k \to M$, and nullary operations to an f are empty functions h to $\mathbb{R}^k$ such that fh is the empty function. As we have just one empty function, it is evidently contractible. This yields unitality. Moreover

$$\mathbb{E}_M \simeq \mathrm{Btop}(k) \times_{\mathrm{Btop}(k)} B_M \simeq B_M$$

which is a Kan complex because of proposition 3.5.7. So there exist a reduced family of operads $\mathcal{O}^{\otimes} \to \mathcal{N}(\mathrm{Fin}_*) \times C$ and an assembly $\mathcal{O}^{\otimes} \to \mathbb{E}_M^{\otimes}$ that induces an homotopy equivalence $C \to B_M$. If one feels uncomfortable with the fact that C is not equal to B_M, we can obtain an homotopy equivalent family by homotopy pulling back along $B_M \to C$, and the new operad will be equivalent to the old. □

Proposition 3.5.6. *Fibers of the family corresponding to $\mathbb{E}_M$ are equivalent to $\mathbb{E}_k$.*

Proof. For $x \in B_M$, take the morphism $\mathbb{E}_k \to \mathbb{E}_M$ induced by $\mathbb{E}_k \to \mathrm{Btop}(k)^{\otimes}$ and the constant x map $\mathbb{E}_k \to B_M^{\amalg}$. Note that $\mathbb{E}_k^{\otimes} \to \mathbb{E}_M^{\otimes} \to \mathrm{Btop}(k)^{\otimes}$ is such that both $\mathbb{E}_k^{\otimes} \to \mathrm{Btop}(k)^{\otimes}$, $\mathbb{E}_M^{\otimes} \to \mathrm{Btop}(k)^{\otimes}$ are approximations. It follows ([10] 2.3.3.18) that $\mathbb{E}_k \to \mathbb{E}_M$ is an approximation. Moreover, since $\mathbb{E}_k$ is reduced, theorem 2.7.5 ensures that it factors through the reduced family $\mathcal{O}^{\otimes} \to \mathbb{E}_M$ of 3.5.5. Since $\mathbb{E}_k \to \mathcal{O}^{\otimes} \to \mathbb{E}_M$ is constant in B_M, $\mathbb{E}_k \to \mathcal{O}^{\otimes}$ factors through $\mathcal{O}_x^{\otimes}$. Being the approximation a problem of liftings, if an approximation factors then the resulting map is still an approximation. It follows that $\mathbb{E}_k \to \mathcal{O}_x^{\otimes}$ is an approximation between reduced operads, and thus ([10], §2.3.3.24) it is an equivalence of ∞ operads. □

Remark 30. The same reasoning of the above proposition applies to $\mathbb{E}_k^{\otimes} \to \mathrm{Btop}(k)^{\otimes}$, where B_M is substituted by $\mathrm{Btop}(k)$. This yields that $\mathrm{Btop}(k)^{\otimes}$ is the assembly of a $\mathrm{Btop}(k)$ reduced family of operads, all equivalent to $\mathbb{E}_k$.

Proposition 3.5.7. *B_M is equivalent to* $\mathrm{Sing}(M)$.

A proof can be found in [10], §5.4.5.2.

Corollary 3.5.7.1. *$\mathbb{E}_{\mathbb{R}^k}$ is operadically equivalent to $\mathbb{E}_k$.*

In the next chapter, we will prove that the theory of $\mathbb{E}_M$ algebras can be substituted by a more rigid version, namely prefactorization algebras: instead of having embeddings $\mathbb{R}^k \to M$ as objects, we will have opens in M that are homeomorphic to $\mathbb{R}^n$, and map sets will be really easier than here ("multinclusions" of opens!).

Chapter 4

Factorization Homology

Factorization algebras are a modern approach to generalize the local-to-global behavior of (co)sheaves in the ∞ setting, where one also allows sections on small opens to combine via a general $\otimes$ to give sections on big opens. In the following, we prove that the theory of (locally constant) prefactorization algebras defined over opens $\simeq \mathbb{R}^n$ is equivalent to the theory of $\mathbb{E}_M$ algebras.

Henceforth, we are led to produce an invariant similar to (co)homology, namely the (derived) global sections. One can think about this as a form of generalized (co)sheaf cohomology; here we look for multiplicative rather than additive invariants. This is achieved by means of factorization homology, that is defined as the (global sections of) left operadic Kan extension of the prefactorization algebra, along the inclusion of opens homeomorphic to $\mathbb{R}^n$ to all the opens. In order to give an example of significance of this new construction, we show that it recovers the Hochschild homology of an associative algebra in case of $M = S^1$.

At last, we note that the treatment of strict (pre)factorization algebras can be useful in its own. See for example [7] for an excellent treatment on the interplay of factorization algebras and Quantum Field Theory in its modern development.

4.1 Strict prefactorization algebras

Given a manifold M and a cover $\mathcal{U}$, we define $\mathcal{U}^{\otimes}$ the strict operad having

- as objects opens $U \in \mathcal{U}$;
- as multimaps, $\mathrm{Mul}_{\mathcal{U}^{\otimes}}(U_1, \ldots, U_n, V)$ is the singleton if U_i are disjoint and contained in V, and empty otherwise.

A strict $\mathcal{U}$-prefactorization algebra P is thus a functor $P : \mathcal{U}^{\otimes} \to C$, where C is a strict symmetric monoidal category regarded as an operad (see example 2). If not specified, a prefactorization algebra is taken with respect to $\mathcal{U}(M) := \{U \subset M \text{ open }\}$. The other main considered $\mathcal{U}$ is $\mathrm{Disk}_M := \{U \simeq \mathbb{R}^n : U \in \mathcal{U}(M)\}$. In this case we simply say Disk_M algebra.

Remark 31. Note that $P(\emptyset)$ is a commutative algebra in C whenever P is a prefactorization algebra. Indeed, we have an inclusion $\mathrm{Comm} \to$ sending the only object to the empty set.

We now make a few examples to get some confidence.

Example 7. *(associative algebras). Take an associative algebra $A : Ass \to C$. Define a functor $Pf : \mathrm{Disk}_{\mathbb{R}} \to Ass$ sending all the objects to the only object in Ass, and the multimap associated to $(a_1, b_1) \coprod \ldots \coprod (a_n, b_n) \subset V$ to the linear ordering induced by the ordering of the n intervals (a_i, b_i). Then A induces a $\mathrm{Disk}_{\mathbb{R}}$ algebra on $\mathbb{R}$, namely $P_A := A \circ Pf$, such that $P_A(a, b) = A$, and $(a, b) \coprod (c, d) \subset (a, d)$ induces the product operation $A \otimes A \to A$. Note that, in this case, all unary operations are invertible. We will see that this completely characterizes disk algebras over $\mathbb{R}$ that come from associative algebras.*

Example 8. *(symmetric algebra of a cosheaf) Let k be a field. We have a pair of adjoint functors*

$$Forg : Alg_{Comm}(Vec_k^{\otimes}) \leftrightarrows Vec_k : Sym$$

That sends respectively a commutative algebra over k to its underlying vector space, and a vector space to the free symmetric commutative algebra over it. Furthermore, Sym sends direct sums to tensor product. Hence, post-composition with Sym induces a functor

$$cosh(M, Vec_k) \to Alg_{\mathrm{Disk}_M} Alg_{Comm}(Vec_k^{\otimes})$$

By sending a functor $\mathcal{U}(M)^{\otimes} \to Vec_k^{\oplus}$ *to*

$$\mathrm{Disk}_M^{\otimes} \to \mathcal{U}(M)^{\otimes} \to Vec_k^{\oplus} \overset{Sym}{\to} Alg_{Comm}(Vec_k^{\otimes})$$

Explicitly, we have that $\mathrm{Sym}(F)(U_1, \ldots, U_n) := \mathrm{Sym}(F(U_1 \coprod \ldots \coprod U_n)) \simeq \mathrm{Sym}F(U_1) \otimes \ldots \otimes \mathrm{Sym}F(U_n)$*, and maps are determined by the co-restrictions.*

We now pass to the infinity setting, where we develop the main tools.As prefactorization algebras are only defined by means of a certain category $\mathcal{U}^{\otimes}$, the easiest way to homotopically relax the concept is to take the nerve $\mathcal{N}(\mathcal{U}^{\otimes})$.

4.2 $\mathbb{E}_M$ algebras and Disk_M algebras

Definition 4.2.1. An $\mathcal{U}$ prefactorization algebra on a symmetric monoidal category $\mathcal{C}^{\otimes}$ is a functor $\mathcal{N}(\mathcal{U}^{\otimes}) \to \mathcal{C}^{\otimes}$.

We focus by the moment on Disk_M algebras. For every disk $U \in \mathrm{Disk}_M$, fix a parameterization $\psi_U : \mathbb{R}^n \simeq U$. Also, for $U \subset V$ define $\psi_{UV} = (\psi_V|^V)^{-1}\psi_U|^V : \mathbb{R}^n \to \mathbb{R}^n$. Note that there exist a functor $\mathcal{N}(\mathrm{Disk}_M) \to \mathbb{E}_M$ given at topological level by $U \mapsto \psi_U$, $U \subset V \mapsto \psi_{UV}$. We are ready to show the main result of this section:

Theorem 4.2.2. *The map* $\mathcal{N}(\mathrm{Disk}_M^{\otimes}) \to \mathbb{E}_M^{\otimes}$ *induces, for every symmetric monoidal category* $\mathcal{C}^{\otimes}$*, an equivalence*

$$Alg_{\mathbb{E}_M}(\mathcal{C}^{\otimes}) \overset{\simeq}{\to} Alg^{loc}_{\mathcal{N}(\mathrm{Disk}_M)}(\mathcal{C}^{\otimes})$$

Proof. We want to use theorem 2.5.4. We must show that the above map:

1. induces a weak homotopy equivalence of underlying Kan complexes $\mathcal{N}(\mathrm{Disk}_M) \to B_M$;

2. is a weak approximation.

Let's begin from (1). The idea is the following: we can "thicken" Disk_M by adding a B_U over every $U \in \mathrm{Disk}_M$ getting a χ with the same weak homotopy. This thickened version will be equivalent to B_M by the generalized Seifert-Van Kampen theorem.

Formally, consider the relative nerve $\psi : \chi \to \mathcal{N}(\mathrm{Disk}_M)$ of the diagram $\mathcal{N}(\mathrm{Disk}_M) \to \mathcal{S}$ (the ∞ category of spaces) given by $U \mapsto B_U$. Note that the fibers of the relative nerve are weakly contractible because homotopy equivalent to $\mathrm{Sing}(U)$, $U \in \mathrm{Disk}_M$. We have a map $\chi \to B_M$ given by gluing the various inclusions $B_U \to B_M$. In virtue of [11] §3.3.4.5 and §3.3.4.6, this is a weak equivalence iff

$$\mathrm{colim}_{U \in \mathrm{Disk}_M} B_U \simeq B_M$$

Thanks to proposition 3.5.7, we can alternatively show that

$$\mathrm{colim}_{U \in \mathrm{Disk}_M} \mathrm{Sing}(U) \simeq \mathrm{Sing}(M)$$

Since sSets is a simplicial model category, colimits as infinity category are the same as hocolimits as model category. Now the sSet $\chi_x = \mathcal{N}(\{U \in \mathrm{Disk}_M : x \in U\})$ is cofiltered, because any finite number of disks containing x contains a disk containing x. A cofiltered category is weakly contractible, because any finite diagram factors through a category with an initial object, which is weakly contractible. We can therefore apply the generalized Seifert-Van-Kampen Theorem ([10] A.3.1) and get the desired result.

We pass to show (2). Since $\mathbb{E}_M \to \mathrm{Btop}(k)^{\otimes}$ is an approximation, we can show that $\mathcal{N}(\mathrm{Disk}_M) \to \mathrm{Btop}(k)^{\otimes}$ is a weak approximation. To this end, we use criterion

[10], §2.3.3.16: we are reduced to show that for any $U \in (\mathrm{Disk}_M)_{\langle n\rangle}$ and every active $\langle m\rangle \to \langle n\rangle$, the induced map

$$\mathcal{N}(\mathrm{Disk}_M)_{/U} \times_{\mathcal{N}(\mathrm{Fin}_*)_{/\langle n\rangle}} \{\langle m\rangle\} \to (\mathrm{Btop}(k)^{\otimes})_{/U} \times_{\mathcal{N}(\mathrm{Fin}_*)_{/\langle n\rangle}} \{\langle m\rangle\}$$

is a weak homotopy equivalence [1]. As everything splits in $\langle n\rangle$, we can suppose $n = 1$. At this point, note that the domain can be identified with the set A of m-uples $(U_1, \ldots, U_m)$ such that they are disjoint and contained in U. Everything is ordinary, so that the structure induced by $\mathcal{N}(\mathrm{Disk}_M)$ on this tuples is still ordinary, and we get $\mathcal{N}(A)$.

Note that we have a map $\pi : \mathrm{Btop}(k)_{/U} \times_{\mathcal{N}(\mathrm{Fin}_*)_{/\langle 1\rangle}} \{\langle m\rangle\} \to \mathrm{Btop}(k)^m$. We will equivalently show that, after passing to the homotopy fibers over the unique vertex of $\mathrm{Btop}(k)^m$, we get an equivalence.

The homotopy fiber on the right amounts to picking an embedding $\mathbb{R}^k \times \langle m\rangle^0 \to U$, so the right hand side becomes $\mathrm{SingEmb}(\mathbb{R}^k \times \langle m\rangle^0, U)$.

The homotopy fiber on the left consist of tuples $((V_1, \ldots, V_m), (f_1, \ldots, f_m))$, where $f_i : \mathbb{R}^k \to \mathbb{R}^k$ are embeddings. We can think of it as embeddings $f_i : \mathbb{R}^k \to V_i$. We have an evident fibration of this homotopy fiber over $N(A)$, sending $((V_i), (f_i))$ to (V_i). The fiber of this fibration at $(V_1, \ldots V_m)$ are $\prod_i \mathrm{SingEmb}(\mathbb{R}^k, V_i)$. Since these are Kan complexes, we can reason as in 4.4.4 and obtain that the homotopy fiber is weakly equivalent to

$$\mathrm{hocolim}_{(V_1,\ldots,V_m)\in\mathcal{N}(A)} \prod_i \mathrm{SingEmb}(\mathbb{R}^k, V_i)$$

To conclude, the idea is to use the Generalized Seifert Van Kampen Theorem.

Note that $\prod_i \mathrm{SingEmb}(\mathbb{R}^k, V_i)$ is not an open subspace of $\mathrm{SingEmb}(\mathbb{R}^k \times \langle m\rangle^0, U)$ with respect to the compact open topology. That's why, with some topological reasoning, one

[1] On the right we abused with notation: U is meant to be only vertex of $\mathrm{Btop}(k)^{\otimes}_{\langle 1\rangle}$

wants (and can) reduce to use configuration spaces in place of embeddings. So we are left with showing that

$$\mathrm{hocolim}_{(V_1,\ldots,V_m)\in\mathcal{N}(A)} \prod_i \mathrm{SingConf}(\{i\}, V_i) \to \mathrm{SingConf}(\langle m\rangle^0, U)$$

is a weak equivalence. Note that when when (V_i) varies, members of the left hand side describe a cover of the right hand side. Moreover, given a configuration x_i of m points in U, the nerve of the poset $T_x = \{(V_1, \ldots, V_m) : x_i \in V_i\}$ is cofiltered, and thus weakly contractible.

□

Remark 32. We can justify example 7 in the ∞ setting, now that we have this tool. Indeed, coherently homotopically associative algebras ($\mathbb{E}_1$ algebras) coincide with locally constant $\mathrm{Disk}_{\mathbb{R}}$ algebras.

4.3 Factorization Algebras

In the same fashion of sheaves, we want to determine the prefactorization algebras that have a local character. More precisely, we would like to reconstruct sections over a big open via sections on a little open. Given a sheaf F, the descent property tell us that

$$F(U) \to \prod_{i,j} F(U_i \cap U_j) \rightrightarrows \prod_k F(U_k)$$

is an equalization (limit) diagram for every cover $\{U_k\}$ of U. In the same context, for a cosheaf G we have that

$$\coprod_{i,j} G(U_i \cap U_j) \rightrightarrows \coprod_k G(U_k) \to G(U)$$

is a coequalization (colimit) diagram. Hence, for prefactorization algebras P, thought as a generalization of cosheaves for general monoidal product, we should require something

like

$$\bigotimes_{i,j} P(U_i \cap U_j) \rightrightarrows \bigotimes_{k} P(U_k) \to P(U)$$

is a colimit diagram. Unluckily, there are some issues in this definition. The subtle issue is about the homotopy relaxation of this concept. We make here an analogy. Suppose we are given a covering U_i of a space X. Then the colimit of U_i along $U_i \cap U_j$ is homeomorphic to X. Nevertheless, if we change the U_i by homotopy equivalence the result is not homotopically equivalent to the original one. In order to have this property, we should consider all the Cech nerve of the poset encoding the inclusions in the covering, see for instance [12]. The other issue is not evident from this formulation, but we must restrict the type of coverings to some special class, i.e. factorizing covers. This is due to the fact that the problem admits a natural formulation on the Ran space, and factorizing covers are precisely the ones that yields a cover of the Ran space. Let us give some preliminary definitions. In the following, we restrict to C being chain complexes.

Definition 4.3.1. A factorizing cover $\mathcal{U}$ of X is a cover such that for every $x_i \in X$ there exist $U \in \mathcal{U}$ such that $\{x_1, \ldots, x_k\} \subset U$.

Definition 4.3.2. Let $\mathcal{U}$ be a cover of $U \subset X$ and F a prefactorization algebra. The associated Cech complex $\mathfrak{C}(\mathcal{U}, F)$ is defined as follows. Consider $\mathcal{PU} = \{(U_1, \ldots, U_n) : U_i \in \mathcal{U}, \text{disjoint}\}$. For every n, we have a complex $\bar{\mathfrak{C}}(\mathcal{U}, F)_n$ given by

$$\bigoplus_{(\alpha_0,\ldots,\alpha_n)\in\mathcal{PU}^{n+1}} \bigotimes_{\substack{U_{i_0}\in\alpha_0,\\ \ldots,\\ U_{i_n}\in\alpha_n}} F(U_{i_0} \cap \ldots \cap U_{i_n})$$

Moreover, we have simplicial maps $d_k^n : \bar{\mathfrak{C}}(\mathcal{U}, F)_n \to \bar{\mathfrak{C}}(\mathcal{U}, F)_{n-1}$ induced by $F(U_{i_0} \cap \ldots U_{i_n}) \to F(U_{i_0} \cap \ldots \cap \hat{U_{i_k}} \cap \ldots \cap U_{i_n})$. Indeed, the sets $\{U_{i_0} \cap \ldots \cap U_\ell \cap \ldots \cap U_{i_n}\}_{U_\ell \in \alpha}$ are disjoint for fixed i_j's, so their multinclusion inside $U_{i_0} \cap \ldots \hat{U_{i_k}} \ldots \cap U_{i_n}$ induces a tensor map. This simplicial maps induce a boundary map $\partial^n := \sum_k (-1)^k d_k^n$, and we can set $\mathfrak{C}(\mathcal{U}, F)$ as the totalization of the bicomplex $(\bar{\mathfrak{C}}(\mathcal{U}, F)_n, \partial^n)$.

Notice that we have a map $\bar{\mathfrak{C}}(\mathcal{U}, F)_n \to F(U)$ for every cover $\mathcal{U}$ of U; indeed, the multinclusion $\{U_{i_0} \cap \ldots U_{i_n}\} \to U$ induces a map

$$\bigotimes_{i_0 \in \alpha_0, \ldots, i_n \in \alpha_n} F(U_{i_0} \cap \ldots \cap U_{i_n}) \to F(U)$$

that commutes with simplicial maps d_k^n. Thus, we have an induced morphism $\mathfrak{C}(\mathcal{U}, F) \to F(U)$. We are ready to state what a factorization algebra is.

Definition 4.3.3. A factorization algebra F over a manifold M with values in chain complexes is a prefactorization algebra $F : \mathcal{N}(\mathcal{U}(M)^{\otimes}) \to \mathrm{Chain}(k)^{\otimes}$ such that for every factorizing cover $\mathcal{U}$ of $U \subset X$, the map $\mathfrak{C}(\mathcal{U}, F) \to F(U)$ is a quasi iso.

This is [4], definition 4.1.13. Note that this definition adapts to give also $\mathcal{U}$-factorization algebras, where $\mathcal{U}$ is a cover closed for intersections.

For a general symmetric monoidal category with colimits, one can obtain a simplicial object $\bar{\mathfrak{C}}(\mathcal{U}, F)$ and then compute the homotopy colimit over Δ: in chain complexes, this coincides with taking the totalization of the associated bicomplex.

4.4 Factorization homology

As the title of the thesis promised, we can finally define factorization homology.

Definition 4.4.1. Let M be a manifold, $\mathcal{C}^{\otimes}$ a symmetric monoidal category, $A \in \mathrm{Alg}_{\mathbb{E}_M}(C)$. By the theorem 4.2.2, it induces a locally constant disk algebra $A' : \mathcal{N}(\mathrm{Disk}_M^{\otimes}) \to \mathcal{C}^{\otimes}$. Factorization homology

$$\int_M A \in \mathcal{C}^{\otimes}$$

of M with coefficients in A is defined, whenever exist, as the global sections of the left operadic Kan extension of A along $\mathcal{N}(\mathrm{Disk}_M^{\otimes}) \to \mathcal{N}(\mathcal{U}(M)^{\otimes})$. Sometimes one speaks of factorization homology for the whole functor, too.

Remark 33. Suppose that we have an $\mathbb{E}_n^{\otimes}$ algebra and a manifold of dimension n. In case M is framed, one can show that $\mathbb{E}_M^{\otimes} \to \mathrm{Btop}(n)^{\otimes}$ factors through $\mathbb{E}_n$. So we have an induced map $\mathrm{Alg}_{\mathbb{E}_n}(\mathcal{C}^{\otimes}) \to \mathrm{Alg}_{\mathbb{E}_M}(\mathcal{C}^{\otimes})$. If an $\mathbb{E}_M$ algebra has to be thought as a family of $\mathbb{E}_n$ algebras parameterized by disks in M, this functor yields a constant family. It follows that we can talk about the factorization homology of an $\mathbb{E}_n$ algebra in case of framed manifolds.

The main goal of this section is to show the following:

Theorem 4.4.2. *Suppose $\mathcal{C}^{\otimes}$ is sifted complete. Then for every $A \in Alg_{\mathbb{E}_M}(\mathcal{C}^{\otimes})$ factorization homology $\int_M A$ exists.*

Recall what we have seen in the section about operadic Kan extensions. The indexing categories for computing the operadic Kan extensions are of the form $(\mathrm{Disk}_M^{\otimes})_{/U}$, where U is an open in M. The latter is isomorphic to

$$\mathrm{Disj}_U := \{V \subset U : V \simeq \mathbb{R}^n \times S,\ S \text{ finite } \}$$

via the function $(V_1, \dots, V_n) \mapsto V_1 \coprod \dots \coprod V_n$. In order to prove existence, we must exhibit a cofinal map $\mathrm{Disj}_U \to$ something sifted that factors $\mathrm{Disj}_U \to \mathcal{C}^{\otimes}$. Actually, the

fact that opens can't "move" would make the quest of factorizing directly $\mathrm{Disj}_U \to \mathrm{Disk}_M$ impossible. But we wisely defined factorization algebras only for disk algebras that comes from $\mathbb{E}_M$ algebras. This gives us more elasticity.

Let us give the following definition:

Definition 4.4.3. Let Man_k be the topological category of k manifolds, equipped with $\mathrm{Hom}_{\mathrm{Man}_k}(M, N) := \mathrm{Emb}(M, N)$. Define $\mathbb{D}(M)$ as the full subcategory of $\mathrm{Man}_{k/M}$ spanned by $j : N \to M$ with $N \simeq \mathbb{R}^k \times S$, S finite.

Remark 34. In other words, an object of $\mathbb{D}(M)$ is an embedding of n disks in M. The main difference with Disj_M is that $\psi \to \phi$ does not imply $\psi(\mathbb{R}^k \times S) \subset \phi(\mathbb{R}^k \times S')$, because the involved commutation must only hold up to specified isotopy.

Remark 35. Note that we have a map $\mathrm{Disj}_U \to \mathbb{D}(U)$ given by choosing parameterizations of the involved disjoint disks. By choosing parameterizations which are consistent with the ones of theorem 4.2.2, we have a commutative diagram

$$\begin{tikzcd} \mathcal{N}(\mathrm{Disk}^{\otimes}) \arrow[r] & \mathbb{E}_M^{\otimes} \\ \mathcal{N}(\mathrm{Disj}_U) \arrow[r] \arrow[u] & \mathbb{D}(U) \arrow[u] \end{tikzcd}$$

We see the first step for the existence.

Proposition 4.4.4. *The map $\mathcal{N}(\mathrm{Disj}_U) \to \mathbb{D}(U)$ is left cofinal.*

Proof. We use Joyal criterion (1.7.2). Let $\psi : \mathbb{R}^k \times S \to U$ be an element of $\mathbb{D}(U)$. We must show that $C := \mathcal{N}(\mathrm{Disj}_M) \times_{\mathbb{D}(U)} \mathbb{D}(U)_{\psi/}$ is weakly contractible. When we take a fiber product of a left fibration with the nerve of an ordinary category, there is something awesome happening. Note that $C \to \mathcal{N}(\mathrm{Disj}_M)$ is still a left fibration, and by straightening-unstraightening this corresponds to the functor $\chi : \mathcal{N}(\mathrm{Disj}_M) \to \mathrm{Cat}_\infty$ given by $V \mapsto \{V\} \times_{\mathbb{D}(U)} \mathbb{D}(U)_{\psi/}$. In other words, it corresponds to homotopy coherent

diagrams

$$\begin{tikzcd} \mathbb{R}^k \times S \arrow[r, "\psi"] \arrow[d] & U \\ V \arrow[ur, "\psi_{VU}"] & \end{tikzcd}$$

And this is the homotopy fiber H_V of post-composition with ψ_{UV} at the element ψ:

$$\begin{tikzcd} \mathrm{SingEmb}(\mathbb{R}^k \times S, V) \arrow[r, "\psi_{UV*}"] & \mathrm{SingEmb}(\mathbb{R}^k \times S, U) \\ H_V \arrow[u] \arrow[r] & \{\psi\} \arrow[u] \end{tikzcd}$$

Now we can come back with Grothendieck-Lurie unstraightening, and the object C is weakly equivalent to the colimit of χ, because χ factors through $\mathcal{S}$, the ∞ category of spaces (=Kan complexes) (for a more precise statement, see [11], §3.3.4.5). Note that $\mathcal{S}$ comes from a simplicial model category, so that ∞ colimits correspond to hocolimits. Moreover, colimits in $\mathcal{S}$ are stable by base change, so that the colimit of an homotopy fiber correspond to the homotopy fiber of the colimit. To show that an homotopy fiber is weakly contractible, is enough to show that the map is a weak equivalence. So are reduced to show that

$$\mathrm{hocolim}_{V \in \mathrm{Disj}_M} \mathrm{SingEmb}(\mathbb{R}^k \times S, V) \to \mathrm{SingEmb}(\mathbb{R}^k \times S, U)$$

is a weak homotopy equivalence. We would like to apply the Generalized Seifert Van Kampen Theorem, but unluckily $\mathrm{Emb}(\mathbb{R}^k \times S, V)$ is not open in $\mathrm{Emb}(\mathbb{R}^k \times S, U)$ wrt the compact open topology. With some topological reasoning, one can reduce to show that:

$$\mathrm{hocolim}_{V \in \mathrm{Disj}_M} \mathrm{SingConf}(S, V) \to \mathrm{SingConf}(S, U)$$

is a weak homotopy equivalence. Note that $\{\mathrm{Conf}(S, V)\}_{V \in \mathrm{Disj}_M}$ is a cover of $\mathrm{Conf}(S, U)$, because every configuration is contained in a disjoint union of disks. In virtue of Gener-

alized Seifert Van Kampen Theorem, we are left with showing that for any configuration x_i the nerve of the poset $T_x = \{V_i \in \mathrm{Disj}_M : x_i \in V_i\}$ has weakly contractible nerve. This is true because T_x is cofiltered: if x_i is contained in $V_1^i, \ldots, V_n^i$, it is also contained in the intersection. □

In order to apply the existence theorem for operadic Kan extensions, we need the following other lemma, that we do not show:

Proposition 4.4.5. *The ∞ category $\mathbb{D}(M)$ is sifted.*

Remark 36. As an open U in manifold is itself a manifold, we get that $\mathbb{D}(U)$ is sifted for every U.

On balance, we have shown that for any $\mathbb{E}_M$ algebra A - if we call p^*A the associated disk algebra - the indexing category for the operadic Kan extension of p^*A along $\mathcal{N}(\mathrm{Disk}_M^{\otimes}) \to \mathcal{N}(\mathcal{U}(M)^{\otimes})$ factors through a sifted category: this yields the existence of factorization homology.

In case $M^{(n)}$ is framed, note that by remark 33 factorization homology is defined for for $\mathbb{E}_n$ algebras. There is a beautiful result that characterize all the functors $\mathcal{N}(\mathcal{U}(M)^{\otimes}) \to \mathcal{C}^{\otimes}$ that arises as factorization homology of an $\mathbb{E}_n$ algebra: namely, they are exactly all the locally constant factorization algebras on M, see [5], §5.4. In the rest of the paper you can find a beautiful account of formal properties enjoyed by factorization homology, especially in CDGAs (when one use $\mathrm{Alg}_{\mathbb{E}_\infty}(\mathrm{Chain}(k)^{\otimes})$ as coefficients).

4.5 Hochschild homology

There is a classical invariant that one can recover with factorization homology. It is an invariant defined for $\mathbb{E}_1$ algebras A over vector spaces. In this context we obtain a general version for $\mathbb{E}_1$ algebras over arbitrary symmetric monoidal categories, recovering the original definition in case of complexes (see e.g. [13]).

The first observation that we do is the following:

Lemma 18. *Let C be an ∞ category. Then C is sifted iff $C_{/c} \to C$ is left cofinal $\forall c$.*

Proof. By Joyal criterion, the latter condition is equivalent to

$$C_{/c} \times_C C_{/d}$$

weakly contractible for every d. Note that this sSet is made of arrows $x \to c, x \to d$ (because the vertex must coincide. This is the same as

$$C \times_{C \times C} (C \times C)_{/(c,d)}$$

i.e. maps $x \to c, y \to d$ such that $x = y$. Again, by Joyal criterion this is equivalent to left cofinality of $C \to C \times C$, which is the condition of being sifted. $\square$

We want to recover Hochschild homology of an $\mathbb{E}_1$ algebra as its factorization homology over S^1. Note indeed that S^1 is framed, so that factorization homology is defined for $\mathbb{E}_1$ algebras. Recall that factorization homology is computed as the weak operadic colimit of a diagram $\mathbb{D}(S^1) \to \mathcal{C}^{\otimes}$. To reduce his complexity, by the previous lemma we have that $\mathbb{D}(S^1)_{/\psi} \to \mathbb{D}(S^1)$ is left cofinal. Before computing the colimit, let us give a definition.

Definition 4.5.1. For $\in \mathbb{N}$, let σ_n be the permutation $(0 \dots n) \in S_{n+1}$. Note that S_{n+1} act on linear orders on $[n]$ by

$$x \ \rho(\leq) \ y \quad \Leftrightarrow \quad \rho^{-1}(x) \leq \rho^{-1}(y)$$

We define Δ_*^{cyc} as the category which has objects $[n]$, $n \in \mathbb{N}$, such that maps are

$$\mathrm{Hom}([n],[m]) = \{(f,k) : \ f : [n]_{\sigma_n^{k+1} \leq} \to [m] \text{non decreasing}, f(0) = 0\}$$

We call maps in this homset cyclic maps.

Remark 37. We observe that if $(f,k) \in \mathrm{Hom}([n],[m])$, then k is the max of $\sigma_n^{k+1}(\leq)$.

Also, the condition of being ordered from the permuted order, given that $f(0) = 0$, can be reformulated in this way:

- If $0 \leq x < x' \leq k$, then $f(x) \leq f(x')$;
- If $x > k$, then $f(x) = 0$.

Indeed, given the original formulation, the first condition is satisfied and the second comes from $x\sigma_n^{k+1}(\leq)0$ implies $f(x) \leq f(0) \leq 0 \Rightarrow f(x) = 0$. On the other hand, given these two conditions, we should verify:

- If $0 \leq x < x' \leq k$, then $f(x) \leq f(x')$ (true);
- If $0 \leq x \leq k < x'$, then $f(x') \leq f(x)$; true because $f(x') = 0$;
- If $k < x < x'$, then $f(x) \leq f(x')$; true because both are zero.

In particular, if f is not constantly zero, $k = \max_{\leq} f^{-1}(\max \mathrm{Im}(f))$.

Remark 38. Maps in the category Δ_*^{cyc} admits also another description. Namely:

$$\mathrm{Hom}([n],[m]) = \{(f, \leq_i), f : [n] \to [m], f(0) = 0, \leq_i \text{ linear order on } f^{-1}(i)$$

$$|\text{composition in Ass of } \leq_i \text{ along } \leq_{[m]} \text{ yields a cyclic permutation of } \leq_{[n]}\}$$

Indeed, given a bunch of linear orders like that, we have that they are the restriction of some $\sigma_n^k \leq_{[n]}$; k is determined by the formula in the above remark or by the linear order on the fiber of zero if the image is constantly zero; finally, a map is always non decreasing from the composed order to the order.

Definition 4.5.2. We call the cyclic cofaces $\delta_i^n : [n] \to [n-1]$ in Δ_*^{cyc} the maps given by:

- If $i < n$, then $\delta_i^n(j) = j$, if $i \geq j$, $\delta_i^n(j) = j - 1$ if $j > i$;
- $\delta_n^n(j) = j$ for $j \leq n - 1$, $\delta_n^n(n) = 0$.

In practice, δ_i^n contracts the $i, i+1$-th vertices (indices modulo n).

Let's reduce the complexity of $\mathbb{D}(S^1)_{\psi/}$.

Lemma 19. *There exist an equivalence $\mathbb{D}(S^1)_{\psi/} \to \mathcal{N}(\Delta_*^{cyc})$. Furhtermore, after choosing*

- *disjoint embeddings $\phi_n : \mathbb{R} \to S^1 \setminus \psi(\bar{\mathbb{R}}) \simeq \mathbb{R}$ such that $\psi(0), \phi_n(0), \phi_{n+1}(0)$ are in anticlockwise order;*
- *a map $m : \mathbb{R} \times [1] \to \mathbb{R}$ that induces the order $0 \leq 1$;*
- *a map $\check{m} : \mathbb{R} \times [1] \to \mathbb{R}$ that induces the order $1 \leq 0$;*

Then the equivalence sends:

- *$[n]$ to $(\psi, \phi_1, \ldots, \phi_n)$;*
- *For $i < n$, δ_i^n to the map $g : \mathbb{R} \times [n] \to \mathbb{R} \times [n-1]$ such that $\pi_0(g) = \delta_i^n$, with all identities on $j \neq i, i+1$ components and $m : \mathbb{R} \times \{i, i+1\} \to \mathbb{R}$;*
- *For $i = n$, δ_i^n to the map $g : \mathbb{R} \times [n] \to \mathbb{R} \times [n-1]$ such that $\pi_0(g) = \delta_i^n$, with all identities on $j \neq 0, n$ components and $\check{m} : \mathbb{R} \times \{0, n\} \to \mathbb{R}$.*

Proof. We have a map from $\mathbb{D}(S^1)_{\psi/} \to \mathbb{D}(S^1) \to \Delta_*^{cyc}$ given by:

- $(f_i) : \mathbb{R}^k \times [n] \to S^1$ maps to $[n]$, where components are numbered so that $f_0 = \psi$, and $i \leq j$ if $\psi(0), f_i(0), f_j(0)$ are in anticlockwise order;
- Let $g : \mathbb{R}^k \times [n] \to \mathbb{R}^k \times [m]$ be a map $(f_i) \to (f_i')$. Then $g_i : \mathbb{R}^k \times \pi_0(g)^{-1}(i) \to \mathbb{R}$ induces a linear order on fibers of $\pi_0(g)$, and because $(f_i')g$ is isotopic to (f_i), the order organizes in a cyclic permutation of the order on $[n]$. Indeed, two isotopy maps can only induce cyclic permutations of the same ordering, no matter where the order starts. Moreover, $\pi_0(g)$ sends 0 to 0 because it preserve the ψ component.

While proving $\mathbb{E}_1 \simeq \mathrm{Ass}^{\otimes}$, we showed that passing from a multimbedding $\mathbb{R} \times S \to \mathbb{R}$ to the linear order it induces an equivalence, so that the map $\mathbb{D}(S^1)_{\psi/} \to \mathbb{D}(S^1) \to \Delta_*^{cyc}$

is faithful. It is also full: given $\alpha : [n] \to [m]$ and its orderings on fibers, and $(f_i) : \mathbb{R}^k \times [n] \to S^1, (f'_i) : \mathbb{R}^k \times [m] \to S^1$, one can build up a g such that:

- $\pi_0(g) = \alpha$;
- $g|\mathbb{R} \times \alpha^-1(i) \to \mathbb{R}$ by choosing an embedding corresponding to the desired ordering via $\mathbb{E}_1 \simeq \mathrm{Ass}$;

The maps $(f'_i)g$ and (f_i) will be isotopic becase they induce cyclic permutations of the same ordering on $[n]$.

The fact that the equivalence admits a description as in the statement comes from the fact that it defines a section of the given equivalence $\mathbb{D}(S^1)_{\psi/} \to \Delta^{cyc}_*$. □

We now prove that astonishingly Δ^{cyc}_* is equivalent to the more familiar Δ^{op}.

Lemma 20. *Δ^{cyc}_* is equivalent to Δ^{op}, and cofaces maps are sent to cyclic cofaces maps.*

Proof. The proof uses all the previous remarks and a max-min-in-fibers correspondence. Define $F : \Delta^{cyc}_* \to \Delta^{op}$ as

- $F([n]) = [n]$;
- $F(f, k)(x) = \max f^{-1}(y)$, where $y = \max[0, z] \cap \mathrm{im} f$. The first max is taken with respect to the linear order on the fiber. [2].

We prove that Ff is non decreasing. Suppose $x < x'$. if $(x, x'] \cap \mathrm{im} f = \emptyset$, then $[0, x] \cap \mathrm{im} f = [0, x'] \cap \mathrm{im} f$ and we are done. Otherwise, take $w' = \max(x, x'] \cap \mathrm{im} f$, and $w = \max[0, x]$. As the order on the domain is obtained by composing the linear orders on the fibers along the order on the codomain, $f(a) > f(b)$ implies $a > b$. So

$$Ff(x') = \max f^{-1}(w') > \max f^{-1}(w) = Ff(x)$$

[2] To lighten the notation, we drop the dependence from k and we write Ff in place of $F(f, k)$

and we are done.

On the other hand, define $G : \Delta^{op} \to \Delta_*^{cyc}$ as

- $G([n]) = [n]$;
- $Gg(y) = (G_0 g(y), k)$, where $k = \max g$ and $G_0 g(y)$: [3]

$$= 0 \text{ if } y > k$$

$$= \min g^{-1}(z), \text{ where } z = \min[y, \infty) \cap \mathrm{im} g, \text{otherwise}$$

We prove that Gg is in Δ_*^{cyc}. Indeed, $Gg(0) = \min g^{-1}(z), z = \min \mathrm{im} g$, and this is 0. In order to prove that it is cyclic, in virtue of previous remarks, it is enough to show that $0 \leq x < x' \leq \max g$, then $Gg(x) \leq Gg(x')$. As above, if $[x, x') \cap \mathrm{im}(g)$ we are done, because $[x, \infty) \cap \mathrm{im} g = [x', \infty) \cap \mathrm{im} g$. Otherwise, let $z' = \min[x', \infty) \cap \mathrm{im} g$ and $z = \min[x, x') \cap \mathrm{im} g$. Then, as g is non decreasing

$$Gg(x) = \min g^{-1}(z) < \min g^{-1}(z') = Gg(x')$$

We now prove that they are inverses constructions. It will be painful but finite. We verify that $FGg = g$. The other direction is completely analogous. Recall that

$$FGg(x) = \max Gg^{-1}(y) : \quad y = \max[0, x] \cap \mathrm{im} Gg$$

1. We prove that $Gg(g(s)) = \min g^{-1} g(s)$ for every s. Indeed, $\min[g(s), \infty) \cap \mathrm{im} g = g(x)$, so that $Gg(g(s)) = \min g^{-1} g(s)$.
2. We prove that $y \leq Gg(g(x))$. Indeed, set $Gg(\alpha) = \max[0, x] \cap \mathrm{im} g$. Note that we can suppose $\alpha \leq \max \mathrm{im}(g)$: indeed, if $\alpha > \max g$, then $Gg(\alpha) = 0$ and we can take

[3] In the following, to lighten the notation, we denote $G_0 g$ by Gg also.

$\alpha = 0$. Thus wlog $Gg(\alpha) = \min g^{-1}(\beta)$, where $\beta = \min[\alpha, \infty) \cap \mathrm{im} g$. The fact that $Gg(\alpha) = \min g^{-1}(\beta) \leq x$ gives that $\beta \leq g(x)$. In particular

$$Gg(\alpha) = \min g^{-1}(\beta) \leq \min \min g^{-1}(g(x)) = Gg(g(x))$$

3. We prove that $y \geq Gg(g(x))$. As y is the max of $[0, x] \cap \mathrm{im} Gg$, it is enough to prove that $Gg(x) \leq x$. This follows from 1.

4. We are left with showing that $g(x)$ is the maximum in its fiber, i.e. $\max Gg^{-1}(Gg(g(x)) = g(x)$. Indeed, suppose that $Gg(\alpha) = Gg(g(x))$. Also, suppose $Gg(\alpha) = \min g^{-}1(g(\gamma))$, for $g(\gamma) = \min[\alpha, \infty] \cap \mathrm{im} g$. Using 1, we have that $\min g^{-1} g(\gamma) = \min g^{-1} g(x)$. In particular, $g(\gamma) = g(x)$. It follows that $g(x) \geq \alpha$.

A simple calculation using the functor G yield that the image of coface maps are the cyclic coface maps. □

Putting together the pieces we get:

Lemma 21. *$\mathbb{D}(S^1)_{/\psi}$ is equivalent to $\mathcal{N}(\Delta^{op})$. Moreover, after choosing:*

- *disjoint embeddings $\phi_n : \mathbb{R} \to S^1 \setminus \psi(\bar{\mathbb{R}}) \simeq \mathbb{R}$ such that $\psi(0), \phi_n(0), \phi_{n+1}(0)$ are in anticlockwise order;*

- *an embedding $m : \mathbb{R} \times [1] \to \mathbb{R}$ that induces the order $0 \leq 1$;*

- *an embedding $\check{m} : \mathbb{R} \times [1] \to \mathbb{R}$ that induces the order $1 \leq 0$;*

the equivalence $\mathcal{N}(\Delta^{op}) \to \mathbb{D}(S^1)_{/\psi}$ maps

$$[n] \quad \mapsto \quad (\psi, \phi_1, \ldots, \phi_n) : \mathbb{R}^k \times [n] \to S^1$$

$$d_i^{op} : [n] \to [n-1] \quad \mapsto \quad \partial_i : \mathbb{R}^k \times [n] \to \mathbb{R}^k \times [n-1]$$

where $\pi_0(\partial_i) : [n] \to [n-1]$ *contracts the* i*-th and the* $(i+1)$*-term (*$n+1$ *must be read zero), and*

$$\partial_i|\mathbb{R} \times \{i, i+1\} = m \text{ if } i \neq n, \check{m} \text{ otherwise}$$

$$\partial_i|\mathbb{R} \times \{j\} = Id, \quad j \neq i, i+1$$

Corollary 4.5.2.1. *Factorization homology over* S^1 *gives the Hochschild homology.*

Proof. Take an $\mathbb{E}_1$ algebra $A : \mathbb{E}_1^{\otimes} \to \mathcal{C}^{\otimes}$. The associated locally constant $\mathrm{Disk}_{\mathbb{R}}$ algebra is given by $p^*(A)(U) = A$ for U a disk, and tensor maps encoded by the multiplication in A. Global factorization homology (that is, global sections of the operadic Kan extension to all the opens) is computed by the colimit of $\mathbb{D}(S^1) \to \mathcal{C}^{\otimes}$, informally sending $(f_1, \ldots, f_n)$ to $p^*A(\mathrm{im} f_0) \otimes \ldots \otimes (\mathrm{im} f_n) = A^{\otimes n}$.

By lemma 18, we can equivalently compute this as the colimit of $\mathbb{D}(S^1)_{/\psi} \to \mathbb{D}(S^1) \to \mathcal{C}^{\otimes}$. By lemma 21, we can equivalently compute the latter as $\mathcal{N}(\Delta^{op}) \to \mathbb{D}(S^1)_{/\psi} \to \mathcal{C}^{\otimes}$ by the functor

$$[n] \mapsto (\psi, \phi_1, \ldots, \phi_n) \mapsto A^{\otimes(n+1)}$$

In particular, if we take $\mathcal{C}^{\otimes}$ to be chain complexes, we can compute this hocolimit as the totalization of the associated bicomplex. Then the simplicial complex gives a chain complex by

$$\partial(a_0 \otimes a_1 \otimes \ldots \otimes a_n) = \sum_{k=0}^{n} (-1)^k \partial_k(a_0, a_1, \ldots, a_n) =$$

$$= \sum_{k=0}^{n-1} (-1)^k (a_0 \otimes \ldots \otimes a_k a_{k+1} \otimes \ldots \otimes a_n) + (-1)^n (a_n a_0 \otimes \ldots \otimes a_{n-1})$$

which is precisely the Hochschild differential. Note that the multiplication comes from the fact that the various ∂_i couples two opens $\{i, i+i\}$ via m (or $\check{m}$ in the last case), and this induces the multiplication. $\square$

See these notes [13] for a series of interesting results and computations in Hochschild homology.

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